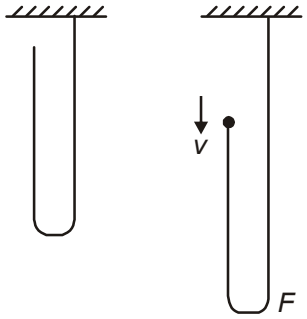
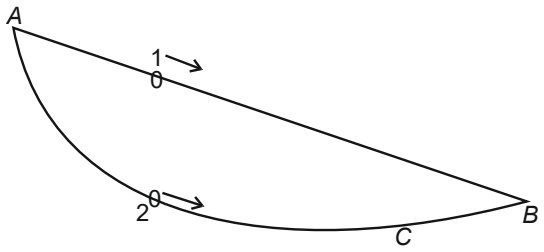
Physics

KINEMATICS

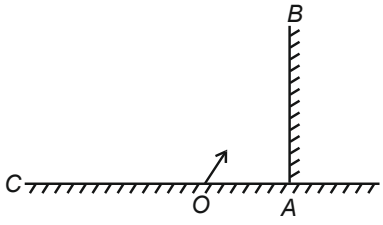
1. One end of a rope is fixed at a point on the ceiling the other end is held close to the first end so that the rope folded. The second end is released from this position. Find the speed at which the fold at *F* is descending at the instant the free end of the rope is going down at speed *v*.



2. Two frictionless ropes connect points *A* and B in vertical plane. Bead 1 is allowed to slide along the straight rope *AB* and bead 2 slides along the curved rope *ACB*. Which bead will reach *B* in less time?



3. A ball is projected from point *O* on the ground. It hits a smooth vertical wall *AB* at a height *h* and rebounds elastically. The ball finally lands at a point *C* on the ground. During the course of motion, the maximum height attained by the ball is *H*.



(a) Find the ratio $\frac{h}{H}$ if $\frac{OA}{OC}=\frac{1}{3}$

(b) Find the magnitude of average acceleration of the projectile for its entire course of flight if it was projected at an angle of 45° to the horizontal.

4. A rocket prototype is fired from ground at time $t = 0$ and it goes straight up. Take the launch point as origin and vertically upward direction as positive x direction. The acceleration of the rocket is given by

$$a = \frac{g}{2} - kt^2 \; ; \; 0 < t \le t_0$$

$$= -g \; ; \qquad t > t_0$$

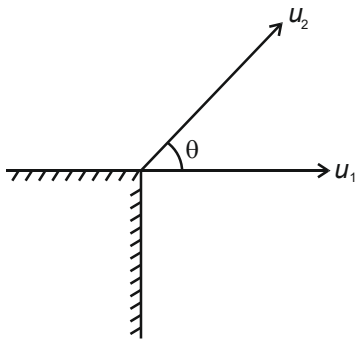
where $t_0 = \sqrt{\frac{3g}{2k}}$

(a) Find maximum velocity of the rocket during the up journey.

(b) Find maximum height attained by the rocket.

(c) Find total time of flight.

5. Two balls are projected simultaneously from the top of a tall building. The first ball is projected horizontally at speed u_1 = 10 m/s and the other one is projected at an angle $\theta = \tan^{-1}\left(\frac{4}{3}\right)$ to the horizontal with a velocity u_2. [g = 10 m/s^2]



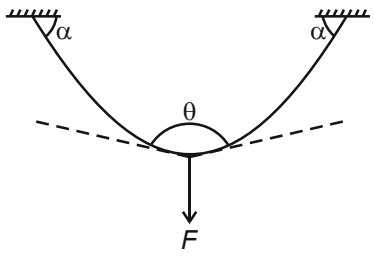
(a) Find minimum value of u_2 (= u_0) so that the velocity vector of the two balls can get perpendicular to each other at some point of time during their course of flight.

(b) Find the time after which velocities of the two balls become perpendicular if the second one was projected with speed u_0.

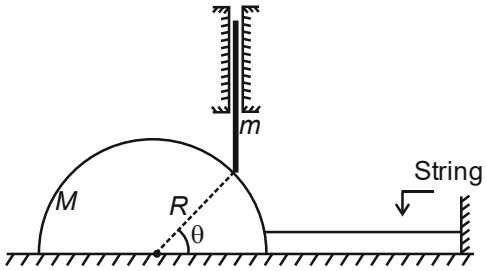
6. A particle rotates in a circle with angular speed ω_0. A retarding force decelerates it such that angular deceleration is always proportional to square root of angular velocity. Find the mean angular velocity of the particle averaged over the whole time of rotation.

NEWTON'S LAWS

1. A rope of mass m is hung from a ceiling. The centre point is pulled down with a vertical force F. The tangent to the rope at its ends makes an angle α with horizontal ceiling. The two tangents at the lower point make an angle of θ with each other. Find θ.



2. A hemisphere of mass M and radius R rests on a smooth horizontal table. A vertical rod of mass m is held between two smooth guide walls supported on the sphere as shown. There is no friction between the rod and the sphere. A horizontal string tied to the sphere keeps the system at rest.



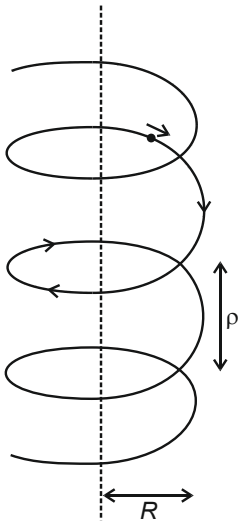
(a) Find tension in the string.

(b) Find the acceleration of the hemisphere immediately after the string is cut.

3. A long helix made of thin wire is held vertical. The radius and pitch of the helix are R and ρ respectively. A bead begins to slide down the helix.

(a) Find the normal force applied by the wire on the bead when the speed of the bead is v.

(b) Eventually, the bead acquires a constant speed of v_0. Find the coefficient of friction between the wire and the bead.

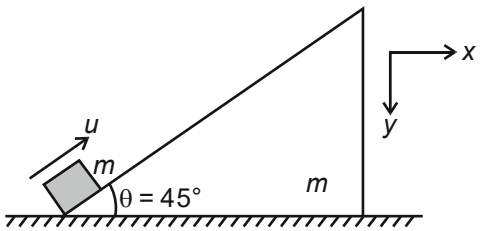


4. A wedge of mass m is kept on a smooth table and its inclined surface is also smooth. A small block of mass m is projected from the bottom along the incline surface with velocity u. Assume that the block remains on the incline and take $\theta = 45°$, $g = 10\ m/s^2$.

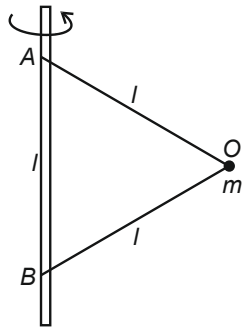
(a) Find the acceleration of the wedge and the x and y components of acceleration of the block.

(b) Draw the approximate path of the block as observed by an observer on the ground. At what angle does the block hit the table?

(c) Calculate the radius of curvature of the path of the block when it is at the highest point.



5. A particle of mass m is attached to a vertical rod with two inextensible strings AO and BO of equal lengths l. Distance between A and B is also l. The setup is rotated with angular speed ω with rod as the axis.



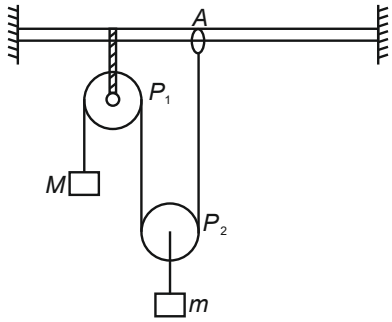
(a) Find the values of ω for which the particle remains at point B.

(b) Find the range of values of ω for which tension (T_1) in the string AO is greater than mg but the other string remains slack.

(c) Find the value of ω for which tension (T_1) in string AO is twice the tension (T_2) in string BO.

(d) Assume that both strings are taut when the string AO breaks. What will be nature of path of the particle moment after AO breaks?

6. The arrangement shown in figure is in equilibrium with all strings vertical. The end A of the string is tied to a ring which can be slide slowly on the horizontal rod. Pulley P_1 is rigidly fixed but P_2 can move freely. A mass m is attached to the centre of pulley P_2 through a thread. Pulleys and strings are massless.

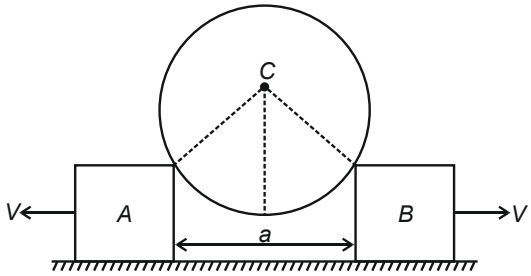


(a) Which block will move up as A is moved slowly to the right?

(b) Will the block of mass m have horizontal displacement?

(c) Is it possible, for a particular position of A, that M has no acceleration but m does have an acceleration? If this happens when string from P_2 to A makes an angle θ with vertical, find the acceleration of m at the instant.

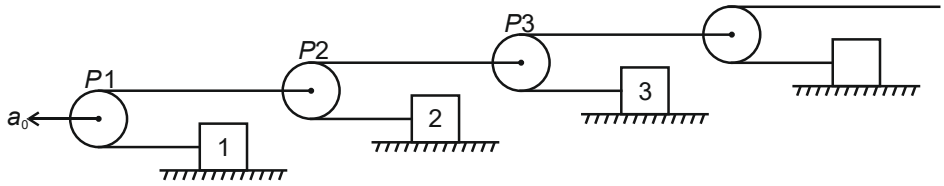
7. A smooth spherical ball of mass M = 2 kg is resting on two identical blocks A and B as shown in the figure. The blocks are moved apart with same horizontal velocity V = 1 m/s in opposite directions (see figure).



(a) Find the normal force applied by each of the blocks on the sphere at the instant separation between the blocks is $a = \sqrt{2}R$; $R = 1.0$ m being the radius of the ball.

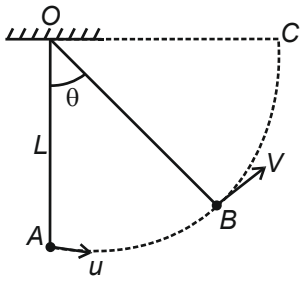
(b) How much force must be applied on each of the two blocks (when $a = \sqrt{2}R$) so that they do not have any acceleration. Assume that the horizontal surface is smooth.

8. In the figure all pulleys ($P1$, $P2$, $P3$ …) are massless and all the blocks (1, 2, 3 …) are identical, each having mass m. The system consist of infinite number of pulleys and blocks. Strings are light and inextensible and horizontal surfaces are smooth. Pulley $P1$ is moved to left with a constant acceleration of a_0. Find the acceleration of block1. Assume the strings to remain horizontal.



WORK, ENERGY AND POWER

1. A pendulum bob is projected form its lowest position with velocity (u), in horizontal direction, that is just enough to make the string horizontal (position OC). At angular position θ, at point B, the speed (V) of the bob was observed to be half its initial projection speed (u).

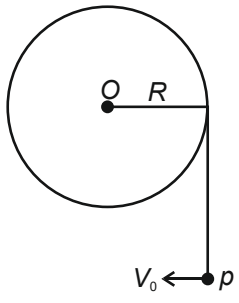


(a) Find θ

(b) Plot variation of magnitude of tangential acceleration with θ.

(c) Let the travel time from A to B be t_1 and that from B to C be t_2. Looking at the graph obtained in part (b), tell which is larger t_1 or t_2?

2. A light thread is tightly wrapped around a fixed disc of radius R. A particle of mass m is tied to the end P of the thread and the vertically hanging part of the string has length πR. The particle is imparted a horizontal velocity $V = \sqrt{\frac{4\pi gR}{3}}$. The string wraps around the disc as the particle moves up. At the instant the velocity of the particle makes an angle of $\theta = 60°$ with horizontal, calculate.

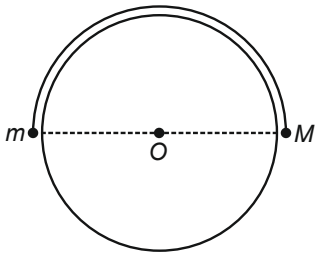


(a) Speed of the particle

(b) Tension in the string

3. Two particles of masses M and m ($M > m$) are connected by a light string of length πR.

The string is hung over a fixed circular frame of radius R.



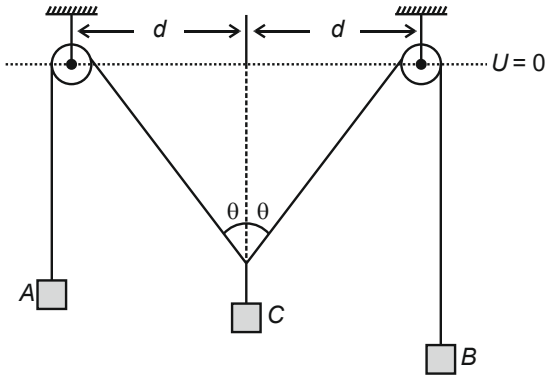
Initially the particles lie at the ends of the horizontal diameter of the circle (see figure). Neglect friction.

(a) If the system is released, and if m remains in contact with the circle, find the speed of the masses when M has descended through a distance $R\theta$ ($\theta < \pi$).

(b) Find the reaction force between the frame and m at this instant.

(c) Prove that m_1 will certainly remain in contact with the frame, just after the release, if $3m > M$.

4. In the arrangement shown in the fig. all the three blocks have equal mass m. The length of the strings connecting A to C and B to C is L each. Assume the gravitational potential energy of any mass at the level of the pulleys to be zero. Neglect dimension of the pulley and treat the strings to be massless. Distance between the pulleys is $2d$.



(a) Write the potential energy of the system as a function of angle θ.

(b) Knowing that potential energy of the system will be maximum or minimum in equilibrium position, find value of θ for equilibrium.

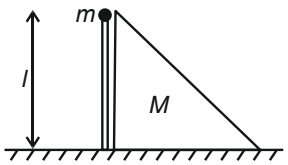
(c) Tell if the equilibrium is stable or unstable.

5. A particle of mass m is attached to an end of a light rigid rod of length a. The other end of the rod is fixed, so that the rod can rotate freely in vertical plane about its fixed end. The mass m is given a horizontal velocity u at the lowest point.

(a) Prove that when the radius to the mass makes an angle θ with the upward vertical the horizontal component of the acceleration of the mass (measured in direction of u) is $[g(2 + 3\cos\theta) - u^2/a]\sin\theta$

(b) If $4ag < u^2 < 5ag$, show that there are four points at which horizontal component of acceleration is zero. Locate the points.

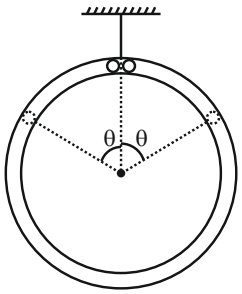
6. A weightless rod of length l with a small load of mass m at one of its end is held vertical with its lower end hinged on a horizontal surface. The load touches a wedge of mass M in this position. A slight jerk towards right sets the system in motion (see figure), with rod rotating freely in vertical plane about its lower end. There is no friction.



(a) For what mass ratio $\frac{M}{m}$ will the rod form an angle $\theta = \frac{\pi}{3}$ with the vertical at the moment the load separates from the wedge?

(b) What is speed of the wedge at that moment? Neglect friction.

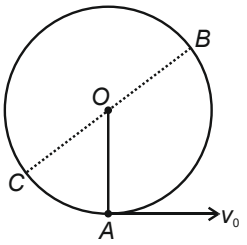
7. A tube of mass *M* hangs from a thread and two balls of mass *m* slide inside it without friction (see figure). The balls are released simultaneously from the top of the ring and slide down on opposite sides. θ defines the positions of balls at any time as shown in figure.



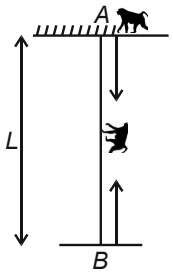
(a) Show that ring will start to rise if $m \geq \frac{3M}{2}$.

(b) If $M = 0$, find the angle θ at which the tube begins to rise.

8. A heavy particle is attached to one end of a light string of length *l* whose other end is fixed at *O*. The particle is projected horizontally with a velocity v_0 from its lowest position A. When the angular displacement of the string is more than 90°, the particle leaves the circular path at *B*. The string again becomes taut at *C* such that B, O, C are collinear. Find v_0 in terms of *l* and *g*.

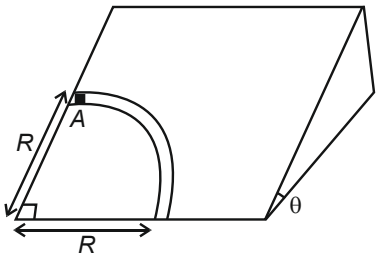


9. *AB* is a vertically suspended elastic cord of negligible mass and length *L*. Its force constant is $k = \frac{4mg}{L}$. There is a massless platform attached to the lower end of the cord. A monkey of mass *m* starts from top end *A* and slides down the cord with a uniform acceleration of $\frac{g}{2}$. Just before landing on the platform, the monkey loses grip on the cord. After landing on the platform the monkey stays on it. Calculate the maximum extension in the elastic cord.



MOMENTUM AND CENTER OF MASS

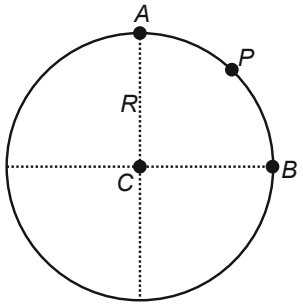
1. A wooden wedge of mass $10m$ has a smooth groove on its inclined surface. The groove is in shape of quarter of a circle of radius R = 0.55 m. The inclined face makes an angle $\theta = \cos^{-1}\left(\frac{\sqrt{11}}{5}\right)$ with the horizontal. A block 'A' of mass m is placed at the top of the groove and given a gentle push so as to slide along the groove. There is no friction between the wedge and the horizontal ground on which it has been placed. Neglect width of the groove.



(a) Find the magnitude of displacement of the wedge by the instant the block A reaches the bottom of the groove.

(b) Find the velocity of the wedge at the instant the block A reaches the bottom of the groove.

2. A disc of mass M and radius R lies on a smooth horizontal table. Two men, each of mass $\frac{M}{2}$, are standing on the edges of two perpendicular radii at A and B.

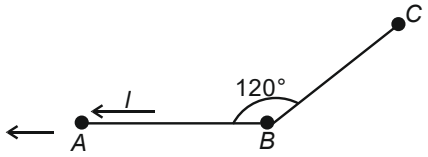


Find the displacement of the centre of the disc if

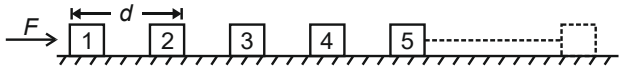
(a) The two men walk radically relative to the disc so as to meet at the centre C.

(b) The two men walk along the circumference to meet at the midpoint(P) of the arc AB.

3. Three particles A, B and C have masses m, $2m$ and m respectively. They lie on a smooth horizontal table connected by light inextensible strings AB and BC. The string are taut and $\angle ABC = 120°$. An impulse is applied to particle A along BA so that it acquires a velocity u. Find the initial speeds of B and C.



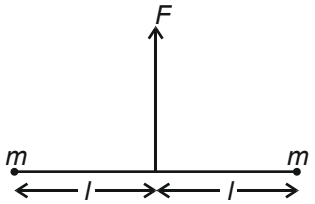
4. A large number of small identical blocks, each of mass m, are placed on a smooth horizontal surface with distance between two successive blocks being d. A constant force F is applied on the first block as shown in the figure.



(a) If the collisions are elastic, plot the variation of speed of block 1 with time.

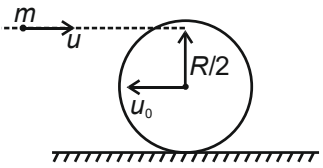
(b) Assuming the collisions to be perfectly inelastic, find the speed of the moving blocks after n collisions. To what value does this speed tend to if n is very large.

5. Two small balls, each of mass m are placed on a smooth table, connected with a light string of length $2l$, as shown in the figure. The midpoint of the string is pulled along y direction by applying a constant force F. Find the relative speed of the two particles when they are about to collide. If the two masses collide and stick to each other, how much kinetic energy is lost.



6. A mass m moving with speed u in x direction collides elastically with a stationary mass $2m$. After the collision, it was found that both masses have equal x components of velocity. What angle does the velocity of mass $2m$ make with the x axis?

7.

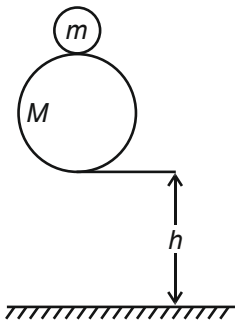


A heavy ball of radius R is travelling on a smooth horizontal surface with a velocity of u_0 towards left. A horizontally moving small ball of mass m strikes it at a height $\frac{R}{2}$ above the centre while travelling with velocity u towards right.

(a) After collision the small ball moves in vertically upwards direction with velocity u. Prove that this can happen only if $u > \sqrt{3}\,u_0$

(b) Find the velocity of small ball after collision if the collision is elastic and the balls are smooth.

8. Two elastic balls of masses M and m ($M >> m$) are placed on top of each other with a small gap between them. The balls are dropped on to the ground with the bottom of the lower ball at height h above the ground. The lower ball has a radius R and the upper ball has negligible dimension.

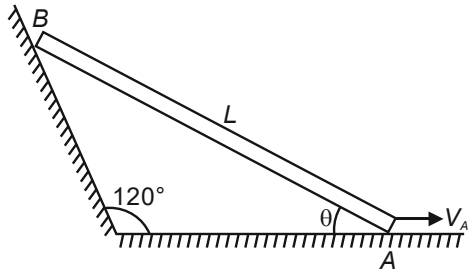


(a) Up to what height the ball of mass m will bounce above the ground?

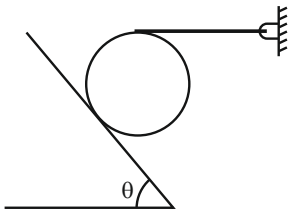
(b) Does the result obtained above violates the low of conservation of mechanical energy?

ROTATIONAL MOTION

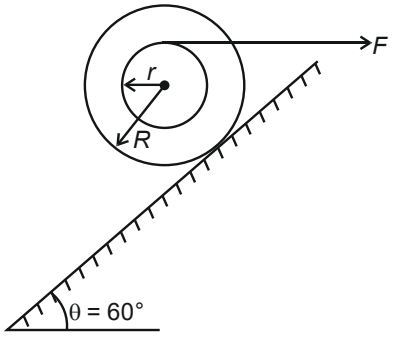
1. A wall is inclined to a horizontal surface at an angle of 120° as shown. A rod *AB* of length *L* = 0.75 m is sliding with its two ends *A* and *B* on the horizontal surface and on the wall respectively. At the moment angle θ = 20° (see figure), the velocity of end *A* is v_A = 1.5 m/s towards right. Calculate the angular speed of the road at this instant. [Take cos40° = 0.766]



2. A horizontal stick of mass *m* has its right end attached to a pivot on a wall, while its left end rests on the top of a cylinder of mass *m* which in turn rests on an incline plane inclined at an angle θ. The stick remains horizontal. The coefficient of friction between the cylinder and both the plane and the stick is μ. Find the minimum value of μ as function of θ for which the system stays in equilibrium.

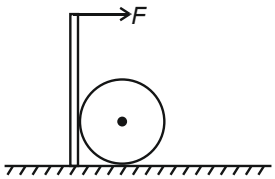


3. A spool is kept in equilibrium on an incline plane as shown in figure. The inner and outer radii of the spool are in ratio $\frac{r}{R}=\frac{1}{2}$. The force applied on the thread (wrapped on part of radius *r*) is horizontal. Find the angle that the force applied by the incline on the spool makes with the vertical. [Take $\tan^{-1}\left(\frac{\sqrt{3}}{5}\right) \simeq 19°$]



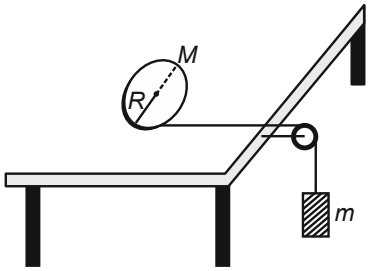
4. A uniform rod of mass *M* and length *L* is hinged at its lower end on a table. The rod can rotate freely in vertical plane and there is no friction at the hinge. A ball of mass *M* and radius $R=\frac{L}{3}$ is placed in contact with the vertical rod and a horizontal force *F* is applied at the upper end of the rod.

   (a) Find the acceleration of the ball immediately after the force starts acting.



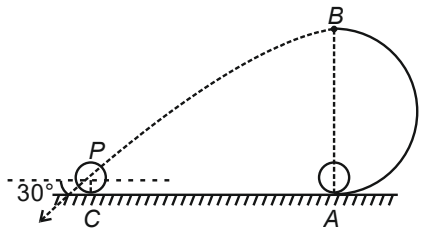
   (b) Find the horizontal component of hinge force acting on the rod immediately after force *F* starts acting.

5. A light thread has been tightly wrapped around a disc of mass M and radius R. The disc has been placed on a smooth table, lying flat as shown.



The other end of the string has been attached to a mass m as shown. The system is released from rest. If $m = M$, which point of the disc will have zero acceleration, immediately after the system is released?

6. A disc of radius $r = 0.1$ m is rolled from a point A on a track as shown in the figure. The part AB of the track is a semi-circle of radius R in a vertical plane. The disc rolls without sliding and leaves contact with the track at its highest point B. Flying through the air it strikes the ground at point C. The velocity of the center of mass of the disc makes an angle of 30° below the horizontal at the time of striking the ground. At the same instant, velocity of the topmost point P of the disc is found to be 6 m/s. (Take $g = 10$ m/s^2).

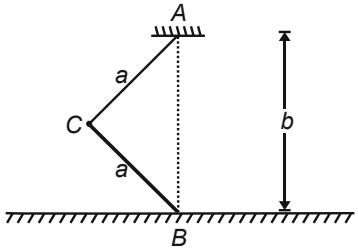


(a) Find the value of R.

(b) Find the velocity of the center of mass of the disc when it strikes the ground.

(c) Find distance AC.

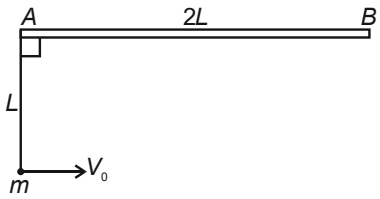
7. A uniform rod BC with length a is attached to a light string AC. End A of the string is fixed to the ceiling and the end B of the rod is on a smooth horizontal surface. B is exactly below point A and length AB is $b(a < b < 2a)$. The system is released from rest and the rod begins to slide. Find the speed of the centre of the rod when the string becomes vertical.



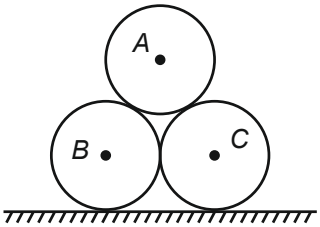
8. A uniform rod of mass m and length $2L$ on a smooth horizontal surface. A particle of mass m is connected to a string of length L whose other end is connected to the end 'A' of the rod. Initially the string is held taut perpendicular to the rod and the particle is given a velocity v_0 parallel to the initial position of the rod.

(a) Calculate the acceleration of the centre of the rod immediately after the particle is projected.

(b) The particle strikes the centre of the rod and sticks to it. Calculate the angular speed of the rod after this.



9. Three identical cylinders have mass M each and are placed as shown in the figure. The system is in equilibrium and there is no contact between B and C. Find the normal contact force between A and B.



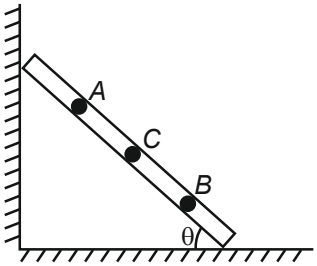
10. A ladder of mass M and Length L stays at rest against a smooth wall. The coefficient of friction between the ground and the ladder is μ.

(a) Let F_{wall}, W and F_g be the force applied by wall, weight of the ladder and force applied by ground on the ladder. Argue to show that the line of action of these three forces must intersect.

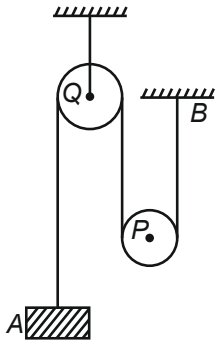
(b) Using the result obtained in (a) show that line of action of F_g makes an angle $\tan^{-1}(2\tan\theta)$ with the horizontal ground where θ is the angle made by the ladder with the ground.

(c) Find the smallest angle that the ladder can make with the ground and not slip.

(d) You climb up the ladder, your presence makes the ladder more likely to slip. Where are you at A or B? C is the centre of mass of the ladder.

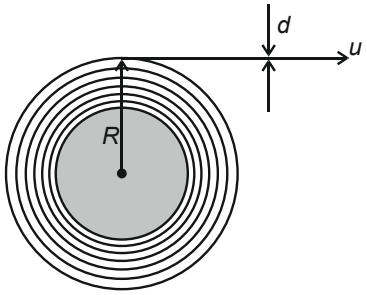


11.

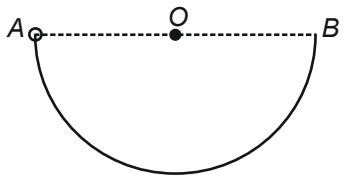


A light thread is wrapped tightly a few turns around a disc P of mass M. One end of the thread is fixed to the ceiling at B. The other end of the thread is passed over a massless pulley (Q) and carries a block of mass M. All segment of the thread (apart from that on the pulley and disc) are vertical when the system is released. Find the acceleration of block A. On which object – the block A or the ceiling at B – does the thread exert more force?

12. A disc is free to rotate about an axis passing through its centre and perpendicular to its plane. The moment of inertia of the disc about its rotation axis is I. A light ribbon is tightly wrapped over it in multiple layers. The end of the ribbon is pulled out at a constant speed of u. Let the radius of the ribboned disc be R at any time and thickness of the ribbon be $d(<< R)$. Find the force (F) required to pull the ribbon as a function of radius R.



13. A uniform semicircular wire is hinged at '*A*' so that it can rotate freely in vertical plane about a horizontal axis through '*A*'. The semicircle is released from rest when its diameter *AB* is horizontal.



Find the hinge force at '*A*' immediately after the wire is released.

GRAVITATION

1. An astronaut on the surface of the moon throws a piece of lunar rock (mass *m*) directly towards the earth at a great speed such that the rock reaches the earth.

   Mass of the earth = *M*, Mass of the moon = $\frac{M}{81}$

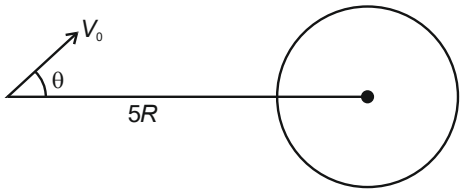
   Radius of the earth = *R*, Distance between the centre of the earth and the moon = 60*R*

   (a) In the course of its journey calculate the maximum gravitational potential energy of the rock

   (b) Find the minimum possible speed of the rock when it enters the atmosphere of the earth.

2. A spy satellite S_1, travelling above the equator is taking pictures at quick intervals. The satellite is travelling from west to east and is ready with picture around the whole equator in 8 hours. Another similar satellite S_2, travelling in the same plane is travelling from east to west and is able to take pictures around the whole equator in 6 hours. Find the ratio of radii of the circular paths of the satellite S_1 and S_2.

3. An asteroid was fast approaching the earth. Scientists fired a rocket which hit the asteroid at a distance of 5 *R* from the centre of the earth (*R* = radius of the earth). Immediately after the hit the asteroid's velocity (V_0) was making an angle $\theta = 30°$ with the line joining the centre of the earth to the asteroid. The asteroid just grazed past the surface of the earth. Find V_0 [Mass of the earth = *M*]



4. A satellite is orbiting around the earth in a circular orbit. Its orbital speed is V_0. A rocket on board is fired from the satellite which imparts a thrust to the satellite directed radially away from the centre of the earth. The duration of the engine burn is negligible so that it can be considered instantaneous. Due to this thrust a velocity variation ΔV is imparted to the satellite. Find the minimum value of the ratio $\frac{\Delta V}{V_0}$ for which the satellite will escape out of the gravitational field of the earth.

5. Earth is rotating about its axis with angular speed ω_0 and average density of earth is ρ. It is proposed to make a space elevator by placing a long rod with uniform mass density extending from just above the surface for the earth out to a radius nR (R is radius of the earth). Prove that the rod can remain above the same point on the equator all time if, $n^2 + n = \frac{8\pi G\rho}{3\omega^2}$, where ρ is density of the earth.

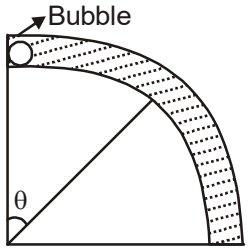
6. A body is projected up from the surface of the earth with a velocity half the escape velocity at an angle of 30° with the horizontal. Neglecting air resistance and earth's rotation, find

   (a) The maximum height above the earth's surface to which the body will rise.

   (b) Will the body move around the earth as a satellite?

FLUIDS

1. A bent tube contains water. An air bubble is trapped inside the liquid. The tube is held vertical (as shown) and is moved horizontally with an acceleration (a) such that the bubble moves to position θ shown in the diagram. Find the direction and magnitude of a.

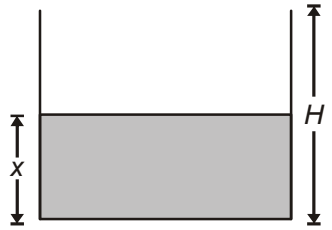


2. A cylindrical container has mass M and height H. The centre of mass of the empty container is at height $\frac{H}{2}$ from the base. A liquid, when completely filled in the container, has mass $\frac{M}{2}$.

   This liquid is poured in the empty container.

   (a) How does the centre of mass of the system (container + liquid) move as the height (x) of liquid column changes from zero to H? Explain your answer qualitatively. Draw a graph showing the variation of height of centre of mass of the system (x_{cm}) with x.

   (b) Find the height of liquid column x from which the centre of mass is at its lowest position.

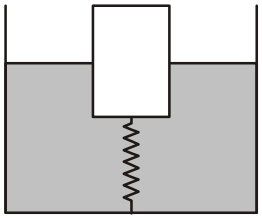


3. A cylindrical container contains water. A cubical block is floating in water with its lower surface connected to a spring

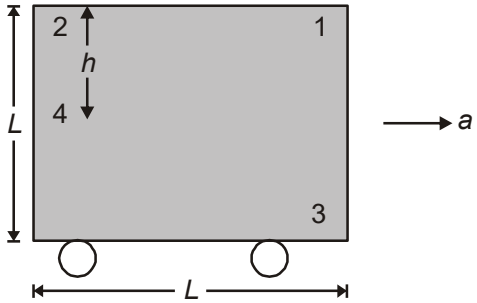
   (a) Suppose that the spring is in relaxed state. Now, if the whole container is accelerated vertically upwards, will the spring get compressed?

   (b) Suppose that the spring is initially compressed. Now, what will happen to the state of the spring when the container is accelerated upwards?

(c) Assume that mass of the block is 1 kg and initially the spring (force constant k = 100 N/m) is compressed by 5 cm. When the container is accelerated up by an acceleration of 5 m/s^2, the spring has a total compression of 6 cm. Calculate the change in volume of block submerged inside water when the container gets accelerated. Density of water is 10^3 kg/m^3.



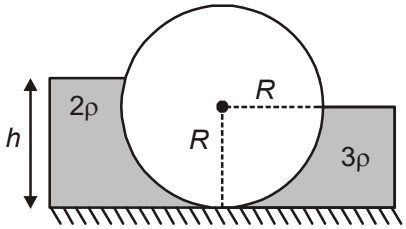
4. A cubical container of side length L is filled completely with water. The container is closed. It is accelerated horizontally with acceleration a. Density of water is ρ.



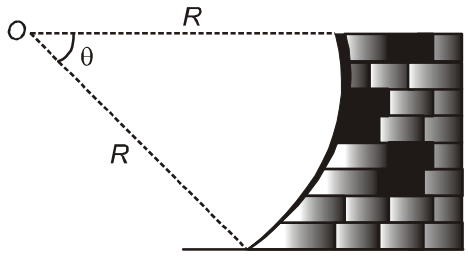
(a) Assuming pressure at point 1 [upper right corner] to be zero, find pressure at point 2 [upper left corner]

(b) Pressure at point 4, at a distance h vertically below point 2, is same as pressure at lower right corner 3. Find h.

5. (i) In the figure shown, the heavy cylinder (radius R) resting on a smooth surface separates two liquids of densities 2ρ and 3ρ. Find the height 'h' for the equilibrium of cylinder.



(ii) The cross section of a dam wall is an arc of a circle of radius R = 20 m subtending on angle of θ = 60° at the centre of the circle. The centre (O) of the circle lies in the water surface. The width of the dam [i.e., dimension perpendicular to the figure] is b = 10m. Neglect atmospheric pressure in following calculations.



(a) Calculate the vertical component of force (F_v) applied by water on the curved dam wall.

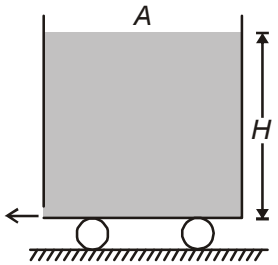
(b) Calculate the horizontal component of force (F_H) applied by water on the curved dam wall.

(c) Calculate the resultant force applied by the water on the curved dam wall.

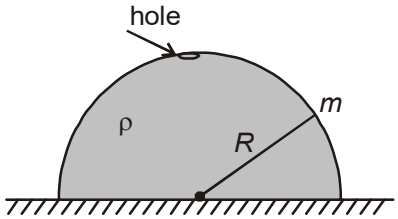
6. An open tank of cross sectional area A contains water up to height H. It is kept on a smooth horizontal surface. A small orifice of area A_0 is punched at the bottom of the wall of the tank. Water begins to drain out. Mass of the empty tank may be neglected.

   (i) Prove that the tank will move with a constant acceleration till it is emptied. Find this acceleration.

   (ii) Find the final speed acquired by the tank when it is completely empty.



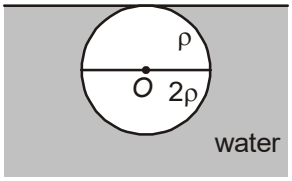
7. A hemispherical bowl of radius R is placed upside down on a flat horizontal surface. There is a small hole at the top of the inverted bowl. Through the hole, a liquid of density ρ is poured in. Exactly when the container gets full, water starts leaking from between the table and the edge of the container. Find the mass (m) of the container.



8. A spherical ball of radius R is made by joining two hemispherical parts. The two parts have density ρ and 2ρ. When placed in a water tank, the ball floats while remaining completely submerged.

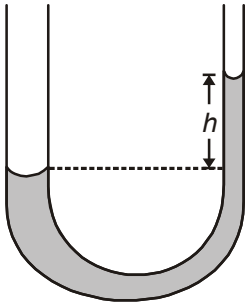
   (a) If density of water is ρ_0, find ρ.

   (b) Find the time period of small angular oscillations of the ball about its equilibrium position. Neglect viscous forces.

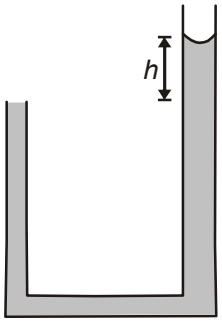


SURFACE TENSION

1. The radii of two columns in a U tube are r_1 and r_2 ($r_1 > r_2$). A liquid of density ρ is filled in it. The contact angle of the liquid with the tube wall is θ. If the surface tension of the liquid is T then plot the graph of the level difference (h) of the liquid in the two arms versus contact angle θ. Plot the graph for angle θ changing from 0° to 90°. Assume the curved surface of meniscus to be part of a sphere.



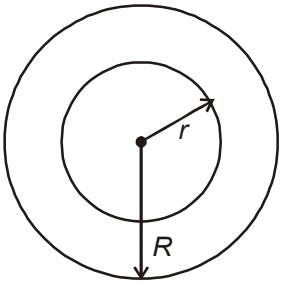
2. Two capillaries of small cross section are connected as shown in the figure. The right tube has cross sectional radius R and left one has a radius of $r(< R)$. The tube of radius R is very long where as the tube of radius r is of short length. Water is slowly poured in the right tube. Contact angle for the tube wall and water is $\theta = 0°$. Let h be the height difference between water surface in the right and left tube. Surface tension of water is T and its density is ρ.



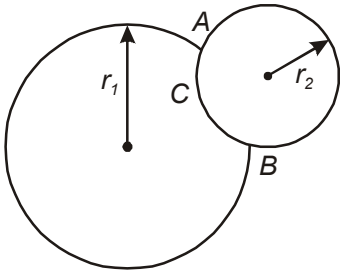
(a) Find the value of h if the water surface in the left tube is found to be flat.

(b) Find the maximum value of h for which water will not flow out of the left tube.

3. A soap bubble of radius r is formed inside another soap bubble of radius R ($> r$). The atmospheric pressure is P_0 and surface tension of the soap solution is T. Calculate change in radius of the smaller bubble if the outer bubble bursts. Assume that the excess pressure inside a bubble is small compared to P_0.



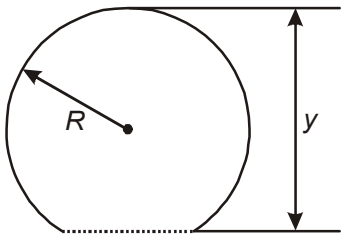
4. Two soap bubbles of radii r_1 and r_2 are attached as shown. Find the radius of curvature of the common film ACB.



5. (a) In the last question find the angle between the tangents drawn to the bubble surfaces at point A.

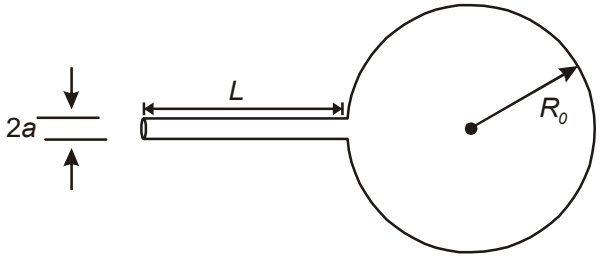
(b) In the above question assume that $r_1 = r_2 = r$. What is the shape of the common interface ACB? Find length AB in this case.

(c) With $r_1 = r_2 = r$ the common wall bursts and the two bubbles form a single bubble find the radius of this new bubble. It is given that volume of a truncated sphere of radius R and height y is $\frac{\pi}{3}y^2(3R-y)$ [see figure]



6. A soap bubble is blown at the end of a capillary tube of radius *a* and length *L*. When the other end is left open, the bubble begins to deflate. Write the radius of the bubble as a function of time if the initial radius of the bubble was R_0. Surface tension of soap solution is *T*. It is known that volume flow rate through a tube of radius *a* and length *L* is given by Poiseuille's equation

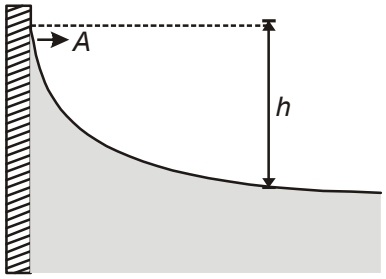
$$Q = \frac{\pi a^4 \Delta P}{8\eta L}$$



Where ΔP is pressure difference at the two ends of the tube and η is coefficient of viscosity. Assume that the bubble remains spherical.

7. A liquid having surface tension *T* and density ρ is in contact with a vertical solid wall. The liquid surface gets curved as shown in the figure. At the bottom the liquid surface is flat.

The atmospheric pressure is P_0.

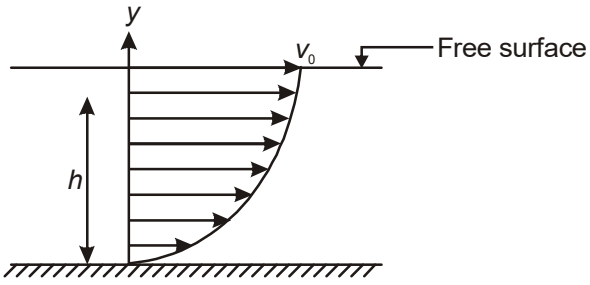


(i) Find the pressure in the liquid at the top of the meniscus (i.e. at *A*)

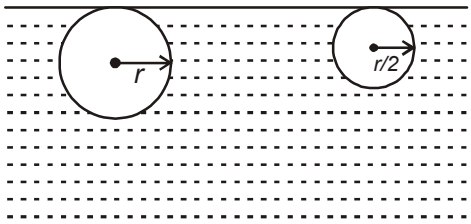
(ii) Calculate the difference in height (*h*) between the bottom and top of the meniscus.

VISCOSITY

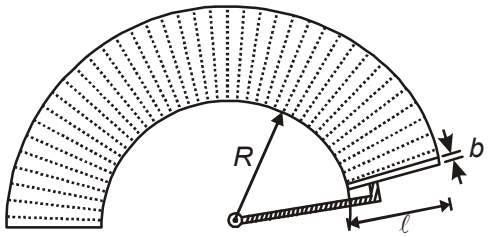
1. A liquid is flowing through a horizontal channel. The speed of flow (*v*) depends on height (*y*) from the floor as $v = v_0\left[2\left(\frac{y}{h}\right) - \left(\frac{y}{h}\right)^2\right]$. Where *h* is the height of liquid in the channel and v_0 is the speed of the top layer. Coefficient of viscosity is η. Calculate the shear stress that the liquid exerts on the floor.



2. Two balls of radii *r* and $\frac{r}{2}$ are released inside a deep water tank. Their initial accelerations are found to be $\frac{g}{2}$ and $\frac{g}{4}$ respectively. Find the velocity of smaller ball relative to the larger ball, a long time after the two balls are released. Coefficient of viscosity is given to be η.



3. A car windshield wiper blade sweeps the wet windshield rotating at a constant angular speed of ω. *R* is the radius of innermost arc swept by the blade. Length and width of the blade are ℓ and *b* respectively. Coefficient of viscosity of water is η. Calculate the torque delivered by the motor to rotate the blade assuming that there is a uniform layer of water of thickness *t* on the glass surface.



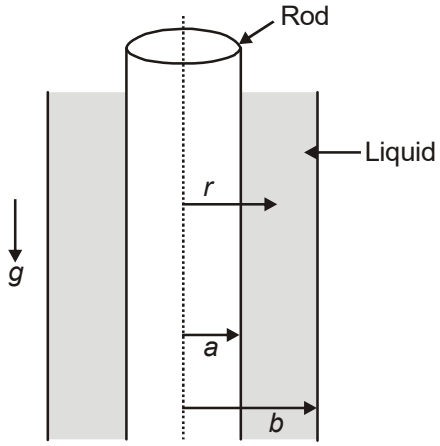
4. A vertical steel rod has radius *a*. The rod has a coat of a liquid film on it. The liquid slides under gravity. It was found that the speed of liquid layer at radius *r* is given by

$$v = \frac{\rho g b^2}{2\eta}\ln\left(\frac{r}{a}\right) - \frac{\rho g}{4\eta}(r^2 - a^2)$$

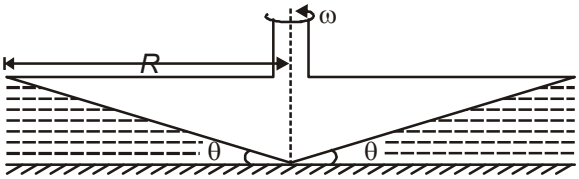
Where *b* is the outer radius of liquid film, η is coefficient of viscosity and ρ is density of the liquid

(i) Calculate the force on unit length of the rod due to the viscous liquid?

(ii) Set up the integral to calculate the volume flow rate of the liquid down the rod. [you may not evaluate the integral]

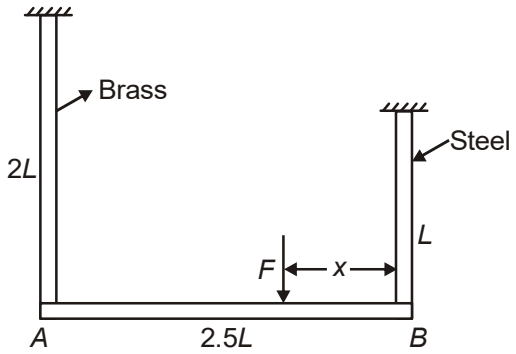


5. A viscometer (an instrument used to study characteristics of a non-ideal fluid) consists of a flat plate and a rotating cone. The cone has a large apex angle and the angle θ shown in figure is very small (typically less than 0.5°). The apex of the cone just touches the plate and a liquid fills the narrow gap between the plate and the cone. The cone has a base radius *R* and is rotated with constant angular speed ω. Consider the liquid to be ideal and take its coefficient of viscosity to be η. Calculate the torque needed to drive the cone.



ELASTICITY

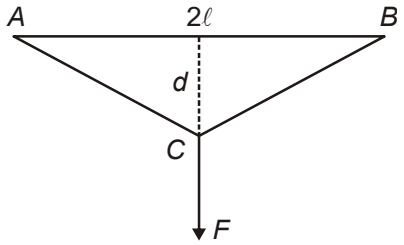
1. A very stiff bar (AB) of negligible mass is suspended horizontally by two vertical rods as shown in figure. Length of the bar is 2.5 L. The steel rod has length L and cross sectional radius of r and the brass rod has length $2L$ and cross sectional radius of $2r$. A vertically downward force F is applied to the bar at a distance x from the steel rod and the bar remains horizontal. Find the value of x if it is given that ratio of Young's modulus of steel and brass is $\frac{Y_s}{Y_B} = 2$.



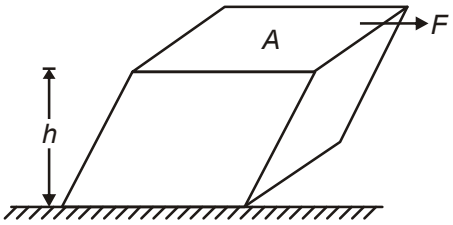
2. A steel wire of radius r is stretched without tension along a straight line with its ends fixed at A and B (figure). The wire is pulled into the shape ACB. Assume that d is very small compared to length of the wire. Young's modulus of steel is Y.

   (a) What is the tension (T) in the wire?

   (b) Determine the pulling force F. Is F larger than T?

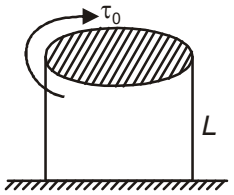


3. A rectangular bar is fixed to a hard floor. Height of the bar is h and its area in contact with the floor is A. A shearing force distorts the bar as shown. Prove that the work done by the shearing force is $W = \left(\frac{\sigma^2}{2\eta}\right) \times$ volume of the bar. Here σ is shear modulus of elasticity. Assume the deformation to be small.



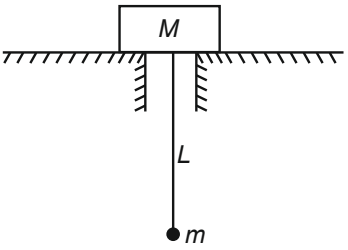
4. Assume that the least load which would break a thread when simply suspended from it is M and that this load produces a strain of 1 percent at the moment of breaking. Also assume that Hooke's law applies to the thread right up to breaking-point. A load of mass m is suspended from a thread of length λ. It is raised to a height and released. Find the least height to which the load must be raised so that it will break the thread when allowed to fall.

5. A metal cylinder of length L and radius R is fixed rigidly to ground with its axis vertical. A twisting torque τ_0 is applied along the circumference at the top of the cylinder. This causes an angular twist of θ_0 (rad) in the top surface. Calculate the shear modulus of elasticity (η) of the material of the cylinder.

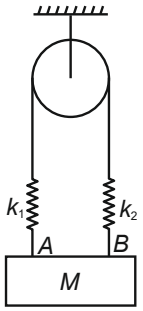


SIMPLE HARMONIC MOTION

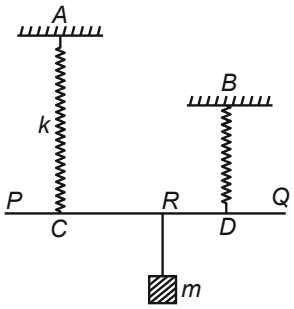
1. A block of mass M rests on a smooth horizontal table. There is a small gap in the table under the block through which a pendulum has been attached to the block. The bob of the simple pendulum has mass m and length of the pendulum is L. The pendulum is set into small oscillations in the vertical plane of the figure. Calculate its time period. The table does not interfere with the motion of the string.



2. In the system shown in the figure the string, springs and pulley are light. The force constant of the two springs are $k_1 = k$ and $k_2 = 2k$. Block of mass M is pulled vertically down from its equilibrium position and released. Calculate the angular frequency of oscillation. The top surface of the block (represented by line AB) always remains horizontal.



3. Two ideal springs of same make (the springs differ in their lengths only) have been suspended from points A and B such that their free ends C and D are at same horizontal level. A massless rod PQ is attached to the ends of the springs. A block of mass m is attached to the rod at point R. The rod remains horizontal in equilibrium. Now the block is pulled down and released. It performs vertical oscillations with time period $T = 2\pi\sqrt{\frac{m}{3k}}$ where k is the force constant of the longer spring.



(a) Find the ratio of length *RC* and *RD*.

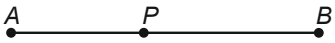
(b) Find the difference in heights of point *A* and *B* if it is given that natural length of spring *BD* is *L*.

4. A particle of mass *m* is constrained to move along a straight line. *A* and *B* are two fixed points on the line at a separation of *L*. When the particle is at some point *P*, between *A* and *B*, it is acted upon by two forces

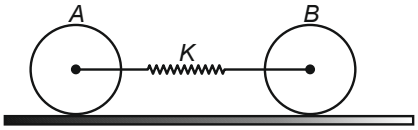
$\vec{F}_1 = \left(\frac{6mg}{L}\right)\overrightarrow{PA}$ and $\vec{F}_2 = \left(\frac{3mg}{L}\right)\overrightarrow{PB}$

At time $t = 0$, the particle is projected from *A* towards *B* with a speed of $\sqrt{gL}$.

At what time '*t*' will the particle reach at *B* for the first time?



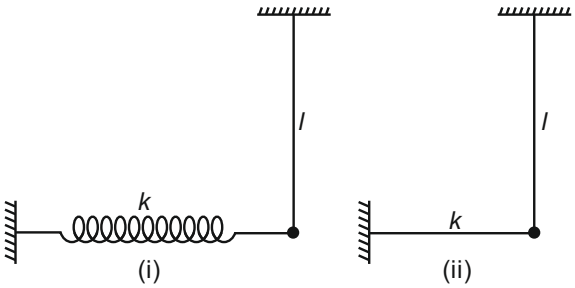
5. Two spheres *A* and *B* of the same mass *m* and the same radius are placed on a rough horizontal surface. *A* is a uniform hollow sphere and *B* is uniform solid sphere. They are tied centrally to a light spring of spring constant *k* as shown in figure. *A* and *B* are released when the extension in the spring is x_0. Friction is sufficient and the spheres do not slip on the surface. Find the frequency and amplitude of SHM of the sphere *A*.



6. A simple pendulum of length *L* has a bob of mass *m*. The bob is connected to light horizontal spring of force constant *k*. The spring is relaxed when the pendulum is vertical (see fig. (i)).

(a) The bob is pulled slightly and released. Find the time period of small oscillations. Assume that the spring remains horizontal.

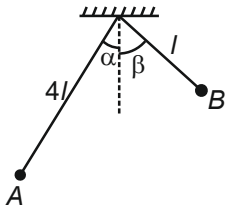
(b) The spring is replaced with an elastic cord of force constant *k*. The cord is relaxed when the pendulum is vertical (see fig (ii)). The bob is pulled slightly and released. Find the time period of oscillations.



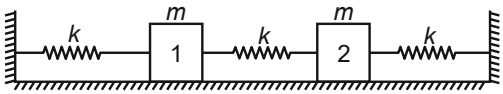
7. Two simple pendulums *A* and *B* have length 4*l* and *l* respectively. They are released from rest from the position shown. Both the angles α and β are small. Calculate the time after which the two string become parallel for the first time if-

(a) $\alpha = \beta$

(b) $\beta = 1.5\ \alpha$



8. Two identical blocks 1 and 2, each of mass *m*, are kept on a smooth horizontal surface, connected to three springs as shown in the figure. Each spring has a force constant *k*. Under suitable initial conditions, the two blocks oscillate in phase and their respective displacement from the mean position is given by

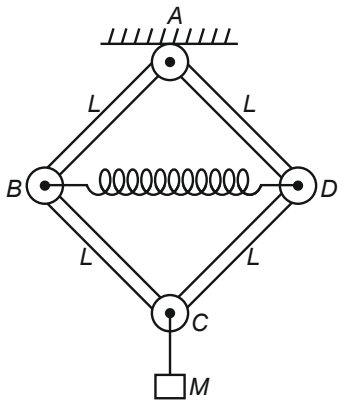


$x_1 = A \sin \omega t$ and $x_2 = A \sin \omega t$

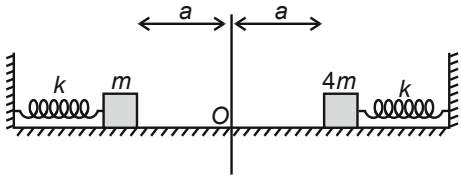
(i) Suggest one such initial condition that will result in such oscillation.

(ii) Find ω

9. Four identical massless rods are connected by hinged joints to form a rhombus of side length *L*. Rods can rotate freely about the joints. The joints *B* and *D* are connected by a massless spring of relaxed length 1.5 *L*. The system is suspended vertically with a load of mass *M* attached at *C* (see fig.). In equilibrium the rods form an angle of 30° with the vertical. Find time period of small oscillations of the load.



10. Two small blocks of mass *m* and 4*m* are connected to two springs as shown in fig. Both springs have stiffness *K* and they are in their natural length when the blocks are at point *O*. Both the blocks are pushed so that both the springs get compressed by a distance *a*. Find the block of mass *m* is released

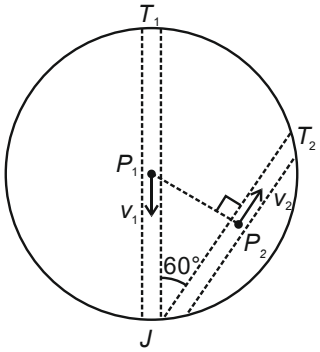


and after it travels through a distance $\left(1-\frac{\sqrt{3}}{2}\right)a$, the second block is also released.

(a) At what distance from point *O* will the two blocks collide?

(b) How much time the two blocks need to collide after the block of mass 4*m* is released?

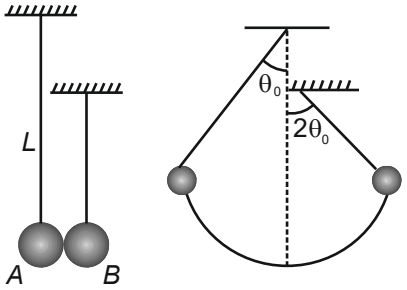
11. Two tunnels - T_1 and T_2 are dug across the earth as shown in figure. One end of the two tunnels have a common meeting point on the surface of the earth. Two particles P_1 and P_2 are oscillating from one end to the other end of the tunnels. At some instant particles are at mid point of their tunnels as shown in figure. Then –



(a) Write phase difference between the particle P_1 and P_2. Can the two particles ever meet?

(b) Write the ratio of maximum velocity of particle P_1 and P_2.

12. Two identical small elastic balls have been suspended using two strings of different length (see fig (a)). Pendulum *A* is pulled to left by a small angle θ_0 and released. It hits ball *B* head on which swings to angle $2\theta_0$ from the vertical. Calculate the time period of oscillation of *A* if its length is known to be *L*.



(a) (b)

13. A pendulum consists of an inextensible thread connected to a solid spherical ball of radius *r*. The distance between the point of suspension and the centre of the ball is *L* (>> *r*). Calculate the percentage difference in the time period of this pendulum to the time period of a simple pendulum of length *L*. How large is this difference for *r* = 5 cm and *L* = 100 cm.

WAVE MOTION

1. (i) Two sinusoidal wave are given as $y_1 = a_1 \sin(\omega t + kx + \delta)$ and $y_2 = a_2 \sin(\omega t - kx)$. They superimpose.

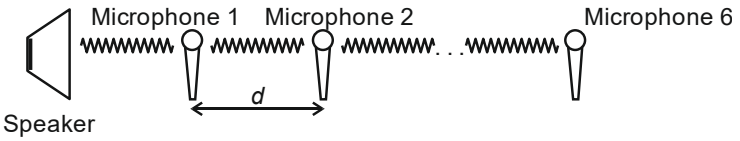
   (a) Calculate the resultant amplitude of oscillation at a position *x*. Is amplitude time dependent?

   (b) Calculate the ratio of maximum and minimum amplitudes observed.

   (ii) A speaker (producing a sound of a single wavelength λ) and a microphone are placed as shown in the figure. The microphone detects the sound and converts it into electrical signal. This way we can obtain the waveform of the sound. Assume that there is no attenuation of the sound. The waveform detected by the microphone is sinusoidal with amplitude *a*. In one experiment 6 microphones are placed in front of the speaker with distance between two neighbouring microphones being $d = \frac{5\lambda}{6}$

   (a) The output from all the 6 microphones is superimposed. What is amplitude of the resultant?

   (b) If large number of microphone are kept with separation *L* between two consecutive ones, how will the combined output change with *L*? Given that $L \neq n\lambda$ (n = 1, 2, 3.....)



2. A string in a guitar is made of steel (density 7962 kg/m^3). It is 63.5 cm long, and has diameter of 0.4 mm. The fundamental frequency is *f* = 247 Hz.

   (a) Find the string tension (*F*).

   (b) If the tension *F* is changed by a small amount ΔF, the frequency *f* changes by a small amount Δf. Show that $\frac{\Delta f}{f} = \frac{1}{2}\frac{\Delta F}{F}$

   (c) The string is tuned with tension equal to that calculated in part (a) when its temperature is 18°C. Continuous playing causes the temperature of the string to rise, changing its vibration frequency. Find Δf if the temperature of the string rises to 29°C. The steel string has a Young's modulus of 2.00×10^{11} Pa and a coefficient of linear expansion of 1.20×10^{-5} (°C)$^{-1}$. Assume that the temperature of the body of the guitar remains constant. Will the vibration frequency rise or fall?

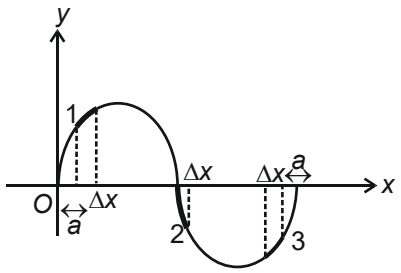
3. A long taut string is connected to a harmonic oscillator of frequency f at one end. The oscillator oscillates with an amplitude a_0 and delivers power P_0 to the string. Due to dissipation of energy the amplitude of wave goes on decreasing with distance x from the oscillator given as $a = a_0e^{-kx}$. In what length of the string $\left(\frac{3}{4}\right)^{th}$ of the energy supplied by the oscillator gets dissipated?

4. The figure shows the $y - x$ graph at an instant for a small amplitude transverse wave travelling on a stretched string. Three elements (1, 2 and 3) on the string have equal original lengths (= Δx). At the given instant

   (i) Which element (among 1, 2 and 3) has largest kinetic energy?

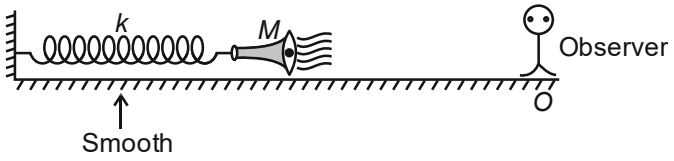
   (ii) Which element has largest energy (i.e., sum of its kinetic and elastic potential energy)

   (iii) Prove that energy per unit length $\frac{\Delta E}{\Delta x}$ of the string is constant everywhere equal to $T\left(\frac{\partial y}{\partial x}\right)^2$ where T is tension the string.



5. A small source of sound has mass M and is attached to a spring of force constant K. It is oscillating with amplitude $A = \frac{V}{20}\sqrt{\frac{M}{K}}$ where V is speed of sound in air. The source of sound produces a sound of frequency f_0 = 399 Hz.

   (a) Find the frequency of sound registered by a stationary observer standing at a distant point O.



   (b) Let Δt_1 be the time interval during which the registered frequency changes from 420 Hz to $\left(399 \times \frac{40}{39}\right)$Hz and Δt_2 be the time interval during which the observed frequency changes from 399 Hz to $\left(399 \times \frac{40}{41}\right)$Hz. Which is larger Δt_1 or Δt_2?

6. A source of sound is located in a medium in which speed of sound is V and an observer is located in a medium in which speed of sound is $2V$. Both the source and observer are moving directly towards each other at velocity $\frac{V}{5}$. The source has a frequency of f_0.

   (a) Find the wavelength of wave in the medium in which the observer is located.

   (b) Find the frequency received by the observer.

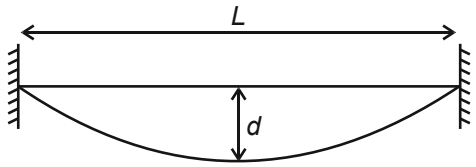
7. A transverse wave $y = A\sin\omega\left(\frac{x}{V_1} - t\right)$ is travelling in a medium with speed V_1. Plane $x = 0$ is the boundary of the medium. For x > 0 there is a different medium in which the wave travels at a different speed V_2. Part of the wave is reflected and part is transmitted. For $x < 0$ the wave function is described as $y_- = A_1\sin\omega\left(\frac{x}{V_1} - t\right) + A_2\sin\omega\left(\frac{x}{V_2} + t\right)$, while for $x > 0$; $y_+ = A_3\sin\omega\left(\frac{x}{V_2} - t\right)$

(a) Using the fact that the wave function must be continuous at $x = 0$, show that $A_1 - A_2 = A_3$

(b) Using the fact that $\frac{\partial y}{\partial x}$ must be continuous at $x = 0$, prove that $\frac{V_2}{V_1}A_1 = A_3 - A_2$

(c) Show that $A_3 = \frac{2V_2A_1}{V_1 + V_2}$ and $A_2 = \left(\frac{V_1 - V_2}{V_1 + V_2}\right)A_1$

8. A wire having mass per unit length μ and length L is fixed between two fixed vertical walls at a separation L. Due to its own weight the wire sags. The sag in the middle is d (<< L). Assume that tension is practically constant along the wire, owing to its small mass. Calculate the speed of the transverse wave on the wire.



9. (i) In a travelling sinusoidal longitudinal wave, the displacement of particle of medium is represented by $s = S(x, t)$. The midpoint of a compression zone and an adjacent rarefaction zone are represented by letter 'C' and 'R' respectively. The difference in pressure at 'C' and 'R' is ΔP and the bulk modulus of the medium is B.

(a) How is $\left|\frac{\partial s}{\partial x}\right|_C$ related to $\left|\frac{\partial}{\partial}\right|$

(b) Write the value of $\left|\frac{\partial s}{\partial x}\right|_C$ in terms of ΔP and B.

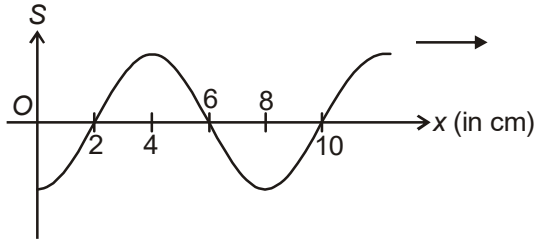
(c) What is speed of a medium particle located mid-way between 'C' and 'R'.

(ii) A standing wave in a pipe with a length of $L = 3$ m is described by $s = A\cos\left(\frac{3\pi x}{L}\right)\sin\left(\frac{3\pi vt}{L}\right)$ where v is wave speed. The atmospheric pressure and density are P_0 and ρ respectively

(a) At $t = \frac{L}{18v}$ the acoustic pressure at $x = \frac{L}{2}$ is 0.2 percent of the atmospheric pressure. Find the displacement amplitude A.

(b) In which overtone is the pipe oscillating?

10. A longitudinal wave is travelling at speed u in positive x direction in a medium having average density ρ_0. The displacement(s) for particles of the medium versus their position (x) has been shown in the figure.



Answer following question for $0 < x \leq 10$ cm

(a) Write x co-ordinates of all positions where the particles of the medium have maximum negative acceleration. What is density at these locations - higher than ρ_0, less than ρ_0 or equal to ρ_0?

(b) Write x co-ordinates of all locations where the particles of the medium have negative maximum velocity. What do you think about density at these positions?

(c) Knowing that the change in density ($\Delta\rho$) is proportional to negative of the slope of s versus x graph, prove that $\frac{d\rho}{dx} \propto -a$, where a is acceleration of the particles at position x. At which point ($0 < x \leq 10$) is $\frac{d\rho}{dx}$ positive maximum.

11. Two sound waves, travelling in same direction can be represented as

$$y_1 = (0.02\text{ mm})\sin\left[(400\pi\text{ rad s}^{-1})\left(\frac{x}{330\text{ms}^{-1}} - t\right)\right]$$

And

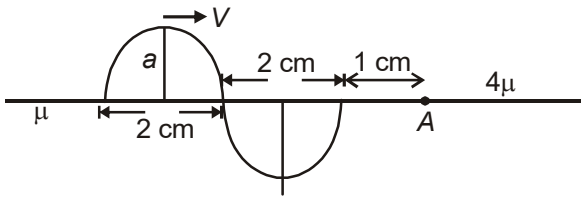
$$y_2 = (0.02\text{ mm})\sin\left[(404\pi\text{ rad s}^{-1})\left(\frac{x}{330\text{ms}^{-1}} - t\right)\right]$$

The waves superimpose.

(i) Find distance between two nearest points where an intensity maximum is recorded simultaneously.

(ii) Find the time gap between two successive intensity maxima at a given point.

12. A taut string is made of two segments. To the left of A it has a linear mass density of μ kg/m and to the right of A its linear mass density is 4 μ kg/m.



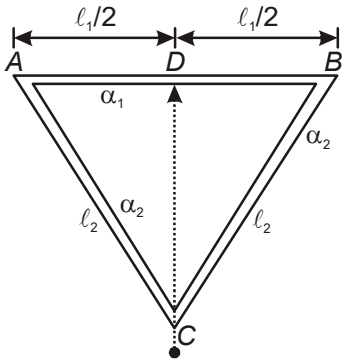
A sinusoidal pulse of amplitude a is travelling towards right on the lighter string with a speed V = 2 cm/s.

Draw the shape of the string after

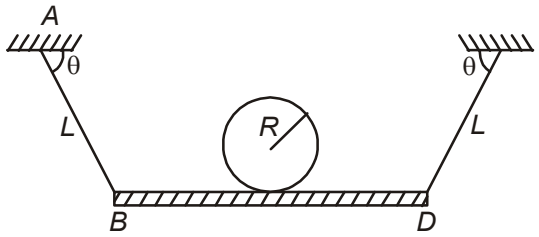
(a) 1 s (b) 2.5 s

TEMPERATURE AND THERMAL EXPANSION

1. In a compensated pendulum a triangular frame ABC is made using two different metals. AB of length ℓ_1 is made using a metal having coefficient of linear expansion α_1. BC and AC of length ℓ_2 each have coefficient of linear expansion α_2. A heavy bob is attached at C. Pendulum can oscillate about the pivot D. Find $\frac{\ell_2}{\ell_1}$ so that distance of bob from the pivot point D does not change with change in temperature.



2. A metal cylinder of radius *R* is placed on a wooden plank *BD*. The plank is kept horizontal suspended with the help of two identical string *AB* and *CD* each of length *L*. The temperature coefficient of linear expansion of the cylinder and the strings are α_1 and α_2 respectively. Angle θ shown in the figure is 30°. It was found that with change in temperature the centre of the cylinder did not move. Find the ratio $\frac{\alpha_1}{\alpha_2}$, if it is know that $L = 4R$. Assume that change in value of θ is negligible for small temperature changes.

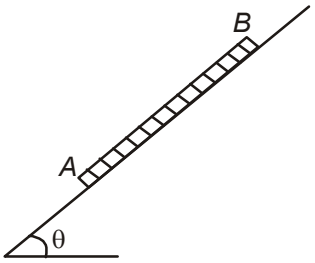


3. A uniform metal rod (*AB*) of mass *m* and length *L* is lying on a rough incline. The inclination of the incline and coefficient of friction between the rod and the incline is $\theta = 37°$ and $\mu = 1.0$ respectively. $\tan 37° = \frac{3}{4}$

   (a) If temperature increases the rod expands. However, there is a point *P* on the rod which does not move. Find the distance of this point from the lower end of the rod.

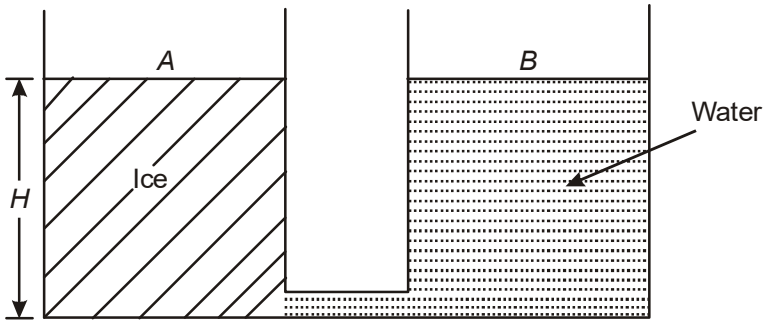
   (b) If the temperature falls the rod contracts. Once again there is a point *Q* which does not move. Find distance of *Q* from the lower end of the rod.

   (c) Will the repeated expansion and contraction cause the rod to slide down?

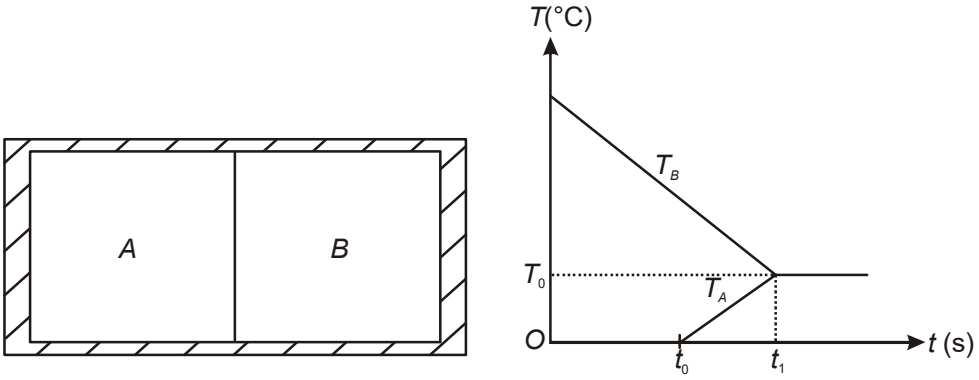


CALORIMETRY

1. Two identical cylindrical containers *A* and *B* are interconnected by a tube of negligible dimensions. Container *A* is filled with an ice block up to height $H = 1.8$ m and container *B* is filled up to same height with water. Ice is at 0°C and water is at 40°C. Due to heat exchange between water and ice, the ice block begins to melt. Assume that the ice block melt in horizontal layers starting from the bottom. The thickness of ice block reduces uniformly over the entire cross section of the container. The ice block moves without friction inside the container and no water enters between the vertical wall of the container and the ice block. Heat is exchanged only between the ice block and the water and there is no heat exchange with containers or atmosphere. Calculate the height of water in container *B* when thermal equilibrium is attained. Relative density and specific latent heat of fusion of ice are 0.9 and 80 cal g^{-1} respectively. Specific heat capacity of water is 1 cal g^{-1} $°C^{-1}$.



2. A well insulated box has two compartments *A* and *B* with a conducting wall between them. 100 g of ice at 0°C is kept in compartment *A* and 100 g of water at 100°C is kept in *B* at time $t = 0$. The temperature of the two parts *A* and *B* is monitored and a graph is plotted for temperatures T_A and T_B versus time (t) [Fig. (b)]. Assume that temperature inside each compartment remains uniform.



(a) (b)

(a) Is it correct to assert that the conducting wall conducts heat at a uniform rate, irrespective of the temperature difference between *A* and *B*?

(b) Find the value of time t_1 and temperature T_0 shown in the graph, if it is known that $t_0 = 200$ s.

Specific heat of ice = 0.5 cal g^{-1} $°C^{-1}$

Specific heat of water = 1.0 cal g^{-1} $°C^{-1}$

Latent heat of fusion of ice = 80 cal g^{-1}

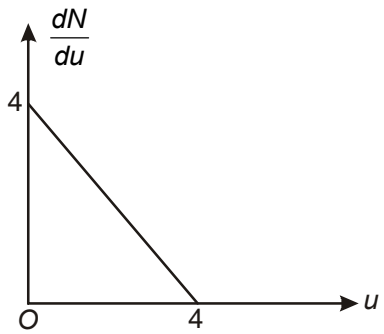
3. A copper calorimeter has mass of 180 g and contains 450 g of water and 50 g of ice; all at 0°C. Dry steam is passed into the calorimeter until a certain temperature (θ) is reached. The mass of the calorimeter and its contents at the end of the experiment increased by 25 g. Assume no heat loss to the surrounding and take specific heat capacities of water and copper to be 4200 J kg^{-1} K^{-1} and 390 J kg^{-1} K^{-1}, respectively. Take specific latent heat of vaporization of water to be 3.36×10^5 J kg^{-1} and 2.26×10^6 J kg^{-1} respectively.

(a) Find the final temperature θ

(b) If steam enters into the system at a steady rate of 5 g min^{-1}, plot the variation of temperature of the system till final temperature θ is attained.

KINETIC THEORY OF GASES AND GAS LAWS

1. A hypothetical gas sample has its molecular speed distribution graph as shown in the figure. The speed (u) and $\frac{dN}{du}$ have appropriate units. Find the root mean square speed of the molecules. Do not worry about units.



2. The Maxwell-Boltzmann distribution of molecular speeds in a sample of an ideal gas can be expressed as

$$f = \frac{4}{\sqrt{\pi}}\left(\frac{m}{2kT}\right)^{3/2} v^2 e^{-\frac{mv^2}{2kT}}.dv$$

where f represent the fraction of total molecules that have speeds between v and $v + dv$. m, k and T are mass of each molecule, Boltzmann constant and temperature of the gas.

(a) What will be value of $\int_{v=0}^{v=\infty} f dv$?

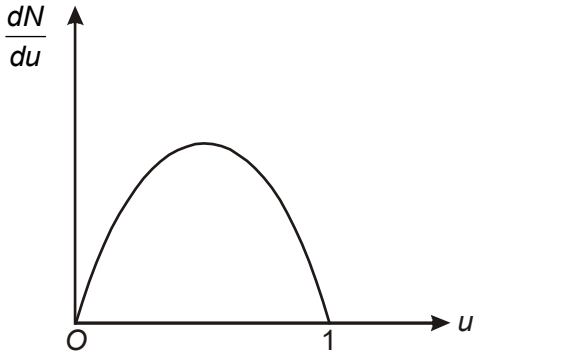
(b) It is given that $\int_0^\infty v^3 e^{-av^2}\, dv = \frac{1}{2a^2}$

Find the average speed of gas molecules at temperature T.

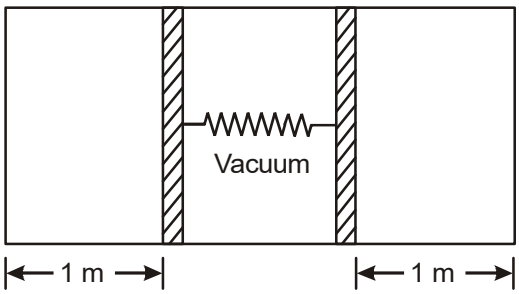
3. The speed distribution of molecules in a sample of a gas is shown in the figure. The graph between $\frac{dN}{du}$ and u is a parabola and total number of molecules in the sample is N_0.

(a) Calculate the rms speed of the molecules

(b) Calculate the total translational kinetic energy of molecules if mass of the sample is 10 g.



4. A cylindrical container is divided into three parts by two tight fitting pistons. The pistons are connected by a spring. The region between the pistons is vacuum and the other two parts have same number of moles of an ideal gas. Initially, both the gas chambers are at temperature T_0 and the spring is compressed by 1 m. Length of both gas chambers is 1 m in this positon. Now the temperature of the left and right chambers are raised to $\frac{4T_0}{3}$ and $\frac{5T_0}{3}$ respectively. Find the final compression in the spring in equilibrium. Assume that the pistons slide without friction.

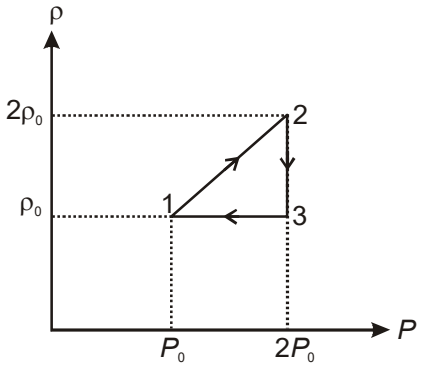


FIRST LAW OF THERMODYNAMICS

1. One mole of a mono atomic gas of molar mass M undergoes a cylic process as shown in the figure. Here ρ is density and P is pressure of the gas.

   (a) Calculate the heat rejected by the gas in one complete cycle.

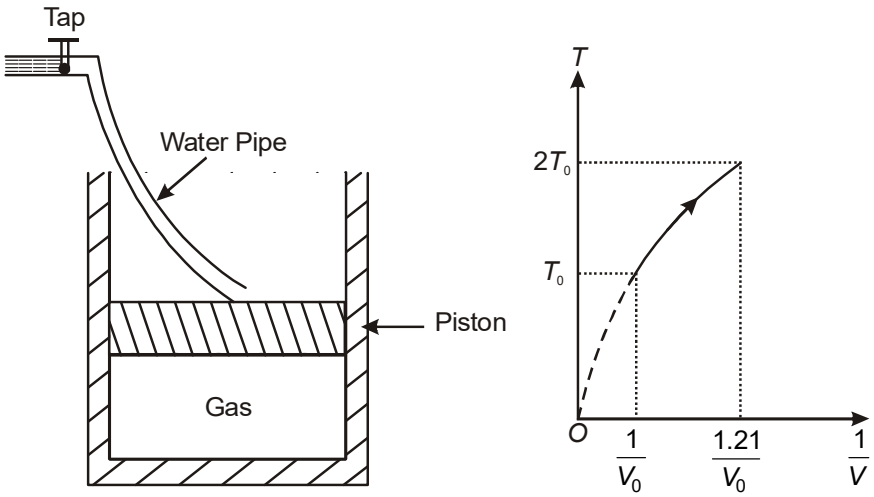
   (b) Find the efficiency of the cycle.



2. A cylindrical container has insulating wall and an insulating piston which can freely move up and down without any friction. It contains a mixture of ideal gases. Originally the gas is at atmospheric pressure P_0 and temperature (T_0). A tap positioned above the container is opened and it supplies water at a constant rate of $\frac{dm}{dt} = 0.25$ kg/s. The water collects above the piston in the container and the gas compresses. The tap is kept open till the temperature of the gas is doubled. During the process the T vs $\frac{1}{V}$ graph for the gas was recorded and found to be a parabola with its vertex at origin as shown in the graph. Area of piston $A = 1.515 \times 10^{-3}$ m^2 and atmospheric pressure = 10^5 N/m^2

   (a) Find the ratio of V_{rms} and speed of sound in the gaseous mixture.

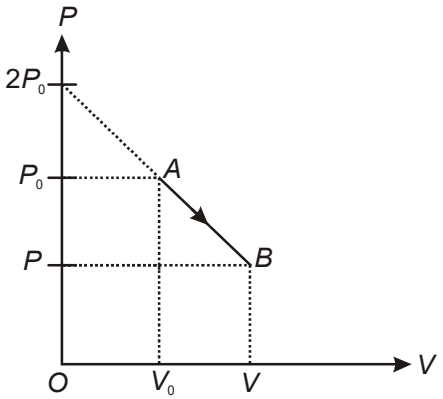
   (b) For how much time the tap was kept open?



3. (a) A polytropic process for an ideal gas is represented by PV^x = constant, where $x \neq 1$. Show that molar specific heat capacity for such a process is given by $C = C_v + \frac{R}{1-x}$.

   (b) An amount Q of heat is added to a mono atomic ideal gas in a process in which the gas performs a work $\frac{Q}{2}$ on its surrounding. Show that the process is polytropic and find the molar heat capacity of the gas in the process.

4. One mole of an ideal gas is expanded from the state $A(P_0, V_0)$ to final state B having volume V. The process follows a path represented by a straight line on the $P - V$ diagram (see figure). Up to what volume (V) the gas shall be expanded so that final temperature is half the maximum temperature during the process.

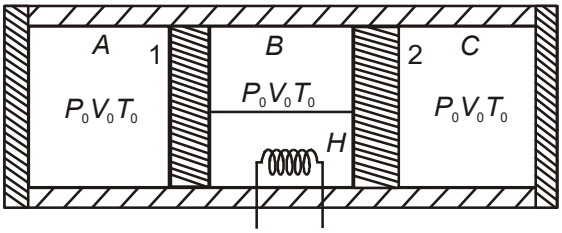


5. An insulated cylinder is divided into three parts A, B and C. Pistons 1 and 2 are connected by a rigid rod and can slide without friction inside the cylinder. Piston 1 is perfectly conducting while piston 2 is perfectly insulating. Equal quantity of an ideal gas is filled in three compartments and the state of gas in every part is same (P_0 V_0 T_0). Adiabatic exponent of the gas is $\gamma = 1.5$. The compartment B is slowly given that heat through a heater H such that the final volume of gas in part C becomes $\frac{4V_0}{9}$

(a) Calculate the heat supplied by the heater.

(b) Calculate the amount of heat flow through piston 1.

(c) If heater were in compartment A, instead of B how would your answers to (a) and (b) change ?

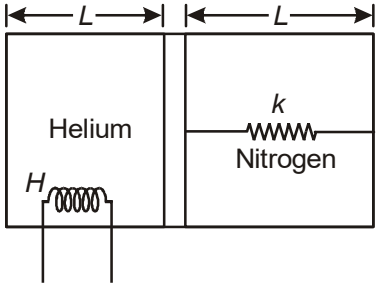


6. An adiabatic cylinder has length $2L$ and cross sectional area A. A freely moving non-conducting piston of negligible thickness divides the cylinder into two equal parts. The piston is connected to the right face of the cylinder with an ideal spring of force constant k. The right chamber contains 28 g nitrogen in which one third of the molecules are dissociated into atoms. The left chamber containes 4 g helium. With piston in equilibrium and spring relaxed the pressure in both chamber is P_0. The helium chamber is slowly given heat using an electric heater (H), till piston moves to right by a distance $\frac{3L}{4}$. Neglect the volume occupied by the spring and the heating coil. Also neglect heat capacity of the spring.

(a) Find the ratio of C_p and C_v for nitrogen gas in right chamber.

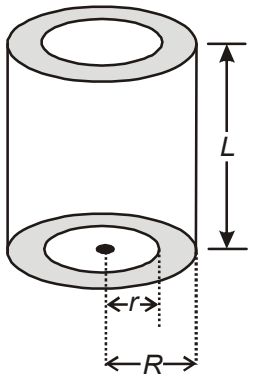
(b) Calculate change in temperature of helium.

(c) Calculate heat supplied by the heater.

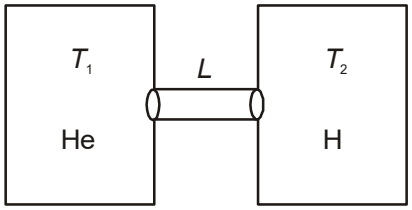


HEAT TRANSFER

1. A thick cylindrical shell made of material of thermal conductivity k has inner and outer radii r and R respectively and its length is L. When the curved surface of the cylinder are lagged (i.e., given insulation cover) and one end is maintained at temperature T_1 and the other end is maintained at $T_2(< T_1)$; the heat current along the length of the cylinder is H. In another experiment the two ends are lagged and the inner wall and outer wall are maintained at T_1 and T_2 respectively. Find the radial heat flow in this case.



2. Two identical adiabatic containers of negligible heat capacity are connected by conducting rod of length L and cross sectional area A. Thermal conductivity of the rod is k and its curved cylindrical surface is well insulated from the surrounding. Heat capacity of the rod is also negligible. One container is filled with n moles of helium at temperature T_1 and the other one is filled with equal number of moles of hydrogen at temperature $T_2(<T_1)$. Calculate the time after which the temperature difference between two gases will becomes half the initial difference.



3. A spherical shell is kept in an atmosphere at temperature T_0. The wavelength corresponding to maximum intensity of radiation for the shell is λ_0. A point source of constant power is switched on inside the shell. The power radiated by the source is $P = 0.4\sigma SeT_0^4$ where S, e and σ are outer surface area of the shell, emissivity of the outer surface of the shell and Stefan's constant respectively. Calculate the new wavelength (λ) corresponding to the maximum intensity of radiation from the shell. Assume that change in temperature (ΔT) of the shell is small compared to the ambient temperature T_0.

4. Consider the sun to be a blackbody at temperature T_s = 5780 K. Radius of the sun is $r_s = 6.96 \times 10^8$ m. The earth-sun distance is $R = 1.49 \times 10^{11}$ m. Assume that 30% of the solar radiation that hits the earth is scattered back into space without absorption.

   (a) Calculate the steady state average temperature of the earth assuming it to be a blackbody. Take $(0.7)^{\frac{1}{4}} = 0.91$

   (b) We know that average temperature of earth is $\simeq$ 288 K. How does this value compare with that obtained in (a)?

   The difference is due to greenhouse effect. Comment on the following statement - "Emissivity of earth is reduced more than absorptivity due to green house effect."

5. *A* and *B* are two sphere made of same material radius of A is double that of *B* and initially they are at same temperature (*T*). Both of them are kept far apart in a room at temperature $T_0 (<T)$. Calculate the ratio of initial rate of cooling (i.e. rate of fall of temperature) of sphere *A* and *B* if

   (a) the spheres are solid,

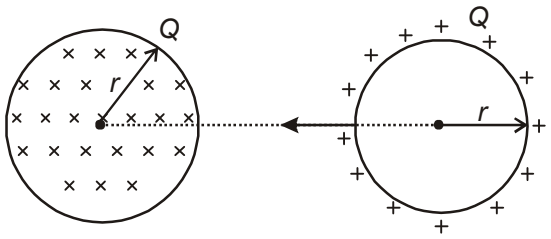
   (b) the spheres are hollow made of thin sheets of same thickness

ELECTROSTATICS

1. There is a ball of radius *r* having uniformly distributed volume charge *Q* on it and there is a spherical shell of radius *r* having uniformly distributed surface charge *Q* on it. The two spheres are far apart.

   (a) A point charge *q* is moved slowly from the centre of the shell (through a small hole in it) to the centre of the ball. Find work done by the external agent in the process.

   (b) The two spheres are brought closer so that their centers are separated by 4*r*. Now calculate the amount of work needed in slowly moving a point charge *q* from the centre of the shell to the centre of the ball. Assume that charge on one ball does not alter the charge distribution of the other. Does your answers in (a) and (b) differ? Why?

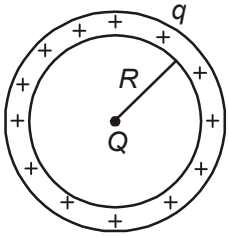


2. A ring of radius *R* has uniformly distributed charge *q*. A point charge *Q* is placed at the centre of the ring.

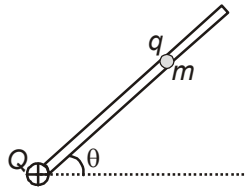
   (a) Find the increase in tension in the ring after the point charge is placed at its centre.

   (b) Find the increase in force between the two semicircular parts of the ring after the point charge is placed at the centre.

   (c) Using the result found in part (*b*) find the force that the point charge exerts on one half of the ring.



3. A smooth fixed rod is inclined at an angle θ to the horizontal. At the bottom end of the rod there is a fixed charge +*Q*. There is a bead of mass *m* having charge *q* that can slide freely on the rod. The equilibrium separation of the bead from fixed charge *Q* is x_0. Find the frequency of oscillation of the bead if it is displaced a little from its equilibrium position.

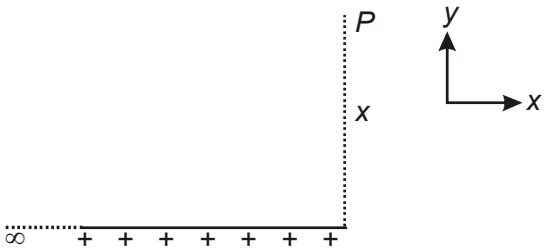


4. (a) There is an infinitely long thread uniformly charged with linear charge density λ C/m. Using Gauss' law, calculate the electric field (E_0) at a distance *x* from the thread.

   (b) Now consider a semi-infinite uniformly charged thread (linear charge density = λ) as shown in figure. Find the *y* component of eletric at point *P* in terms of E_0. Use simple qualitative argument.

(c) For the situation described in (b) calculate the x component of electric field at point P using the method of integration.

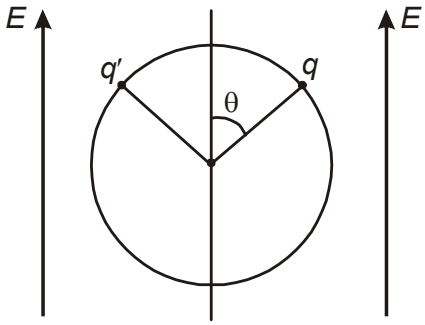
(d) Find the angle that the electric field at P makes with x direction.



5. A uniform non-conducting ring has mass m and radius R. Two point charges q and q' are fixed on its circumference at a separation of $\sqrt{2}R$. The ring remains in equilibrium in air with its plane vertical in a region where exists a uniform vertically upward electric field E. Given $E = \frac{4mg}{7q}$.

(a) Find angle θ in equilibrium position (see figure).

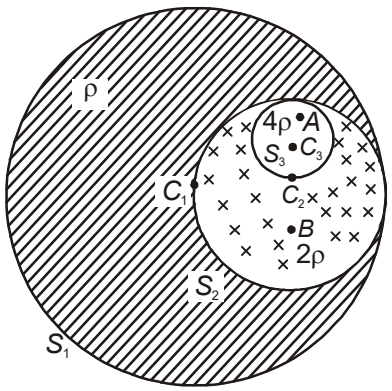
(b) The ring is given a small rotation in the plane of the figure and released. Will it perform oscillations?



6. In the figure shown, spheres S_1, S_2 and S_3 have radii R, $\frac{R}{2}$ and $\frac{R}{4}$ respectively. C_1, C_2 and C_3 are centers of the three spheres lying in a plane. Angle $\angle C_1C_2C_3$ is right angle. Sphere S_3 has a uniformly spread volume charge density 4ρ. The remaining part of S_2 has uniform charge density of 2ρ and the left over part of S_1 has uniform charge density of ρ.

(a) Find electric field at a point A at a distance $\frac{R}{8}$ from C_3 on the line C_2C_3 (see figure).

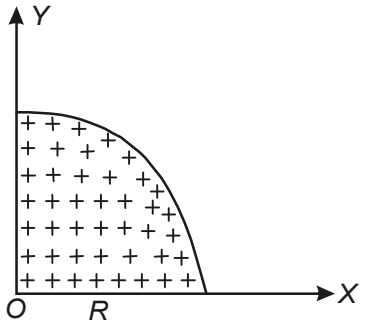
(b) Find electric field at point B at a distance $\frac{R}{4}$ from C_2 on the line C_3C_2 (see figure).



7. The quarter disc of radius R (see figure) has a uniform surface charge density σ.

   (a) Find electric potential at a point (O, O, Z)

   (b) Find the Z component of electric field at (O, O, Z)

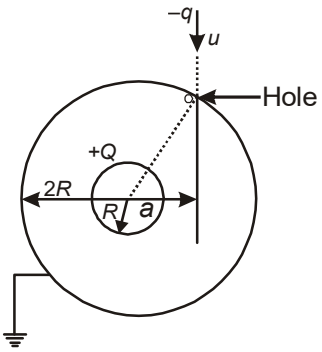


8. Two concentric spherical shells have radii R and $2R$. The outer shell is grounded and the inner one is given a charge $+Q$. A small particle having mass m and charge $-q$ enters the outer shell through a small hole in it. The speed of the charge entering th shell was u and its initial line of motion was at a distance $a = \sqrt{2}R$ from the centre.

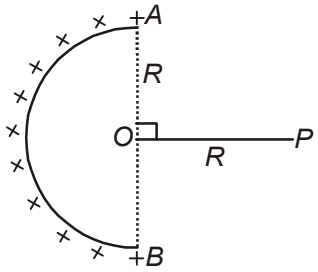
   (a) Find the radius of curvature of the path of the particle immediately after it enters the shell.

   (b) Find the speed with which the particle will hit the inner sphere.

   Assume that distribution of charge on the spheres do not charge due to presence of the charge particle.



9. A semicircular ring of radius R carries a uniform linear charge density of λ. P is a point in the plane of the ring at a distance R from centre O. OP is perpendicular to AB. Find electric field intensity at point P.



10. In a certain region of space the electrostatic field depends only on the coordinates x and y as follows.

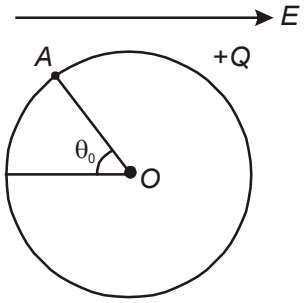
$E = 0$ for $\sqrt{x^2 + y^2} < r_0$

$E = a\left(x\hat{i} + y\hat{j}\right) / \left(x^2 + y^2\right)$ for $\sqrt{x^2 + y^2} > r_0$

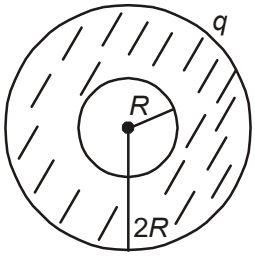
where a is a positive constant, and $\hat{i}$ and $\hat{j}$ are the unit vectors along the X- and Y-axes. Find the charge within a sphere of radius $2r_0$ with the centre at the origin.

11. Two infinite line have linear charge densities $-\lambda$ and $+\lambda$. They are parallel to z axis passing through x axis at points $x = -a$ and $x = a$ respectively. Show that the equipotential surface having potential $\frac{\lambda \ln(2)}{4\pi\varepsilon_0}$ is a cylinder having radius $2\sqrt{2}a$.

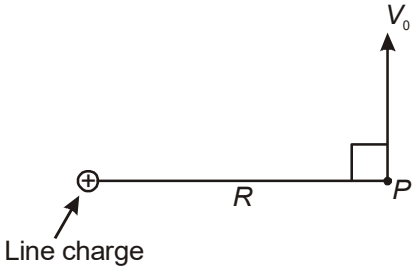
12. A conducting sphere of radius R having charge Q is placed in a uniform external field E. O is the centre of the sphere and A is a point on the surface of the sphere such that AO makes an angle of $\theta_0 = 60°$ with the opposite direction of external field. Calculate the potential at point A due to charge on the sphere only.



13. A conducting shell has inner radius R and outer radius $2R$. A charge $+q$ is given to the spherical shell.
    (a) Find the electric field at a point which is at a distance x from the centre of the shell. Give your answer for three cases.
        (i) $x < R$ (ii) $R < x < 2R$ (iii) $x > 2R$
    (b) Find the electric potential in all the three cases mentioned in (a).
    (c) Find field and potential in all the three cases mentioned in (a) after a point charge $-q$ is introduced at the centre of the shell.
    (d) Write the electrical potential energy of the system consisting of the shell and the point charge at its centre.
    (e) Find the electrostatic force that the shell exerts on the point charge.
    (f) Now, another point charge $+q$ is placed at a distance $4R$ from the centre of the shell. Find electric field and potential in following cases.
        (i) $x < R$
        (ii) $R < x < 2R$



14. An infinite line charge is perpendicular to the plane of the figure having linear charge density λ. A particle having charge Q and mass m is projected in the field of the line charge from point P. The point P is at a distance R from the line charge and velocity given to the particle is perpendicular to the radial line at P (see figure).



(a) Find the speed of the particle when its distance from the line charge grows to ηR ($\eta > 1$).

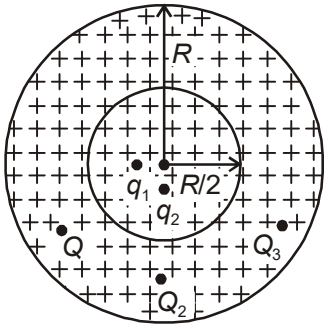
(b) Fine the velocity component of the particle along the radial line (joining the line charge to the particle) at the instant its distance becomes ηR.

15. (a) Calculate the electrostatic self energy of a uniformly charged sphere of radius R having charge Q.

(b) Divide the above sphere (mentally) into two regions-spherical concentric part having radius $\frac{R}{2}$ and the remaining annular part $\left(\text{between } \frac{R}{2} \text{ and } R\right)$. Denote the point charges in sphere of radius $R/2$ by q_1, q_2, q_3 ... etc.

The charges in annular part be denoted by Q_1, Q_2, Q_3 ... etc.

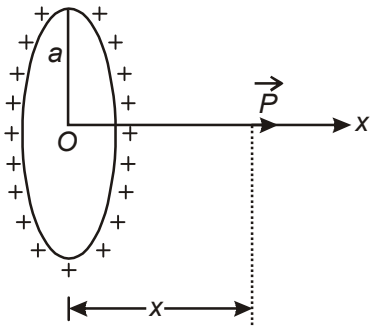
Calculate the electrostatic interaction energy for all pairs like $[(Q_i, Q_j) + (q_i + q_j)]$.



16. A short electric dipole (dipole moment P) is placed on the axis of a uniformly charged ring at a distance x from the centre as shown in figure. Radius of the ring is a and charge on it is Q.

(i) Write the force on the dipole when $x = \frac{a}{2}$ and when $x = a$. Why the direction of force at two points is different?

(ii) Is the force on the dipole zero if $x = 0$? If not, where will you place the dipole so that force on it is zero.



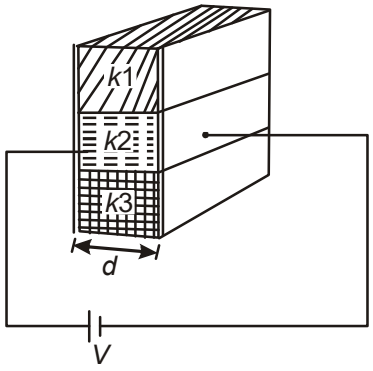
CAPACITOR

1. A parallel plate capacitor of plate area A and spacing between the plates d is filled with three dielectrics as shown in the figure. The dielectric constants of the three dielectrics are $K_1 = K$, $K_2 = 2K$, $K_3 = 3K$. The capacitor is connected to a cell of emf V.

(a) Write the ratio of maximum to minimum charge density on the surface of the capacitor plate.

(b) Calculate the surface charge density of bound (induced) charge on the middle dielectric.

(c) If the three dielectrics occupy equal volume between the plates, calculate the capacitance of the capacitor.

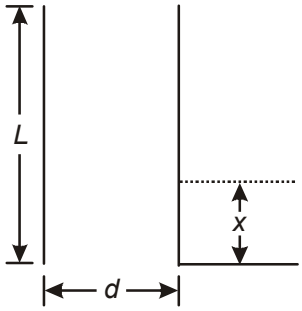


2. A parallel plate capacitor has square plates of side length L. Plates and kept vertical at separation d between them. The space between the plates is filled with a dielectric whose dielectric constant (K) changes with height (x) from the lower edge of the plates as $K = e^{\beta x}$ where β is a positive constant. A potential difference of V is applied across the capacitor plates.

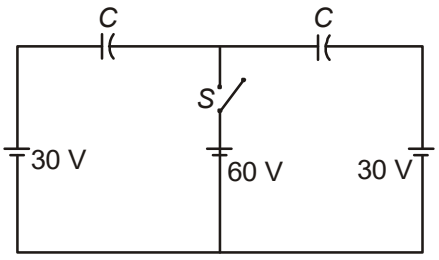
(i) Plot the variation of surface charge density (σ) on the positive plate of the capacitor versus x.

(ii) Plot the variation of electric field between the plates as a function of x.

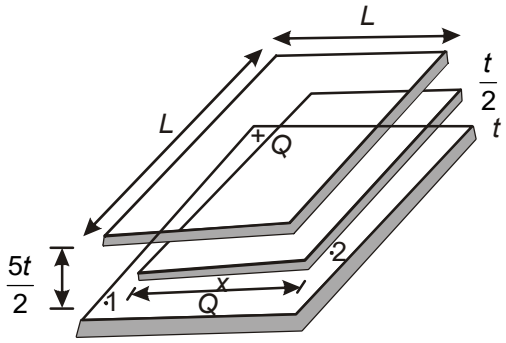
(iii) Calculate the capacitance of the capacitor.



3. Find heat dissipated in the circuit after switch S in closed. $C = 2\ \mu F$.



4. Two square metal plates have sides of length L and thickness t (<< L). They are arranged parallel to each other with their inner faces at a separation of $\frac{5}{2}t$ .One of the plates is given a charge $-Q$ and the other one is given a charge $+Q$. A third rectangular metal plate of sides L and x, having thickness $\frac{t}{2}$ is inserted between the plates as shown. The third plate is equidistant from the two plates and parallel to them. Neglect edge effects.



(a) Find the charge density on lower plate at points 1 and 2 shown in figure.

(b) Find potential difference between the upper plate and the middle plate.

(c) Find electric field between the two outer plates in space where the third plate is not present (i.e., at a point above point 1.)

(d) Find the capacitance of the system across two outer plates.

5. A hollow spherical conductor of radius R has a charge Q on it. A small dent on the surface decreases the volume of the spherical conductor by 2%. Assume that the charge density on the surface does not change due to the dent and the electric field in the dent region remains same as other points on the surface.

(a) ΔE is the electrostatic energy stored in the electric field in the shallow dent region and E is the total electrostatic energy of the spherical shell. Find the ratio $\frac{\Delta E}{E}$.

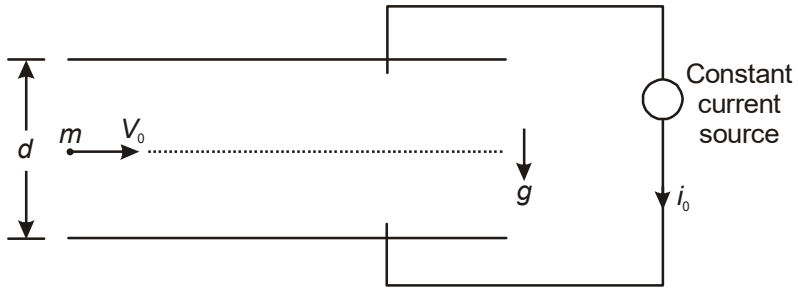
(b) Using the ratio obtained in part (a) calculate the percentage change in capacitance of the sphere due to the dent.

6. A particle of mass m and charge $+q$ enters horizontally with speed V_0 midway between the horizontal plates of a parallel plate capacitor at time $t = 0$. Separation between the capacitor plates is 'd' and it starts getting charged, by a constant current source, at time $t = 0$. Plate area of the capacitor is A. It was found that the particle just misses (to hit) the lower plate. Assume that the plates are quite long and acceleration due to gravity is g.

(a) Give a rough sketch of the path of the particle.

(b) Find the constant current (i_0) supplied by the source to the capacitor.

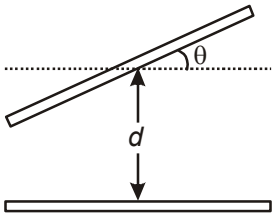
Consider no magnetic force on the charge.



7. One plate of a parallel plate capacitor is tilted by a small angle about its central line as shown in the figure. The tilt angle θ is small. Both the plates are square in shape with side length a and separation between their centers is d. Find the capacitance of the capacitor.

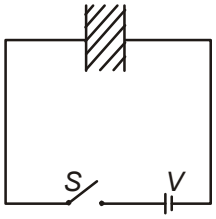
Given : $\ell n(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + ...$

$\ell n(1-x) = -\left(x + \frac{x^2}{2} + \frac{x^3}{3} + ...\right)$



CURRENT ELECTRICITY

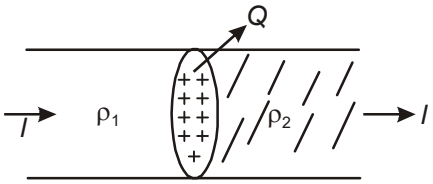
1. A parallel plate capacitor has plate area *A* and separation between the plates equal to *d*. A material of dielectric constant *K* and resistivity ρ is filled between the plates. The switch is closed to connect the capacitor to a cell of emf *V*.



(a) Write the steady state current in the circuit and charge on the capacitor.

(b) When the circuit is in steady state, switch *s* is opened (at *t* = 0). Write charge on the capacitor as function of time (*t*) after this.

2. Two cylindrical rods, of different material, are joined as shown. The rods have same cross section (*A*) and their electrical resistivities are ρ_1 and ρ_2. When a current *I* is passed through the rods, a charge (*Q*) gets piled up at the junction boundary. Assuming the current density to be uniform throughout the cross section, calculate Q.

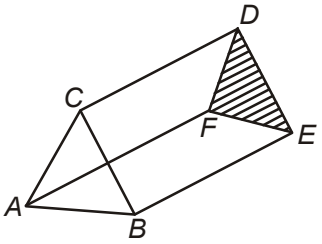


Under what condition the charge *Q* is negative?

3. A prism shaped network of resistors has been shown in the figure. Each arm (like *AB*, *AC*, *CD*, *DF*...) has resistance *R*. Find the equivalent resistance of the network between

(a) *A* and *B*

(b) *C* and *D*

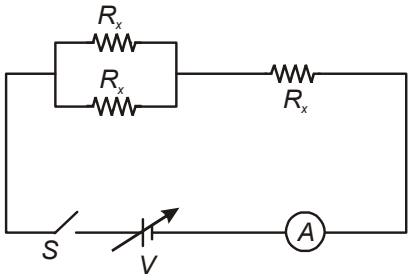


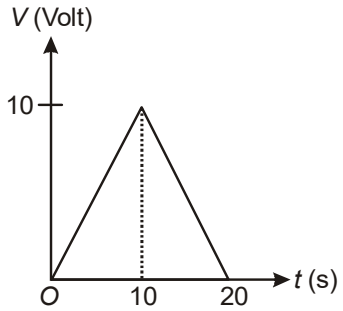
4. In the circuit shown, each resistor has a resistance R_X which depends on the voltage V_X across it.

   For $V_X \leq 1$ V, $R_X = 1\ \Omega$

   and for $V_X > 1$ V, $R_X = 2\ \Omega$

   The emf (V) of the source, changes with time (t) after the switch is closed at $t = 0$. The variation of V with time is depicted in the graph. Plot the variation of ammeter reading with time.

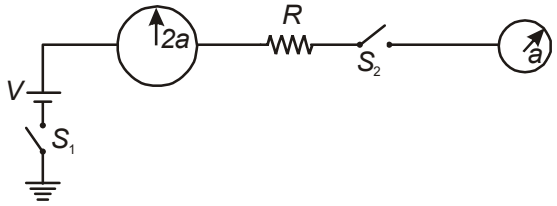




5. In the Fig. two neutral spherical conductors of radii $2a$ and a are separated by a large distance. Initially, switch S_1 is kept closed and S_2 is open. Now S_1 is opened and S_2 is closed at $t = 0$

   (a) Find the rate of fall in potential of the conductor of radius $2a$ as a function of time.

   (b) Find the heat dissipated after S_2 is closed.



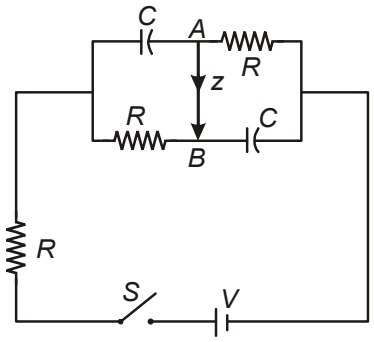
6. In the circuit shown in the fig, the switch 'S' is closed at time $t = 0$. The current in branch AB is represented by z and is taken to be positive when it is from A to B.

   (a) Write the value of z immediately after the switch is closed.

   (b) Write the value of z infinite time after the switch is closed.

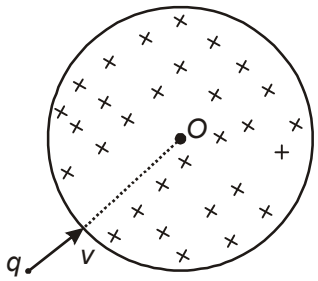
   (c) Write z as a function of time (t) and plot the variation of z with time.

   (d) At what time t_0 the current z become zero?



MOTION OF CHARGE IN ELECTROMAGNETIC FIELD

1. Figure shows a circular region of radius $R = \sqrt{3}\ m$ which has a uniform magnetic field B = 0.2 T directed into the plane of the figure. A particle having mass m = 2 g, speed v = 0.3 m/s and charge q = 1 mC is projected along the radius of the circular region as shown in figure. Calculate the angular deviation produced in the path of the particle as it comes out of the magnetic field. Neglect any other force apart from the magnetic force.



2. A particle of mass m and charge q is projected into a region having a uniform magnetic field B_0. Initial velocity (v_0) of the particle is perpendicular to the magnetic field. Apart from the magnetic force the particle faces a frictional force which has a magnitude of $f = kv$ where v is instantaneous speed and k is a positive constant.

   (a) Find the radius of curvature of the path of the particle after it has travelled through a distance of $x_0 = \frac{mv_0}{2k}$.

   (b) Plot the variation of radius of curvature of the path of the particle with time (t).

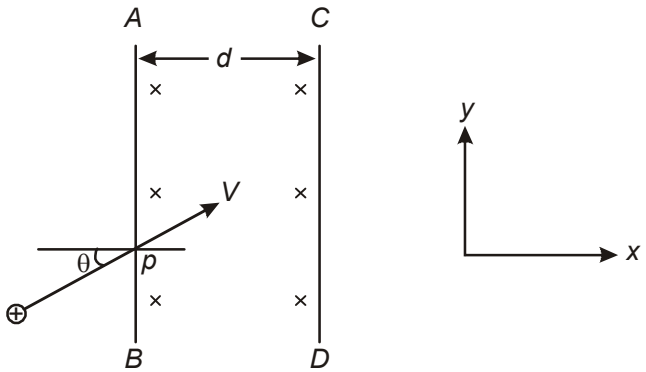
3. AB and CD are two parallel planes perpendicular to the X axis. There is a uniform magnetic field (B) in the space between them directed in negative Z direction. Width of the region having field is d and rest of the space is having no field. A particle having mass m and charge $+q$ enters the region with a velocity V making an angle θ with the X direction as shown.

   (a) Find the values of d for which the particle will come out of the magnetic field crossing CD.

   (b) For $d = \left(\frac{\sqrt{2}-1}{2}\right)\frac{mv}{qB}$ and $\theta = \frac{\pi}{6}$ find the angular deviation in the path of the particle.

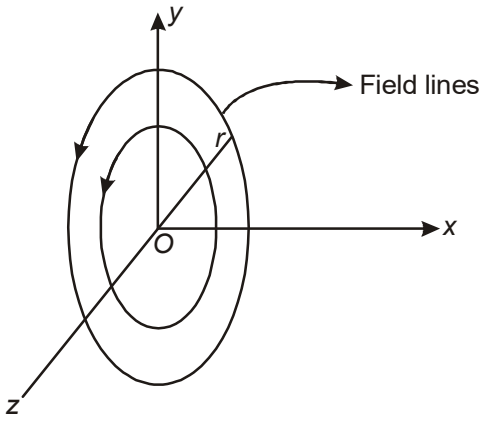
   (c) Find the deviation in path of the particle if

$$d = \frac{5mv}{4qB}(1-\sin\theta)$$



4. A long uniform cylindrical beam of radius R consists of positively charged particles each of charge q, mass m and velocity V along positive x direction. The axis of the beam is the x-axis. The beam is incident on a region having magnetic field in y-z plane. The magnetic field in the region is confined to $0 \le x \le \Delta x$ (Δx is small). The field lines are circular in yz plane as shown. The magnitude of the field is given by

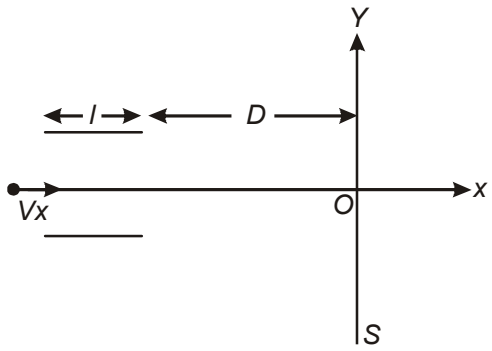
   $B = B_0 r$ where B_0 is a constant and r is distance from the origin.



   Show that this magnetic field acts as a converging lens for the ion beam and obtain the expression for focal length. Neglect the divergence in the beam caused due to electromagnetic interaction of the charge particles.

5. (a) A charge particle travelling along positive x direction with speed V_x enters a region of width ℓ having a uniform electric field E in positive y direction. A screen is kept, at a distance D(>>ℓ) from the region of the field, in yz plane. Find the y co-ordinate of the point where the particle strikes the screen. Charge and mass of the particle is $+q$ and m respectively.

   (b) The electric field in the region is replaced with a uniform magnetic field B in negative z direction. Now calculate the y co-ordinate of the point on the screen where the particle hits it. Assume deflection due to field to be small and $D >> \ell$.



   (c) Now the field region is filled with a uniform electric (E) and magnetic field (B) both directed in positive y direction. A beam of protons and some other positive ion enters the region travelling along x direction. The particle hit the screen along two curved paths. Explain.

   Draw the two curves in yz plane and point out which one represents the protons.

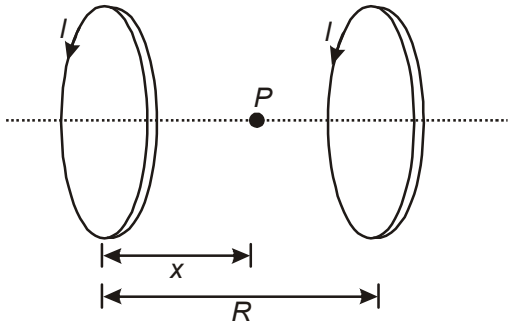
6. In a region of space a uniform magnetic field exist in positive z direction and there also exists a uniform electric field along positive y direction. A particle having charge $+q$ and mass m is released from rest at the origin. The particle moves on a curve known as cycloid. If a wheel of radius R were to roll on the X axis, a fixed point on the circumference of the wheel would generate this cycloid. Find the radius R. It is given that strength of magnetic and electric fields are B_0 and E_0 respectively.

MAGNETIC EFFECT OF CURRENT

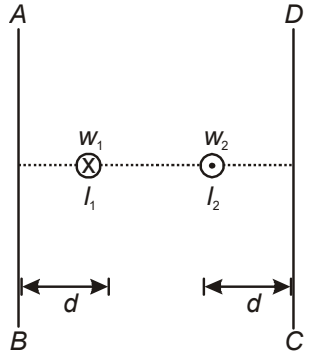
1. Two identical coils having radius R and number of turns N are placed co-axially with their centres separated by a distance equal to their radius R. The two coils are given same current I in same direction. The configuration is often known as a pair of Helmholtz coil.

   (i) Calculate the magnetic field (B) at a point (P) on the axis between the coils at a distance x from the centres of one of the coils.

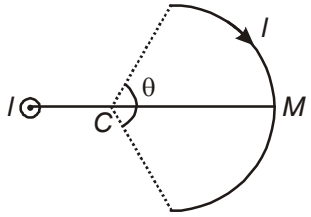
   (ii) Prove that $\frac{dB}{dx} = 0$ and $\frac{d^2B}{dx^2} = 0$ $\left[\text{In fact } \frac{d^3B}{dx^3} \text{ is also equal to zero}\right]$ at the point lying midway between the two coils. What conclusion can you draw from these results?



2. In the figure shown w_1 represent the cross section of an infinitely long wire carrying current I_1 into the plane of the fig. AB is a line of length L and the wire w_1 is symmetrically located with respect to the line. The line integral $\int_A^B \vec{B}\cdot\vec{dl}$ along the line from A to B is equal to $-a_0$ where a_0 is a positive number. Another long wire w_2 is placed symmetrically with respect to AB (see fig) and the value of $\int_A^B \vec{B}\cdot\vec{dl}$ becomes zero. Consider a line DC to the right of w_2. The line is parallel to AB and has same length. The two wires fall on perpendicular bisector of both lines. If $\int_C^D \vec{B}\cdot\vec{dl} = 2a_0$ with both wires w_1 and w_2 present, calculate the ratio of current $\frac{I_2}{I_1}$ in the two wires.



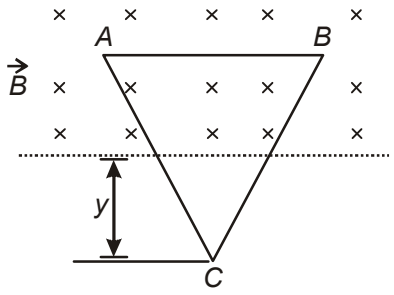
3. A current carrying conductor is in the shape of an arc of a circle of radius R subtending an angle θ at the centre (C). A long current carrying wire is perpendicular to the plane of the arc and is at a distance $2R$ from the midpoint (M) of the arc on the line joining the points M and C. Current in the arc as well as straight wire is I. Find the magnetic force on the arc.



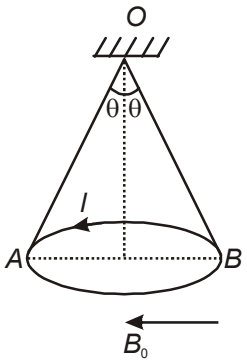
4. A current carrying loop is in the shape of an equilateral triangle of side length *a*. Its mass is *M* and it is in vertical plane. There exists a uniform horizontal magnetic field *B* in the region shown.

   (a) The loop is in equilibrium for $y_0 = \frac{\sqrt{3}}{4}a$, Find the current in the loop.

   (b) The loop is displaced slightly in its plane perpendicular to its side *AB* and released. Find time period of its oscillations. Neglect emf induced in the loop. Express your answer in terms of *a* and *g*.



5. A uniform ring of mass *M* and radius *R* carries a current *I* (see figure). The ring is suspended using two identical strings *OA* and *OB*. There exists a uniform horizontal magnetic field B_0 parallel to the diameter *AB* of the ring. Calculate tension in the two strings. [Given $\theta = 60°$]

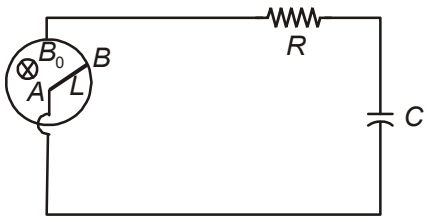


ELECTROMAGNETIC INDUCTION

1. A perfectly conducting ring of radius *L* is kept fixed on a horizontal surface. A vertical uniform magnetic field B_0 exists in the region. A conducting rod (*AB*) of length *L* is hinged at the centre of the ring at *A* and its other end (*B*) touches the ring. The ring and the end *A* of the rod are connnected to an external circuit having resistance *R* and capacitance *C*. The rod is made to rotate at a constant angular speed ω_0. Neglect friction and self inductance of the circuit.

   (i) Find work done by the external agent in rotating the rod of the time the capacitor acquires a charge q_0.

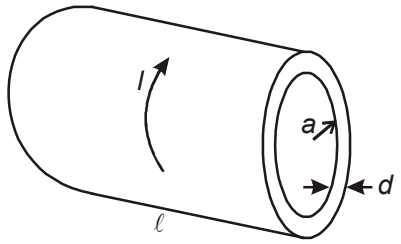
   (ii) Find heat generated in resistance *R* by the time the capacitor acquires a charge q_0.



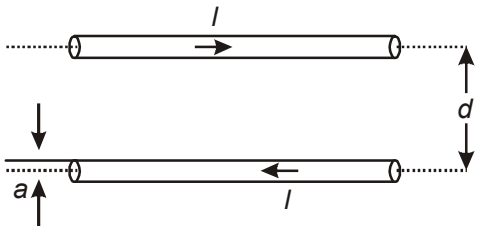
2. A hollow cylinder made of material of resistivity ρ has length ℓ, radius *a* and wall thickness *d* ($\ell >> a >> d$). A current *I* flows through the cylinder in tangential direction and is uniformly distributed along its length.

(a) Find the emf developed along the circumference of the cylinder if the current changes at the rate $\frac{dI}{dt} = \beta$

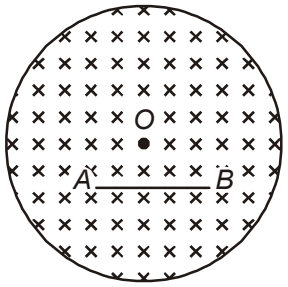
(b) Assume that no external source is present and the current at time $t = 0$ is I_0. The current decays with time. Write I as a function of time.



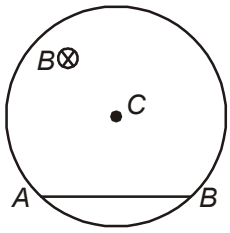
3. A long straight wire of cross sectional radius *a* carries a current *I*. The return current is carried by an identical wire which is parallel to the first wire. The centre to centre distance between the two wires is *d*. Find the inductance (*L*) of a length *x* of this arrangement. Neglect magnetic flux inside the wires.



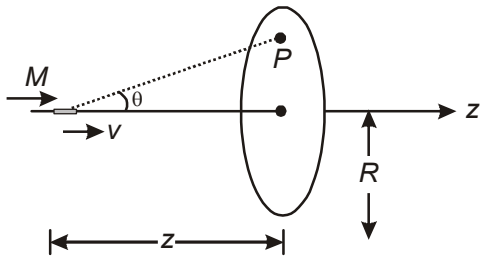
4. A cylindrical region of radius *R* is filled with a uniform magnetic field *B* as shown in the figure. A metal wire (*AB*) of length *L* is placed inside the field such that its ends are symmetrically located with respect to the centre (*O*) of the circular cross section of the region. If the magnetic field is changed at a rate $\frac{dB}{dt}$ the emf induced in the metal wire is ε. Find change in value of ε if the wire is displaced by a small distance ΔL parallel to its own length. Assume that the wire remains inside the field region.



5. A cylindrical volume of radius *R* has a uniform axial magnetic field *B*, which is increasing at a rate of $\frac{dB}{dt} = \alpha\ \text{Ts}^{-1}$. A chord (*AB*) of the circular cross section of the cylindrical region has length *L*. Calculate the line integral of induced electric field $\left(\int_A^B \vec{E}\vec{dl}\right)$ as one moves along the chord from *A* to *B*. Try to find the answer without actually performing the integration. Is the value of integral same if one moves along the arc from *A* to *B*?



6. A short bar magnet having magnetic dipole moment M is moving along the axis of a fixed conducting (non-magnetic) ring of radius R. The axis of the ring is along z direction.

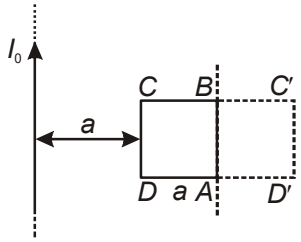


(a) Write the z component of magnetic field due to the magnetic dipole at a point P in the plane of the ring, at the instant the magnet is at a distance z from the centre of the ring. Position of point P can be defined in terms of angle θ as shown.

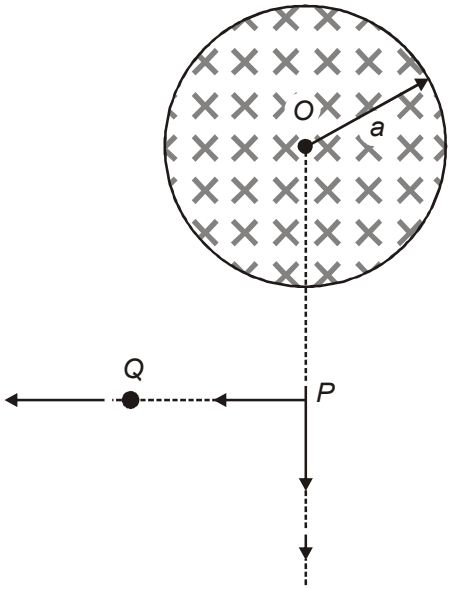
(b) Write the magnetic flux through the ring due to the magnetic field produced by the magnet as a function of z.

(c) Write the magnitude of emf induced in the ring at the instant shown if speed of the magnet at the moment is v.

7. Figure shows a square conducting frame and a long wire-both lying in the same plane. The side length of the square loop is 'a' and it is at a distance 'a' from the long wire which is having a steady current I_0. The inductance and resistance of the square loop are L and R respectively. The loop is turned by 180° about its side AB so as to bring it to final position $ABC'D'$ at rest. Calculate the net charge that flow past a side of the loop.



8. There exists a uniform magnetic field perpendicular to the plane of the figure in a cylindrical region of radius a. The magnetic field is increasing at a constant rate of α Ts^{-1}. A particle having charge q is at a point P outside the field region. The particle is slowly moved to infinity. Calculate work done by the external agent on the particle if

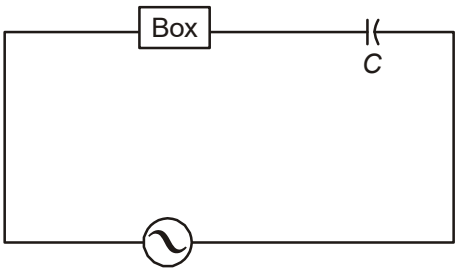


(a) The particle is moved in radial direction OP.

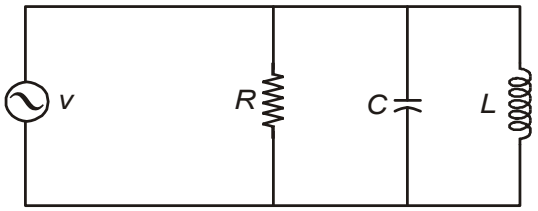
(b) The particle is moved in a direction perpendicular to OP along PQ.

ALTERNATING CURRENT

1. A box has a large electric circuit inside it. When it was connected to an ac generator it was found that it was putting a lot of load on the generator and the power factor of the box was $\frac{1}{\sqrt{2}}$. A capacitor of capacitance *C* was connected in series with the box and the power factor of the circuit became equal to the ideal value. Find the impedance of the box. The generator has an angular frequency of ω.



2. In the circuit shown the source voltage is given as $v = V_0 \sin\omega t$. Find the current through the source as a function of time.



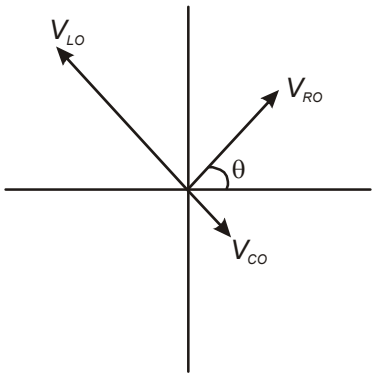
3. In a series *LCR* circuit the phasors corresponding to voltage across resistance, capacitor and inductor at an instant are as shown in the Figure and have amplitudes of V_{RO} = 4 volt, V_{CO} = 3 volt, and V_{LO} = 6 volt

   (a) Is the source frequency larger than or lesser than the resonance frequency? Does the current (*I*) lead or lag the source voltage?

   (b) Find the voltage amplitude of the source.

   (c) If V_R phasor makes θ = 53° at time *t* = 0, write the source voltage as function of time. Take angular frequency of the source to be ω

   $\left[\sin 53° = \frac{4}{5}\right]$

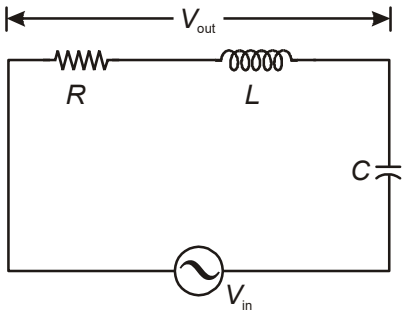


4. A series *LCR* circuit having resistance *R*, capacitance *C* and inductance *L* has a voltage source of angular frequency ω and voltage V_{in}. Output voltage (V_{out}) is taken as voltage across the resistor and inductor combined.

   (a) Find $\eta = \frac{V_{out}}{V_{in}}$

(b) Find η in the limit of large $\omega\left(\omega >> \frac{1}{RC}, \frac{1}{\sqrt{LC}}, \frac{R}{L}\right)$

(c) Find η in the limit of small $\omega\left(\omega << \frac{1}{RC}, \frac{1}{\sqrt{LC}}, \frac{R}{L}\right)$

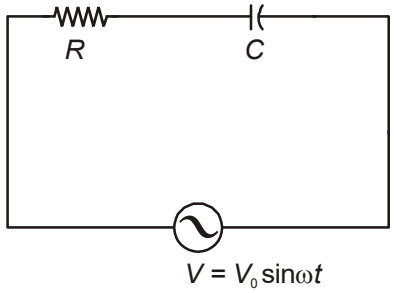


5. A resistance *R* and a capacitor having capacitance *C* are connected to an alternating source having emf $V = V_0\sin(\omega t)$. It is given that $\omega = \frac{1}{\sqrt{3}RC}$

(a) Plot the variation of power supplied by the source as a function of time. Mark the maximum and minimum values of power in the graph.

(b) How does the plot change if capacitor is removed and only *R* remains connected to the source?

(c) Plot the graph when only *C* remains connected to source and *R* is removed.

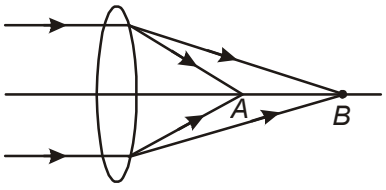


GEOMETRICAL OPTICS

1. The refractive index of light in glass varies with its wavelength according to equation

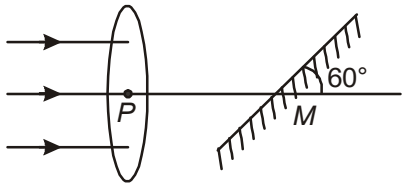
$$\mu(\lambda) = a + \frac{b}{\lambda^2}$$

where *a* and *b* are positive constants.

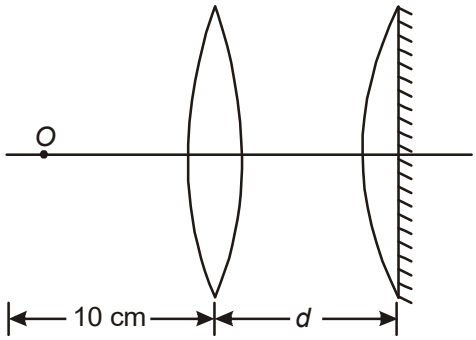


A nearly monochromatic parallel beam of light is incident on a thin convex lens as shown. The wavelength of incident light is $\lambda_0 \pm \Delta\lambda$ where $\Delta\lambda << \lambda_0$. The light gets focused on the principal axis of the lens over a region *AB*. If the focal length of the lens for a light of wavelength λ_0 is is f_0, find the spread *AB*.

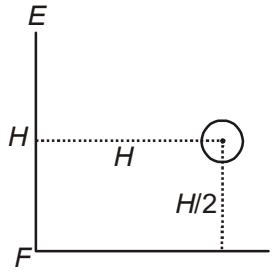
2. A horizontal parallel beam of light passes through a vertical convex lens of focal length *f*. The optical centre of the lens is *P*. A small plane mirror is placed at point *M* inclined at 60° to the axis of the lens. Distance $PM = \frac{f}{2}$. The mirror reflects the light passing through the lens and froms an image at point *I*. Find distance *PI*.



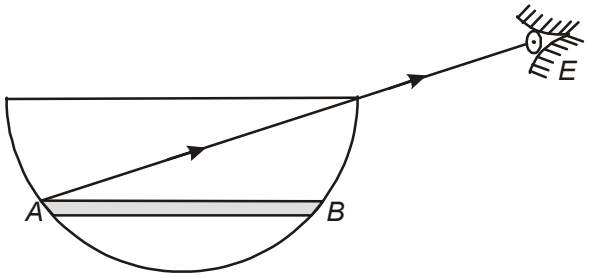
3. A convex lens of focal length 20 cm and another planoconvex lens of focal length 40 cm are placed co-axially. The plane surface of the planoconvex lens is silvered. An object *O* is kept on the principal axis at a distance of 10 cm from the convex lens (see Figure). Find the distance *d* between the two lenses so that final image is formed on the object itself.



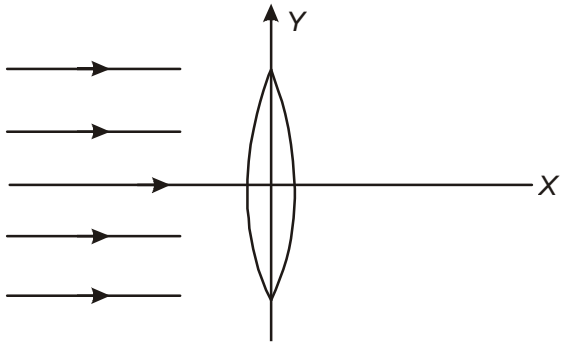
4. In the Figure *FE* is a man of height *H* standing on a floor. *E* is eye of the man and *F* is his foot. The distance between eye and the head is negligible. A steel ball of radius *r* is suspended in front of him. The distance of the ball from the man is *H* and height of the centre of the ball from the floor is $\frac{H}{2}$. It is given that $r << H$. The surface of the ball acts like a mirror and the man sees his image in it. Calculate the angle subtended by the image at the eye of the man.



5. A stick is placed inside a hemispherical bowl as shown in Figure. The stick is horizontal and has a length of 2*a*. Eye of an observer is located at *E* such that it can just see the end *A* of the stick. A liquid is filled upto edge of the bowl and the end *B* of the stick becomes visible to the observer. Radius of the bowl is *R*. Find the refractive index (μ) of the liquid.

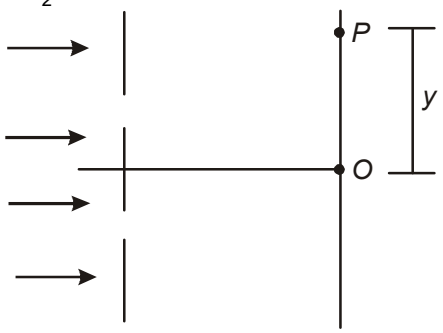


6. Monochromatic light rays parallel to the principal axis (then *x* axis) are incident on a convex lens of focal length *f*. If the lens oscillates such that it tilts up to a small angle θ on either side of the *y* axis, then find the distance between the extreme positions of the image. Take $\sec\theta \simeq 1 + \frac{\theta^2}{2}$ for small θ.



WAVE OPTICS

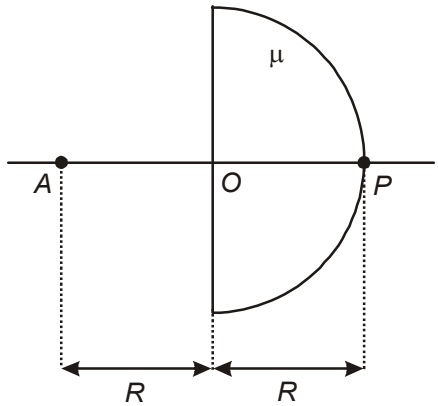
1. In Young's double slit experiment, when the slit plane is illuminated with light of wavelength λ_1, it was observed that point *P* is closest point from central maximum *O*, where intensity was 75% the intensity at *O*. When the light of wavelength λ_2 is used, point *P* happens to be the nearest point from *O* where intensity is 50% of that at *O*. Find the ratio $\frac{\lambda_1}{\lambda_2}$.



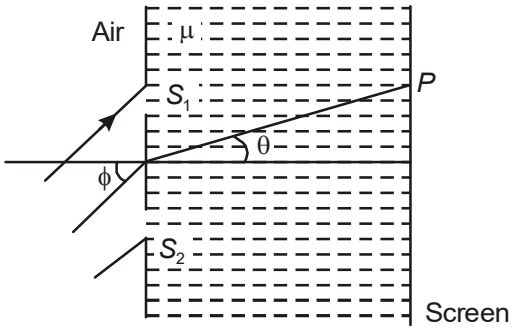
2. A point source (*A*) is kept on the axis of a hemispherical paper weight made of glass of refractive index $\mu = \frac{3}{2}$. The distance of the point source from the centre (*O*) of the sphere is *R* where *R* is radius of the hemisphere. Use paraxial approximations for answering following questions.

   (a) Find the change in radius of curvature of the wavefronts just after they enter the glass at *O*.

   (b) Find the radius of curvature of the wavefronts at point *P* just outside the glass.



3. Light is incident at an angle ϕ with the normal to a vertical plane containing two narrow slits (S_1 and S_2) at separation d. The medium to the left of slit plane is air and wavelength of the incident light is λ. The medium to the right of the slit plane has refractive index μ. Find all values of angular position (θ) of a point P where we will observe constructive interference. Wavelength of incident light is λ.

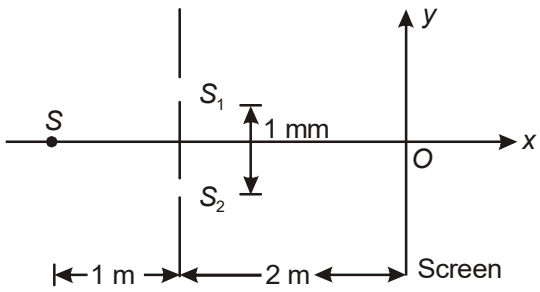


4. In a Young's double slit experiment set up source S of wavelength λ = 500 nm illuminates two symmetrically located slits S_1 and S_2. The source S oscillates about its shown position parallel to the screen according to the equation y = (0.5 mm) sin(πt)

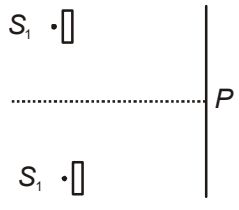
   Where t is time in second. Distances are as marked in the figure.

   (a) Write the y co-ordinate (y') of central maximum as a function of time

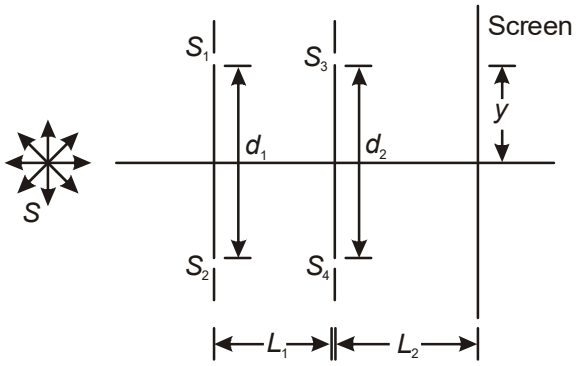
   (b) Find least value of time (t) at which the intensity becomes maximum at a point on the screen that is exactly in front of slit S_1.



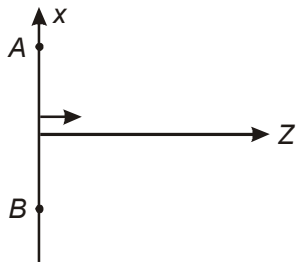
5. In Young's double slit experiment, the upper slit is covered by a thin glass plate of refractive index 1.4 while the lower slit is covered by another glass plate having same thickness as the first one but having refractive index 1.7. Interference pattern is observed using light of wavelength 5400 Å. It is found that the point P on the screen where the central maximum fell before the glass plates were inserted, now has $\frac{3}{4}$ the original intensity. It is also observed that what used to be 5th maximum earlier, lies below the point P while the 6th minimum lies above P. Calculate the thickness of glass plates. Absorption of light by glass plate may be neglected.



6. In the arrangement shown, S is a point source of monochromatic light. S_1 and S_2 are two slits located symmterically with respect to the source with separation between them d_1. Parallel to this slit plane there are two more slits (S_3 and S_4) at separation d_2. These slits are also symmetrically located with respect to S. A screen is at a distance L_2 from this slit plane. How does the intensity on the screen change with y and d_1?

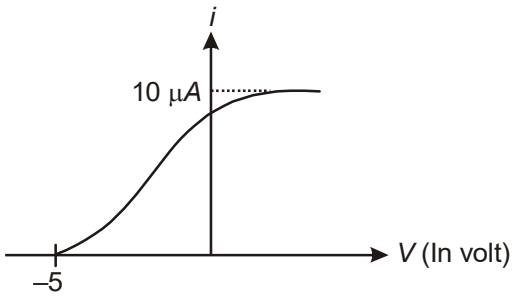


7. The refractive index of a medium changes as $\mu = \mu_0\left[1-\frac{x^2+y^2}{d^2}\right]^{\frac{1}{2}}$ where μ_0 is the refractive index on the *z* axis. A plane wavefront (*AB*) is incident along *z* axis as shown in the figure. Draw the wavefront at a later time Δt. Is the wavefront getting focused?



WAVE PARTICLE DUALITY AND ATOMIC PHYSICS

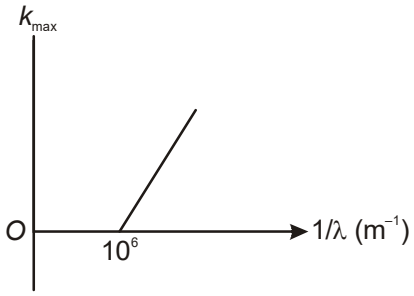
1. In a photoelectric experiment a metal plate of work function ϕ = 2.0 eV is irradiated with beam of monochromatic light. The photoelectric current (*i*) versus the applied potential difference (*V*) graph is as shown in the figure. It is known that the efficiency of photoemission is 10^{-3} %. Calculate the power of light incident on the metal plate.

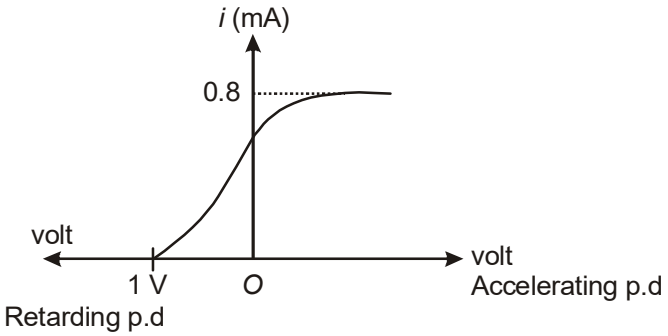


2. In photoelectric experiment set up, the maximum kinetic energy (k_{max}) of emitted photoelectrons was measured for different wavelength (λ) of light used. The graph of k_{max} vs $\frac{1}{\lambda}$ was obtained as shown in first figure. In the same setup, keeping the wavelength of incident light fixed at λ, the applied potential difference was varied and the photoelectric current was recorded. The result has been shown in graph in second figure.

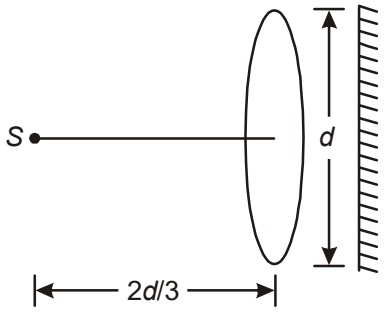
   (a) Find λ in Å

   (b) Taking the photo efficiency to be 2% (i.e. percentage of incident photons which produce photoelectrons) find the power of light incident on the emitter plate in the experiment. [Take *hc* = 12400 eV Å]

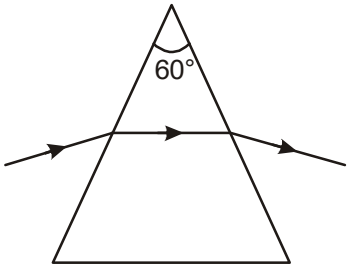




3. A point source (S) of light having power 500 W is kept at the focus of a lens of aperture diameter d. The focal length of the lens is $\frac{2d}{3}$. Assume that 40% of the incident light energy is transmitted through the lens and the complete transmitted light is incident normally on a perfectly reflecting surface placed behind the lens. Calculate the force on the reflecting surface.



4. An equilateral glass prism kept on a table has refractive index of $\mu = \sqrt{2}$. It is illuminated by a narrow laser beam having power P_0 and wavelength λ. The path of the laser beam inside the prism is parallel to the base of the prism. Calculate change in weight of the prism due to the incident laser beam.



5. Assume that structure of hydrogen atom is governed by classical mechanics. An electron is circulating around a proton and it is radiating energy at a rate given by $\frac{dE}{dt} = \frac{e^2a^2}{6\pi c^3 \in_0}$ where a is the acceleration of the electron.

   Assume that speed of the electron is v(<< c).

   (a) Estimate the fraction of kinetic energy lost by the electron per revolution in terms of v. Make suitable assumptions.

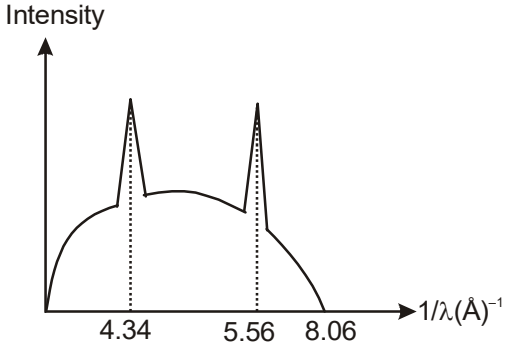
   (b) Is the energy loss per revoution large? Is it safe to assume that orbit is circular during a small time interval?

6. The *X*-ray spectrum of a metallic target has been shown in figure

   (a) What is the accelerating potential difference for bombarding electrons?

   (b) Two characteristic *X*-rays have been shown in the figure one of them is k_α *X*-ray and the other one is k_β *X*-ray. What is wavelength of k_α *X*-ray?

   (c) Find the atomic number of the target atom.



7. A hydrogen atom in ground state is moving with a kinetic energy of 30 eV. It collides with a deuterium atom in ground state at rest. The hydrogen atom is scattered at right angle to its original line of motion. Assume that energy of nth state in both the atoms is given by $E_n = -\frac{13.6}{n^2}$eV and the mass of deuterium is twice that of hydrogen. Write the maximum and minimum possible kinetic energy of deuterium after collision.

NUCLEAR PHYSICS

1. A free neutron at rest, decays into three particles: a proton, an electron and an anti neutrino.

$$^1_0n \rightarrow {}^1_1p + {}^0_{-1}e + \bar{v}$$

   The rest masses are: m_n = 939.5656 Me*V*/c^2

   m_p = 938.2723 MeV/c^2 m_e = 0.5109 MeV/c^2

   In a particular decay, the antineutrino was found to have a total energy(including rest mass energy) of 0.0004 MeV and the momentum of proton was found to be equal to momentum of electron. Find the kinetic energy of the electron.

2. A radioactive sample consists of two isotopes one of them decays by α-emission with a half life of τ_1 = 405 s and the other one decays by β emission with half life of τ_1 = 1620 s. At time t = 0, probabilities of getting α and β particles from the sample are equal.

   (a) If a particle coming out of the sample is detected at t = 1620 s, what is the probability that it is α particle?

   (b) Find the time when total number of surviving nuclei is half the initial quantity.

   Given : If $x^4 + 4x - 2.5 = 0$ then $x = 0.59$

   $\log_{10} 2 = 0.301$; $\log_{10} 5.9 = 0.774$

3. The uranium - 238 decay series (also known as $4n + 2$ series. Can you tell why?) ends at a stable isotope $_{82}Pb^{206}$. Lead -206 is found in a certain uranium ore due to distintegration of uranium.

   (a) How many alpha and beta particles are emitted in the series decay of one $_{92}U^{238}$ nucleus into $_{82}Pb^{206}$?

   (b) What is the age of uranium ore if it now contains 8g of Pb^{206} for every 10 g of U^{238}? Half life of U^{238} is 4.5 billion year.

   [Take $\ell n2 = 0.6930$; $\ell n\ (1.9243) = 0.6546$]

4. Moderator is used in a nuclear reactor to slow down fast neutrons generated after a fission event in the reactor. Consider a fast neutron (mass m_n) moving at speed v. The neutron hits a moderator nucleus (Mass M), which is orginally static, and gets scattered eleastically by 180°.

   (a) Find the ratio of kinetic energy of the neutron after and before collision.

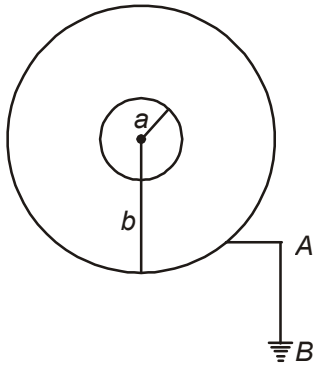
   (b) Show that when $M >> m_n$, the neutron does not lose much energy and therefore, a moderator should have a small mass number.

5. A β active radioactive source is in the form of a conducting sphere of radius a. It is surrounded by a concentric conducting shell of radius $b(> a)$. The shell is grounded. β particles are emitted with kinetic energy ranging from E_1 to E_2 ($> E_1$).

   (a) Find the maximum potential that will be acquired by the sphere of radius a.

   (b) Find the total charge that will flow through the grounding wire AB.

   (c) Find the final maximum charge on the outer sphere.



6. Nuclei of radioactive element A are being produced at rate t^2 at time t. The element A has decay constant equal to λ. If N is the number of active nuclei of element A at any time t then $\frac{dN}{dt}$ was found to be minimum at $t = t_0$. Find N at time $t = t_0$.

❑ ❑ ❑

1. (a) From among the elements, choose the following :

Cl, Br, F, O, Al, C, Li, Cs, and Xe

(i) The element with highest electron gain enthalpy.

(ii) The element with lowest ionisation enthalpy.

(iii) The element whose oxide is amphoteric.

(iv) The element which has smallest radius.

(v) The element whose atom has 6 electrons in the outermost shell.

(vi) The element which belongs to zero group.

(vii) The element which forms largest number of compounds.

(viii) The element which shows diagonal relationship with Mg.

(ix) The element which is in liquid state under ordinary atmospheric conditions.

(b) The elements Na, Mg, Al, Si, P, S, Cl and Ar are arranged in the periodic table in the increasing order of their atomic numbers.

(i) Which element is the most electronegative?

(ii) Which element is the most electropositive?

(iii) Which element is the least reactive?

(iv) Which elements are gases at room temperature?

(v) Which element is the most abundant metal?

(vi) Which element is the strongest oxidising agent?

(vii) Which element has the electronic configuration?

$1s^2, 2s^2, 2p^6, 3s^2 3p_x^1 3p_y^1 3p_z^1$

(viii) Which element shows +6 oxidation state?

(c) Pick up from the elements Na, Cl, Si and Ar.

(i) The element with highest ionisation enthalpy.

(ii) The element with highest electron gain enthalpy.

(iii) The element with smallest size.

(iv) The element with largest radius.

(v) The element with lowest ionisation enthalpy.

(d) A, B and C are the elements of a short period of a periodic table containing one, two and three electrons in their outermost shell, respectively.

Arrange A, B and C in the increasing order of :

(i) Their ionisation enthalpy

(ii) Basic character of their oxides

(iii) Covalent character in their chlorides and

(iv) Melting points of their chlorides

2. Ionisation enthalpy and electron gain enthalpy of fluorine and 17.42 and 3.45 eV respectively. Calculate the electronegativity of flurine.

3. The electron gain enthalpy of chlorine is 3.7 eV. How much energy in kcal is released when 2g of chlorine is completely converted to Cl^- ion in a gaseous state?

(1 eV = 23.06 kcal mol^{-1})

4. Calculate the electronegativity of silicon using Allred-Rochow method. Covalent radius of silicon is 1.175 Å.

5. The correct order of relative stability of half filled and completely filled shells is

(a) $p^3 < d^5 < d^{10} < p^6$ (b) $d^5 > p^3 > d^{10} > p^6$

(c) $p^3 < d^{10} < d^5 < p^6$ (d) $d^5 < p^3 < d^{10} < p^6$

6. In which of the following arrangements, the sequence is not strictly according to the property written against it?

(a) $CO_2 < SiO_2 < SnO_2 < PbO_2$: increasing oxidising power

(b) HF < HCl < HBr < HI : increasing acid strength

(c) $NH_3 < PH_3 < AsH_3 < SbH_3$: increasing basic strength

(d) B < C < O < N : increasing first ionisation energy

7. The amount of energy released when 10^6 atoms of iodine in vapour state are converted to I^- ions is 4.9×10^{-13}J. What is the electron affinity of iodine in eV per atom?

(a) 2.0 (b) 2.5

(c) 2.75 (d) 3.06

8. Match the following lists and select the correct answer.

List-I	**List-II**
(A) $1s^2, 2s^2 2p^6, 3s^2 3p^6, 4s^1$	(p) *d*-block element
(B) $1s^2, 2s^2 2p^6, 3s^2 3p^6$	(q) Halogen
(C) $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^6, 4s^2$	(r) Alkali metal
(D) $1s^2, 2s^2 2p^5$	(s) Noble gas

	A	B	C	D
(a)	p	q	r	s
(b)	r	s	p	q
(c)	p	r	q	s
(d)	q	s	r	p

9. Which is not arranged in the correct sequence?

(a) d^5, d^4, d^2, d^1 — Increasing magnetic moments

(b) MO, M_2O_3, MO_2, M_2O_5 — Increasing acidic strength

(c) Mn, Cr, V, Sc — Decreasing number of oxidation states

(d) p^6, d^{10}, p^3, d^5 — Decreasing relative stability

10. What is the ΔH of the following reaction?

$$Mg(g) + 2F(g) \longrightarrow Mg^{2+}(g) + 2F^{-}(g) \text{ or } MgF_2(g)$$

If electron affinity of F(g) = –328 kJ mol^{-1} and

First ionisation energy of Mg = 737.7 mol^{-1} and

Second ionisation energy of Mg = 1451 kJ mol^{-1}.

(a) 1532.7 kJ mol^{-1}

(b) 1860.7 kJ mol^{-1}

(c) 2516.7 kJ mol^{-1}

(d) 2844.7 kJ mol^{-1}

11. When ZnS and PbS minerals are present together, NaCN is added to separate them in froth floatation process because

(a) $Pb(CN)_2$ is precipitated while there is no effect on ZnS

(b) ZnS forms soluble complex, $Na_2Zn(CN)_4$

(c) PbS forms soluble complex, $Na_2Pb(CN)_4$

(d) Both (a) and (b)

12. Poling process is used

(a) For the removal of Cu_2O from Cu

(b) For the removal of Al_2O_3 from Al

(c) For the removal of Fe_2O_3 from Fe

(d) In all the above

13. Match the metals of List-I with the process of refining of List-II.

List-I	List-II
(a) Ni	(p) Mond's process
(b) Ti	(q) Van Arkel method
(c) Zr	(r) Distillation
(d) Zn	(s) Pyrometallurgical process

14. Match Column-I with Column-II.

Column-I	Column-II
(a) Cyanide process	(p) Extraction of Ag
(b) Electrolytic reduction	(q) Extraction of Au
(c) Carbon reduction	(r) Extraction of Al
(d) Self reduction	(s) Extraction of Pb
	(t) Extraction of Hg

15. (a) Compare the structures of H_2O and H_2O_2.

(b) Do you expect different products in solution when $AlCl_3$ is treated separately with (i) normal water, (ii) acidified water and (iii) alkaline water?

16. What mass of CaO will be required to remove the hardness of 1000 litres of water containing 1.62 g of calcium bicarbonate per litre?

17. 100 mL tap water containing $Ca(HCO_3)_2$ was titrated with N/50 HCl with methyl orange as indicator. If 30 mL of HCl were required, calculate the temporary hardness as parts of $CaCO_3$ per 10^6 parts of water.

18. 20 mL of H_2O_2 after acidification with dilute H_2SO_4 required 30 mL of N/12 $KMnO_4$ for complete oxidation. Calculate the percentage of H_2O_2 in the solution. Equivalent mass of H_2O_2 = 17.

19. 25 mL of hydrogen peroxide solution were added to excess of acidified potassium iodide solution. The iodine so liberated required 20 mL of 0.1 N sodium thiosulphate solution. Calculate strength in terms of normality, percentage and volume strength.

20. What is the drgree of hardness of sample of water containing 24 mg of $MgSO_4$ (molecular mass 120) per kg of water?

(a) 10 ppm (b) 15 ppm

(c) 20 ppm (d) 25 ppm

21. When a substance A reacts with water, it produces a combustible gas B and a solution of substance C in water. D reacts with this solution of C and produces the same gas B on warning. D can also produce gas B on reaction with dilute H_2SO_4. A imparts a deep golden yellow colour to smokeless flame. A, B, C and D respectively are

(a) Na, H_2, NaOH, Zn (b) K, H_2, KOH, Al

(c) Ca, H_2, $Ca(OH)_2$, Sn (d) CaC_2, C_2H_2, $Ca(OH)_2$, Fe

22. 'A' is a binary compound of a univalent metal. 1.422 g of 'A' reacts completely with 0.321 g of sulphur in an evacuated and sealed tube to give 1.743 g of white crystalline solid 'B', that forms a hydrated double salt 'C' with $Al_2(SO_4)_3$. Identify 'A', 'B' and 'C'.

23. How would you prepare the following?

(a) Sodium thiosulphate from sodium carbonate.

(b) Microcosmic salt from disodium hydrogen phosphate.

(c) Sodium carbonate from sodium chlroride.

(d) Sodium thiosulphate from sodium sulphite.

(e) Sodium cyanide from metallic sodium.

24. Answer the following :
    (i) What is meant by black ash?
    (ii) What is the action of NaOH on ammonium salts?
    (iii) Which electrolyte is used to obtain sodium in Castner's process?
    (iv) What is washing soda?
    (v) Which electrolyte is used in Nelson's cell?
    (vi) What is the intermediate product in Solvay's process?
    (vii) Most abundant alkali metal in the earth's crust is
    (viii) Which chloride of an alkali metal is soluble in alcohol?
    (ix) Which substance is added to sodium chloride as to reduce its fusion temperature during manufacture of sodium?
    (x) What product is formed when carbon monoxide is passed through sodium hydroxide under high pressure?
25. What happens when following are heated?
    (i) Potassium nitrate
    (ii) Sodium carbonate
    (iii) Microcosmic salt
    (iv) Sodium bicarbonate
    (v) Lithium hydroxide
    (vi) Lithium nitrate
    (vii) Lithium carbonate
    (viii) Sodium thiosulphate
26. Give one test each to make distinction between the following paris :
    (i) NH_4Cl and KCl
    (ii) Na_2SO_3 and Na_2SO_4
    (iii) Na_2SO_3 and $Na_2S_2O_3$
    (iv) KNO_3 and KI
    (v) Na_2SO_3 and Na_2CO_3
    (vi) NaCl and KCl
27. When 20.02 g of a white solid x is heated, 4.4 g of an acid gas (A) and 1.8 g of a neutral gas (B) are evolved, leaving behind a solid residue (Y) of weight 13.8 g. (A) turns lime water milky and (B) condenses into liquid which changes anhydrous copper sulphate blue. The aqueous solution of (Y) is alkaline to litmus and give 19.7 g white precipitate (Z) with barium chloride solution. (Z) gives carbon dioxide with an acid. Identify A, B, X, Y and Z.
28. When 16.78 g of a white solid X were heated, 4.4 g of acid gas A that turned lime water milky was driven off together with 1.8 g of a gas B which condensed to a colourless liquid. The solid that remained, Y, dissolved in water to give an alkaline solution, which with excess of barium chloride solution gave a white precipitate, Z. The precipitate effervesced with acid giving off carbon dioxide. Identify A, B and Y and write down the equation for the thermal decomposition of X.

29. An aqueous solution of an halogen salt of potassium reacts with same halogen X to give KX_3, a violet coloured solution which is used in volumetric exercises (iodimetric titrations).

The halogen X is

(a) Fluorine (b) Bromine

(c) Chlorine (d) Iodine

30. The aqueous solution of an unknown sodium salt gives the following reactions.

(i) It gives white turbidity with dilute HCl solution.

(ii) It decolourises a solution of iodine in potassium iodide.

(iii) It gives a white precipitate with $AgNO_3$ solution which changes colours and finally becomes black on standing.

The unknown sodium salt is

(a) Sodium sulphite (b) Sodium sulphide

(c) Sodium bisulphite (d) Sodium thiosulphate

31. Give the names and formula of the compounds indicated in the following statements :

(a) A compound of Ca used in setting fractured bones.

(b) A compound of Mg, S, O and H used as a purgative in medicine.

(c) A compound of Ca and C used for the production of acetylene.

(d) A compound of Ca, C and N used as a fertilizer.

(e) A triatomic compound which on treatment with water gives H_2.

32. 0.3 g of magnesium ribbon was placed in a crucible and heated with the lid on until the magnesium began to burn brilliantly. At the end of the experiment there was 0.45 g of the white powder left. Show that this result does not agree with equation,

$2Mg(s) + O_2(g) \longrightarrow 2MgO(s)$

What might have gone wrong?

33. Beryllium gives a compound with the following percentage composition : Be, 6.1%; N, 37.8%; Cl, 48%; H, 8.1%. One mole of the compound had a mass of 148 g. M(Be) = 9 g mol^{-1}.

(i) What is the molecular formula of the compound?

(ii) In water, 1 mole of the compound reacts with 2 moles of Ag+ ions.

Suggest a structural formula for the compound.

34. Lattice energies of BeF_2, MgF_2, CaF_2 and BaF_2 are –2906, –2610, –2459 and –2367 kJ mol^{-1} respectively. Hydration energies of Be^{2+}, Mg^{2+}, Ca^{2+}, Ba^{2+} and F^- are –2494, –1921, –1577, –1305 and –457 kJ mol^{-1} respectively. Which of the fluorides is soluble in water?

(a) BeF_2 (b) MgF_2

(c) CaF_2 (d) BaF_2

35. The name and formula of the compound of magnesium, chlorine and oxygen used as drying agent is

(a) Magnesium oxychlorite, $Mg(OCl)_2$ (b) Magnesium chlorate, $Mg(ClO_3)_2$

(c) Magnesium perchlorate, $Mg(ClO_4)_2$ (d) None of the above

36. The metallic salt XY is soluble in water.

When the solution is treated with NaOH solution, a white precipitate (A) is formed. This precipitate is soluble in excess of NaOH solution to form a compound (B). When this solution is boiled with solid NH_4Cl, a precipitate of a compound (C) is formed. An aqueous solution of XY upon treatment with $BaCl_2$ solution gave a white precipitate (D) which was insoluble in dilute HCl. The salt XY forms a double salt with potassium sulphate.

37. Compound (X) on reduction with $LiAlH_4$ gives a hydride (Y) containing 21.72% hydrogen along with other products. The compound (Y) reacts with air explosively resulting in boron trioxide. Identify (X) and (Y). Give balanced reactions involved in the formation of (Y) and its reaction with air. (Draw the structure of (Y).

38. Hydride of boron occurs as B_2H_6 but B_2Cl_6 does not exist, This is because:

(a) $p\pi$ — $d\pi$ back bonding is possible in B_2H_6 but not in B_2Cl_6

(b) Boron and hydrogen have almost equal values of electronegativity

(c) Boron and chlorine have almost equal atomic sizes

(d) Small hydrogen atoms can easily fit in between boron atoms but large chlorine atoms do not

39. The number of isomers possible for disubstituted borazine,

$B_3N_3H_4X_2$ is

(a) 3 (b) 4

(c) 6 (d) 5

40. Give the name and structural formula of a boron compound which is optically active.

41. A colourless crystalline solid (BA) dissolves in water. When the solid is heated with concentrated H_2SO_4, reddish brown gas (X) is liberated. A solution of the salt when treated with potassium chromate solution gives a yellow precipitate (Y) insoluble in acetic acid. The same solution gives a white precipitate which dissolves in excess of NaOH to form a compound (Z). Identify (X), (Y), (Z) and the salt (BA).

42. A white solid (A) is not completely soluble in dilute HCl or dilute H_2SO_4 but dissolves in dilute HNO_3 with effervescences. In the evolved gas, a burning magnesium ribbon continues to burn. (A) when heated forms a yellow solid (B) which is an amphoteric oxide (B) when heated in air forms a red coloured compound (C). (C) when dissolved in concentrated HNO_3 gives a brown residue (D). Identify (A) to (D).

43. A white coloured inorganic salt (A) gives the following reactions :

(i) It is soluble in water and the solution has sweet taste. The solution turns black in presence of H_2S.

(ii) The solution gives a white precipitate with dilute HCl which is soluble in hot water.

(iii) The salt when heated gives acetone and a yellow coloured residue.

Identify the salt (A) and give chemical reactions.

44. A substance (X) is dissolved in concentrated hydrochloric acid and on treating the solution with sodium hydroxide solution a white precipitate formed which dissolved in excess, giving a solution with strongly reducing properties.

On heating (X) with sulphur, a brown powder (Y) was formed which dissolved in yellow ammonium sulphide and on adding HCl, a yellow precipitate was formed.

When (X) was strongly heated in air, a white powder (Z) was obtained. When (Z) was fused with sodium hydroxide and extracted with hot water, white crystals were obtained. Identify (X), (Y) and (Z) and write the reactions.

45. An inorganic compound (A) is a strong reducing agent. Its hydrolysis in water gives a white turbidity (B). Aqueous solution of (A) gives a white precipitate (C) with NaOH solution which is soluble in excess of NaOH. (A) reduces auric chloride to produce purple of cassius. (A) also reduces iodine and gives chromyl chloride test. What are (A) to (C) and give the chemical reactions?

46. A colourless gas which burns with blue flame and reduces CuO to Cu is

(a) N_2 (b) CO

(c) CO_2 (d) NO_2

47. A red coloured mixed oxide (X) on treatment with conc. HNO_3 gives a compound (Y). (Y) with HCl produces a chloride (Z) which is insoluble in cold water but soluble in hot water. (Z) can also be formed by treating (X) with conc. HCl. Compounds (X), (Y) and (Z) are

(a) Pb_3O_4, PbO_2, $PbCl_2$ (b) Mn_3O_4, MnO_2, $MnCl_2$

(c) Fe_3O_4, Fe_2O_3, $FeCl_3$ (d) Fe_3O_4, FeO, $FeCl_2$

48. A white coloured inorganic salt gives the following reactions :

(i) It is soluble in water and the solution has sweet taste. The solution turns black in presence of H_2S.

(ii) The salt when heated gives acetone and a yellow coloured residue which is used in paints.

(iii) The solution of the salt gives a white precipitate with dilute HCl which is soluble in hot water.

The inorganic salt is

(a) $Ca(CH_3COO)_2$ (b) $Zn(CH_3COO)_2$

(c) $Pb(CH_3COO)_2$ (d) $Ba(CH_3COO)_2$

49. What are the following compounds? Give their uses

(i) Vitrosil

(ii) Carbogen

(iii) Kieselguhr

(iv) Chrome yellow and chrome red

(v) Butter of tin

(vi) Litharge and Massicot

(vii) Carburated water gas

(viii) Crooke's glass

(ix) Opal glass

50. Complete the following road map.

$$H_2C_2O_4 \xrightarrow[\Delta]{\text{Conc. } H_2SO_4} \underset{\text{Gas}}{(A)} + \underset{\text{Gas}}{(B)} + \underset{\text{Liquid}}{(C)}$$

Gas (A) burns with blue flame and gas (B) turns lime water milky. Gas (A) on oxidation gives gas (B). Liquid (C) produces blue colour when comes in contact with anhydrous copper sulphate.

$$(A) + Cl_2 \longrightarrow (D) \xrightarrow[\Delta]{NH_3} (E)$$

$$(B) \xrightarrow[\Delta]{NH_3} (E)$$

What are the compounds (A) to (E)?

51. Answer the following :

(a) Why elemental phosphorus does not exist as P_2 like N_2?

(b) How NO_3^- and PO_3^- ions are differ structurally?

(c) NCl_3 gets readily hydrolysed white NF_3 does not, why?

(d) Oxides of nitrogen have open chain structures while those of phosphorus have closed chain or cage structures. Why is it so? Illustrate with one structural example for each type of the oxides.

52. An aqueous solution of a gas(X) shows following reactions :

(i) It turns red litmus blue.

(ii) When added in excess to a copper sulphate solution, a deep blue colour is obtained.

(iii) On addition to ferric chloride solution a brownish precipitate soluble in HNO_3 is obtained.

Identify (X) and give equations for the reactions in step (II) and (III).

53. Give the names and formulae of the compounds described below :

(i) A compound of N, H and O which on heating gives laughing gas.

(ii) A compound of N, H and O which on heating gives ntrogen gas.

(iii) A compound of Ca, P and O which is found in bones.

(iv) A compound of N and H which is used as refrigerant.

(v) A compound of N and H which behaves like an acid.

(vi) A compound of N and H which is used as a rocket fuel.

(vii) A compound of N, H, S and O which is used as a fertilizer.

(viii) A compound of N, H and O which is used for making oximes.

(ix) Two neutral oxides of nitrogen.

(x) The oxyacid of phosphorus is used for the preparation of HBr and HI from bromides and iodides respectively.

54. (i) An inorganic iodide (A) on heating with a solution of KOH gives a gas (B) and the solution of a compound (C).

(ii) The gas (B) on ignition in air gives a compound (D) and water.

(iii) Copper sulphate is reduced to the metal on passing (B) through the solution.

(iv) A precipitate of the compound (E) is formed on reaction of (C) with copper sulphate solution.

Identify (A) to (E) and give chemical equations for reactions at steps (I) to (IV).

55. An inorganic compound (A) when heated decomposes completely to give only two gases (B) and (C). (B) is a neutral gas, fairly soluble in water and itself decomposes on heating to two different gases (D) and (E).

(A) when warmed with NaOH gives another gas (F) which turns mercurous nitrate paper black. After sometime the gas (F) ceases to evolve, however its supply is restored by treating residual solution with aluminium powder. Identify (A) to (F) and give necessary equations.

56. Gradual addition of KI solution to $Bi(NO_3)_3$ solution initially produces a dark brown precipitate which dissolves in excess of KI to give a clear yellow solution. Write chemical equations for the above reactions.

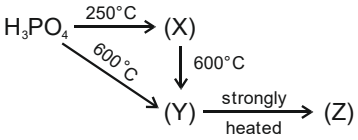
57. (i) Sodium salt of an acid (A) is formed on boiling white phosphorus with NaOH solution.

(ii) On passing chlorine through phosphorus kept fused under water, another acid (B) is formed which on strong heating gives metaphosphorus acid.

(iii) Phosphorus on treatment with conc. HNO_3 gives an acid (C) which is also formed by the action of dilute H_2SO_4 on powdered phosphorite rock.

(iv) (A) on treatment with a solution of $HgCl_2$ first gives a white precipitate of compound (D) and then a grey precipitate of (E).

Identify (A) to (E) and write balanced equations for the reactions at steps (i) to (iv).

58. 

(X), (Y) and (Z) are

(a) $H_4P_2O_7$, HPO_3 and P_4O_{10}

(b) HPO_3, $H_4P_2O_7$ and P_4O_{10}

(c) $H_4P_2O_6$, H_3PO_3 and P_4O_6

(d) $H_4P_2O_6$, HPO_3 and P_4O_6

59. For H_3PO_3 and H_3PO_4, the correct choice is

(a) H_3PO_3 is dibasic and reducing

(b) H_3PO_3 is dibasic and non-reducing

(c) H_3PO_4 is tribasic and reducing

(d) H_3PO_3 is tribasic and non-reducing

60. Explain the following:

(i) Oxygen is a gas while sulphur is a solid.

(ii) Water is liquid while H_2S is a gas.

(iii) Sulphuric acid has high boiling point and viscosity.

(iv) Oxygen almost invariably exhibits an oxidation state of –2 but the other members of the family exhibit negative as well as positive oxidation states of +2, + 4 and + 6.

(v) The tendency to show –2 oxidation state diminishes from sulphur to polonium.

(vi) Sulphur dioxide is more powerful reducing agent in an alkaline medium than in acidic medium.

(vii) Dry SO_2 does not bleach dry flowers.

(viii) H_2S is stronger acid than H_2O.

(ix) Hydrogen sulphide cannot be dried by passing through conc. H_2SO_4.

(x) Nitric acid cannot be used to prepare H_2S.

(xi) In the manufacture of sulphuric acid by the contact process, sulphur trioxide is not directly dissolved in water.

(xii) For the dilution of H_2SO_4, water should not be added to conc. H_2SO_4.

(xiii) Conc. H_2SO_4 cannot be used for drying H_2.

(xiv) The bond angle in H_2S is lower than in H_2O.

(xv) $KMnO_4$ should not be dissolved in conc. H_2SO_4.

(xvi) Ozone layer in the upper regions of atmosphere protects earth from the harmful radiations of sun.

61. A pale yellow substance (A) when heated with conc. HNO_3 evolves a brown coloured gas (B). The substance (A) also dissolves in sodium sulphite solution on heating. A clear solution (C) is formed which on acidification gives a turbid solution and a pungent smelled gas (D) which is formed by the substance (A) in air. The solution (C) decolourises iodide solution. Identify (A) to (D).

62. When sulphur is boiled with Na_2SO_3, a compound (A) is produced. (A) with excess of $AgNO_3$ solution gives a compound (B) which is water soluble and produces a black coloured sulphide (C). Compounds (A), (B) and (C) will be respectively

(a) $Na_2S_2O_3$, $Ag_2S_2O_3$, Ag_2S (b) Na_2SO_4, Ag_2SO_4, Ag_2S

(c) $Na_2S_2O_7$, Ag_2SO_4, Ag_2S (d) Na_2SO_5, Ag_2SO_5, Ag_2SO_4

63. An organic acid (A) reacts with concentrated H_2SO_4 to give a neutral oxide (B), acidic oxide (C) and a diatomic oxide (D). When (D) reacts with chlorine gas, a poisonous gas (E) is evolved. This gas with ammonia gives an organic compound (F). The compounds (A) and (F) are

(a) (A) = $H_2C_2O_4$ and (F) = NH_2CONH_2 (b) (A) = CH_3COOH and (F) = NH_2CONH_2

(c) (A) = $CHCl_3$ and (F) = $H_2C_2O_4$ (d) (A) = CCl_4 and (F) = CH_3CHO

64. Compounds A and B are treated with dilute HCl separately. The gases liberated are Y and Z respectively. Y turns acidified $K_2Cr_2O_7$ paper green while Z turns lead acetate paper black. The compounds A and B are respectively.

(a) Na_2S and Na_2SO_3 (b) Na_2SO_3 and Na_2S

(c) NaCl and Na_2CO_3 (d) Na_2SO_3 and Na_2SO_4

65. Like sulphur, oxygen does not show +4 and +6 oxidation states. The reason is

(a) That oxygen is a gas while sulphur is a solid

(b) That oxygen has high ionisation enthalpies in comparison to sulphur

(c) That oxygen has high electron affinity in comparison to sulphur

(d) That oxygen has no *d*-orbitals in its valence shell

66. Oxygen exhibits +2 oxidation state in

(a) H_2O (b) OF_2

(c) Cl_2O (d) H_2O_2

67. An inorganic compound (X) gives a brick red flame on performing flame test. This compound gives the following test also.

(a) Smells of chlorine when placed in moist air.

(b) If KI and CH_3COOH are added to the suspension in water, a brown colour is obtained.

Identify (X) and write down equations for reactions at steps (A) and (B).

68. Arrange the following :

(i) HClO, $HClO_2$, $HClO_3$, $HClO_4$ in increasing order of thermal stability.

(ii) HClO, $HClO_2$, $HClO_3$, $HClO_4$ in order of increasing acid strength.

(iii) HClO, $HClO_2$, $HClO_3$, $HClO_4$ in order of increasing oxidising power.

(iv) F^-, Cl^-, Br^-, I^- in order of increasing reducing nature.

(v) I_2, HI, HIO_4, ICI in order of increasing oxidation number of iodine.

(vi) HOF, HOCl, HOBr, HOI in order of increasing acid strength.

(vii) F_2, Cl_2, Br_2, I_2 in order of increasing oxidising power.

(viii) HF, HCl, HBr, HI in order of increasing acid strength.

(ix) F, Cl, Br, I in order of increasing electronegativity or reactivity.

(x) F, Cl, Br, I in order of increasing electron affinity.

69. A black powder when heated with NaCl and conc. H_2SO_4 gives off a greenish yellow gas. The gas on passing through liquor ammonia liberates N_2 and on passing through boiling KOH yields compounds one of which when heated with the black powder evolves oxygen. Name the gas and the black powder. Explain the chemical reactions involved with equations.

70. A certain compound (X) shows the following reactions.

(i) When KI is added to an aqueous suspension of (X) containing acetic acid, iodine is liberated.

(ii) When CO_2 is passed through an aqueous suspension of (X), the turbidity transforms to a precipitate.

(iii) When the paste of (X) in water is heated with ethyl alcohol, a product of anaesthetic use is obtained.

Identify (X) and write down chemical equations for reactions at step (i), (ii) and (iii).

71. Match the following :

[A]	
(i) Fluorine	(a) Bromo-carnallite
(ii) Chlorine	(b) Hydrogen bonding
(iii) Bromine	(c) Beckmann's process
(iv) Iodine	(d) Reducing agent
(v) Bleaching powder	(e) Pseudohalogen
(vi) HF	(f) Dennis process
(vii) HI	(g) Sea-weeds
(viii) $(CN)_2$	(h) Deacon's process

[B]	
(i) Moissan	(a) Chlorine
(ii) Scheele	(b) Bleaching powder
(iii) Balard	(c) Iodine
(iv) Courtois	(d) Fluorine
(v) Odling	(e) Bromine

72. Consider the following oxides

1. OF_2
2. Cl_2O
3. Br_2O

The correct sequence of X — O — X bond angle is

(a) 1 > 2 > 3

(b) 3 > 2 > 1

(c) 2 > 1 > 3

(d) 1 > 3 > 2

73. Identify the correct statements

1. On heating $Rb(ICl_2)$ will decompose giving RbCl and ICl.
2. CsI_3 will ionise to give Cs^{+3} and I^-.
3. KBrICl will give KCl and IBr on heating.
4. In the reaction $I^- + I_2 \longrightarrow$, the Lewis acid is I_2.

(a) 1, 3, 4 (b) 1, 2, 3

(c) 2, 3, 4 (d) 1, 2, 3, 4

74. In a given sample of bleaching powder, the percentage of available chlorine is 49. The volume of chlorine obtained if 10 g of the sample is treated with HCl at N.T.P. is

(a) 1.5 litre (b) 3.0 litre

(c) 15.0 litre (d) 150 litre

75. Which one of the following arrangements does not give the correct picture of the trends indicated against it?

(a) $F_2 > Cl_2 > Br_2 > I_2$: Bond dissociation energy

(b) $F_2 > Cl_2 > Br_2 > I_2$: Electronegativity

(c) $F_2 > Cl_2 > Br_2 > I_2$: Oxidising nature

(d) $F_2 > Cl_2 > Br_2 > I_2$: Electron gain enthalpy

76. Select the correct order of acidity

(a) HI > HBr > HCl > HF

(b) $HClO_4 > HBrO_4 > HIO_4$

(c) HClO < HBrO < HIO

(d) $HClO_4 > HClO_3 > HClO_2 > HClO$

Comprehension-1

Bleaching powder is a mixed salt of hydrochloric acid and hypochlorous acid. It has the formula, $CaOCl_2.H_2O$. It is manufactured by the action of chlorine on dry slaked lime at 40°C. There is also a view that bleaching powder is a mixture of calcium hypochlorite and basic calcium chloride $[Ca(OCl)_2 + CaCl_2.Ca(OH)_2.H_2O$.

The amount of chlorine obtained from a sample of bleaching powder by the treatment with excess of dilute acids or CO_2 is called availbale chlorine. A good sample of bleaching powder contains 33-38% of available chlorine. On long standing, it undergoes auto-oxidation and the amount of available chlorine decreases.

The estimation of available chlorine is done volumetrically by (A) Iodometric method or by (B) Arsenite method.

In textile industry, the cotton cloth is mainly bleached with the help of bleaching powder.

77. Maximum percentage of available chlorine on the basis of $CaOCl_2.H_2O$ formula is

(a) 35 (b) 40

(c) 45 (d) 49

78. The percentage of available chlorine in commercial samples of bleaching powder is usually between 33-38%. The low value is due to

(a) Incomplete reaction between slaked lime and Cl_2 during its formation

(b) Impurities present in the original slaked lime

(c) Decomposition of bleaching powder when kept in air

(d) All of the above

79. 3.55 g of bleaching powder when treated with acetic acid and excess of KI liberated iodine which required 60 mL of 0.5 N sodium thiosulphate solution. The percentage of available chlorine in the sample is

(a) 30.0 (b) 25.0

(c) 20.0 (d) 35.0

80. Select the correct matching

List-I	List-II
I : XeF_4	P. Pyramidal
II : XeF_6	Q. T-shape
III : XeO_3	R. Distorted octahedral
IV : $XeOF_2$	S. Square planar

	I	II	III	IV
(a)	S	R	P	Q
(b)	P	Q	R	S
(c)	Q	P	R	S
(d)	S	P	R	Q

81. Match Column-I with Column-II.

Column-I	Column-II
(a) Zn + V.dil. HNO_3	(p) NO
(b) Sn + dil. HNO_3	(q) NO_2
(c) Ag + dil. HNO_3	(r) NH_4NO_3
(d) Pb + conc. HNO_3	(s) $Sn(NO_3)_2$
	(t) $Zn(NO_3)_2$

82. Match the process in List-I with compounds in List-II and the catalyst used in List-III.

List-I	List-II	List-III
(a) Haber's process	(p) Manufacture of chlorine	(u) Iron powder
(b) Deacon's process	(q) Manufacture of nitric acid	(v) Pt gauge
(c) Ostwald's process	(r) Manufacture of vanaspati ghee	(w) Copper (II) chloride
(d) Hydrogenation of vegetable oils	(s) Manufacture of ammonia	(x) Finely divided nickel

83. Explain the following :

(i) Acidified $K_2Cr_2O_7$ solution turns green when sodium sulphite is added to it.

(ii) Partial roasting of sulphide ore is done in the metallurgy of copper.

(iii) Zinc becomes dull in moist air.

(iv) In the metallurgy of iron, limestone is added to the ore.

(v) A little acid is always added in the preparation of aqueous ferrous sulphate solution.

(vi) The addition of NaOH solution to a solution of zinc chloride produces a white precipitate which dissolves on further addition of NaOH.

(vii) The addition of NH_4OH to $ZnSO_4$ solution produces white precipitate but no precipitate is formed if it contains NH_4Cl.

(viii) Mercuric chloride and stannous chloride cannot exist as such if present together in an aqueous solution.

(ix) Zinc and not copper is used for the recovery of silver from complex $[Ag(CN)_2]^-$.

(x) Why is chalcocite roasted and not calcined in the extraction of copper?

(xi) Copper implements were known to mankind much earlier than iron implements.

(xii) Copper sulphate dissolves in NH_4OH solution but $FeSO_4$ does not.

(xiii) A ferrous salt turns brown in air.

(xiv) A ferrous salt decolourises acidified $KMnO_4$ solution.

(xv) Cadmium is used to extinguish nuclear fire.

(xvi) Copper hydroxide is soluble in ammonium hydroxide but not in sodium hydroxide.

84. Explain the following :

(i) Magnesium oxide is used for lining of steel making furnace.

(ii) Cast iron is hard but pure iron is soft in nature.

(iii) Fe^{3+} is more stable iron Fe^{2+}.

(iv) Anhydrous $FeCl_3$ cannot be obtained by heating hydrated ferric chloride.

(v) The compounds of iron are coloured and paramagnetic.

(vi) Mercuric chloride is called corrosive sublimate.

(vii) The colour of mercurous chloride changes from white to black when treated with ammonia solution.

(viii) Excess of carbon is added in the zinc metallurgy.

(ix) Zinc readily liberates hydrogen from cold dilute H_2SO_4.

(x) The compounds of Zn, Cd and Hg are usually, white.

(xi) A dark blue precipitate is formed when sodium hydroxide solution is added to copper sulphate solution. The precipitate darkens on heating.

(xii) Cuprous chloride is insoluble in water and dilute HCl but dissolves in concentrated HCl.

(xiii) CuS is not precipitated by passing H_2S through copper sulphate solution containing KCN.

(xiv) Silver nitrate solution is kept in dark coloured bottles.

85. Explain the following :

(i) Copper (I) salts are not known in aqueous solutions.

(ii) Ferric iodide is very unstable but ferric chloride is not.

(iii) Silver bromide is used in photography.

(iv) Anode mud in copper refining contains silver and gold.

(v) Silver fluoride is fairly soluble in water while other silver halides are insoluble.

(vi) Copper is regarded as transition metal though it has completely filled *d*-orbital (d^{10})

(vii) Out of cobalt and zinc salts, which is attracted in a magnetic field?

(viii) The species $[CuCl_4]^{-2}$ exist but $[CuI_4]^{2-}$ does not.

(ix) Copper is largely used in electrical wiring.

(x) A solution of K_2CrO_4 changes colour on being acidified.

86. A metal chloride (x) shows the following reactions.

(a) When H_2S is passed in an acidified solution of (x) a black precipitate is obtained.

(b) The black precipitate is not soluble in ammonium sulphide.

(c) The solution of stannous chloride is added to an aqueous solution of (x), a white precipitate is obtained which turns grey on addition of more stannous chloride.

(d) When aqueous solution of KI is added to an aqueous solution of (x), a red precipitate is obtained which dissolves on addition of excess of KI.

Identify (x) and wirte down the equations for the reactions.

87. How will you obtain the following?

(i) Cuprous chloride from copper sulphate.

(ii) Silver nitrate from silver.

(iii) Silver from silver nitrate.

(iv) Mercury from corrosive sublimate.

(v) Zinc chloride from zinc sulphate.

(vi) Anhydrous ferric chloride from iron.

(vii) Ferric chloride from ferrous sulphate.

88. Explain the following giving proper reason.

(i) Anhydrous ferric chloride cannot be obtained by heating hydrated ferric chloride.

(ii) ZnO is used as a white paint inspite of the fact that it has less covering power than white lead.

(iii) $CuSO_4$ is paramgnetic while $ZnSO_4$ is diamagnetic.

(iv) Hydrated zinc chloride cannot be dehydrated on heating?

(v) Zinc is used in the galvanisation of iron.

(vi) Poling process is used in the purification of copper.

(vii) A solution of ferric chloride gives a brown precipitate on standing.

(viii) Transition metals and their compounds act as catalysts.

(ix) In moist air copper corrodes to produce a green layer on the surface.

89. Explain :

(a) Why is $K_2Cr_2O_7$ generally preferred to $Na_2Cr_2O_7$ in volumetric analysis?

(b) Which divalent metal ion has maximum paramagnetic character among the first transition metals? Why?

(c) $HgCl_2$ and $SnCl_2$ cannot exist together in an aqueous solution. Why?

(d) Why does $AgNO_3$ produce a black stain on the skin?

(e) How the colour of $K_2Cr_2O_7$ solution does depend on pH on the solution?

90. Match the following :

[A]

(i) Corrosive sublimate	(a) Hg_2Cl_2
(ii) Vermilion	(b) Impure zinc
(iii) Calomel	(c) K_2HgI_4 + KOH
(iv) Lithopone	(d) Cobalt zincate
(v) Zinc oxide	(e) $Hg(CNS)_2$
(vi) White vitriol	(f) An ore of mercury
(vii) Spelter	(g) An ore of zinc
(viii) Rinmann's green	(h) $HgCl_2$
(ix) Nessler's reagent	(i) Zinc sulphate
(x) Pharash's serpent	(j) ($BaSO_4$ + ZnS)
(xi) Calamine	(k) Red form of HgS
(xii) Cinnabar	(l) A white pigment

[B]

(i) Lunar caustic	(a) Basic copper acetate
(ii) Blue vitriol	(b) Impure copper
(iii) Malachite	(c) Black oxide of copper
(iv) Argentite	(d) Mixture of copper sulphide and iron sulphide
(v) Blister copper	(e) Silver nitrate
(vi) Brass	(f) An ore of silver
(vii) Light sensitive	(g) An ore of copper
(viii) Matte	(h) An alloy of copper
(ix) Verdigris	(i) Copper sulphate
(x) Tenorite	(j) Silver bromide

91. A metal gives two chlorides 'A' and 'B'. 'A' gives black precipitate with NH_4OH and 'B' gives white. With KI 'B' gives a red precipitate soluble in excess of KI. 'A' and 'B' are respectively.

(a) $HgCl_2$ and Hg_2Cl_2 (b) Hg_2Cl_2, $HgCl_2$

(c) $HgCl_2$ and HgCl (d) None of these

92. Write the formulae of the following complexes :

(a) Diamminechlorido (ethylenediamine) nitrito-N-platinum (IV) chloride.

(b) Calcium hexacyanoferrate (II).

(c) Tetraamminecobalt (III) – μ – amino – μ – nitrotetraammine-cobalt (III) nitrate.

(d) Dichlorido bis (ethylenediamine) cobalt (III) sulphate.

(e) Sodium tetrahydroxoaluminate (III).

(f) Potassium ethylenediamine tetraacetate ferrate (II).

(g) Hexaammine chromium (III) hexacyanocobaltate (III).

93. Arrange the following in order of decreasing number of unpaired electrons :

(i) $[Fe(H_2O)_6]^{2+}$

(ii) $[Fe(CN)_6]^{3-}$

(iii) $[Fe(CN)_6]^{2-}$

(iv) $[Fe(H_2O)_6]^{3+}$

(a) (i), (ii), (iii), (iv)
(b) (iv), (ii), (i), (iii)
(c) (ii), (iii), (i), (iv)
(d) (iv), (i), (ii), (iii)

94. Find out the wrong statement for an octahedral complex.

(a) An ion with d^5 configuration has one unpaired electron both in weak and strong fields.

(b) A central metal ion with d^8 configuration has two unpaired electrons.

(c) An ion with d^6 configuration is diamagnetic in a strong field.

(d) In d^4, d^5, d^6 and d^7 configurations, weak and strong field complexes have different number of unpaired electrons.

95. The correct order of hybridization of the central atom in following species NH_3, $[PtCl_4]^{2-}$, PCl_5, BCl_3 is (Atomic Number of Pt = 78)

(a) dsp^2, dsp^3, sp^2 and sp^3
(b) sp^3, dsp^2, sp^3d, sp^2
(c) dsp^2, sp^2, sp^3d, sp^3
(d) dsp^2, sp^3, sp^2, dsp^3

96. Geometry, hybridisation and magnetic moment of the ions $[Ni(CN_4)^{2-}$, $[MnBr_4]^{2-}$ and $[FeF_6]^{3-}$ respectively are

(a) Tetrahedral, square planar, octahedral : sp^3, dsp^2, sp^3d^2 : 5.9, 0, 4.9

(b) Tetrahedral, square planar, octahedral : dsp^2, sp^3, sp^3d^2 : 0, 5.9, 4.9

(c) Square planar, tetrahedral, octahedral : dsp^2, sp^3, d^2sp^3 : 5.9, 4.9, 0

(d) Square planar, tetrahedral, octahedral : dsp^2, sp^3, sp^3d^2 : 0, 5.9, 4.9

97. The reaction $[Fe(CN)_6]^{3-} \longrightarrow [FeF_6]^{3-}$ takes place with.

(a) Decrease in magnetic moment

(b) Increase in magnetic moment

(c) Decrease in coordination number

(d) Increase in coordination number

98. Which of the following can exhibit geometrical isomerism?

(a) $[MnBr_4]^{2-}$
(b) $[Pt(NH_3)_3Cl]^+$
(c) $[PtCl_2.P(C_2H_5)_3]_2$
(d) $[Fe(H_2O)_5NOS]^{2+}$

99. A solution contains 2.675 g of $CoCl_3.6NH_3$ (molar mass = 267.5 g mol^{-1}) is passed through a cation exchanger. The chloride ions obtained in solution were treated with excess of $AgNO_3$ to give 4.305 g of AgCl. The formula of the complex is

(a) $[CoCl(NH_3)_5]Cl_2$

(b) $[Co(NH_3)_6]Cl_3$

(c) $[Co(NH_3)_4Cl_2]Cl$

(d) $[Co(NH_3)_3Cl_3]$

100. Match the complexes in List-I with their informations in List-II.

List-I	List-II
(a) $[Cu(NH_3)_4]SO_4$	(p) dsp^2
(b) $[Pt(NH_3)_2Cl_2]$	(q) Octahedral
(c) $K_4[Fe(CN)_6]$	(r) sp^3d^2
(d) $[Fe(H_2O)_6]Cl_3$	(s) Square planar

101. Match the complexes in the List-I with their coordination number in the List-II.

List-I	List-II
(a) $[Co(en)_3]^{3+}$	(p) 6
(b) $[Ca(EDTA)]^{2-}$	(q) 4
(c) $[Ni(CO)_4]$	(r) 2
(d) $[Ag(NH_3)_2]Cl$	(s) 5
(e) $[Fe(CO)_5]$	

102. Match the comlexes in List-I with their stereoproperties in List-II.

List-I	List-II
(a) $[CoCl_3(NH_3)_3]$	(p) Show facial form
(b) $[Cr(OX)_3]^{3-}$	(q) Cis form is optically active
(c) $[CrCl_2(OX)_2]$	(r) Trans form is optically inactive
(d) $[RhCl_3(Py)_3]$	(s) Show meridian form

103. Write down the IUPAC names of the following complexes :

(i) $(NH_4)_2[FeF_5H_2O]$

(ii) $[Co(NH_3)_5H_2O]Cl_3$

(iii) $[Co(H_2O)(NH_3)_5][Co(C_2O_4)_3]$

(iv) $[Co(H_2O)(NH_3)_5][Cr(SCN)_4(NH_3)_2]_3$

(v) $[Co(H_2O)(NH_3)_5][Co(NO_2)_4(NH_3)_2]_3$

(vi) $K_4[Mo(CN)_8]$

(vii) $[Cr(NH_3)_5(NCS)]\ [ZnCl_4]$

(viii) $[Bi\{SC(NH_2)_2\}_2]Cl_3$

(ix) $[(CH_3)_4N]_2[Co(NCS)_4]$

104. $[V(CO)_6]$ has a magnetic moment corresponding to one unpaired electron. Draw a valence bond structure for this molecule. What is geometrical structure of complex?

105. Give the example of a tetrahedral complex showing optical activity.

106. What do you understand by macrocyclic effect?

107. The value of Δ for $[RhCl_6]^{3-}$ is 243 kJ mol^{-1}. What wavelength of light will promote an electron from the t_{2g} set to the e_g set? What is the colour of the complex?

108. Explain why $[Cu(en)_2]^{2+}$ is less stable than $[Fe(EDTA)]^-$?

109. What is the crystal field splitting energy or crystal field stabilisation energy (CFSE) for d^7 configuration for strong ligand field?

(a) $-1.6\ \Delta_0$ (b) $-2.0\ \Delta_0$

(c) $-2.4\ \Delta_0$ (d) $-1.8\ \Delta_0$

110. In the following complexes of manganese what is the distribution of electrons on *d*-orbitals of manganese?

(i) $[Mn(H_2O)_6]^{2+}$

(ii) $[Mn(CN)_6]^{4-}$

(a) $t_{2g}^3\ e_g^2$ in both

(b) $t_{2g}^5\ e_g^0$ in both

(c) $t_{2g}^3\ e_g^2$ in (i) and $t_{2g}^5\ e_g^0$ in (ii)

(d) $t_{2g}^5\ e_g^0$ in (i) and $t_{2g}^3\ e_g^2$ in (ii)

111. Two solid laboratory reagent 'A' and 'B' give the following reactions :

Compound A :

(i) On strongly heating it gives two oxides of sulphur.

(ii) On adding aqueous NaOH solution to its aqueous solution a dirty green precipitate is obtained, which starts turning brown on exposure to air.

Compound B :

(iii) It imparts green colour to the flame.

(iv) Its solution does not give a precipitate on passing H_2S.

(v) When it is heated with solid $K_2Cr_2O_7$ and concentrated H_2SO_4, a red gas is evolved. The gas when passed in an aqueous solution of NaOH, turns it yellow.

Identify 'A' and 'B'.

112. A certain compound (X) is used in laboratory for analysis, its aqueous solution gives the following reactions.

(i) On adding copper sulphate, a brown precipitate is obtained which turns white on addition of excess of $Na_2S_2O_3$ solution.

(ii) On addition of Ag^+ ion solution a yellow curdy precipitate is obtained which is insoluble in ammonium hydroxide.

Identify (X) and give equations for the reactions at step (i) and (ii).

113. An unknown solid mixture contains one or two of the followiing : $CaCO_3$, $BaCl_2$, $AgNO_3$, Na_2SO_4, $ZnSO_4$ and NaOH. The mixture is completely soluble in water and the solution gives pink colour with phenolphthalein. When dilute hydrochloric acid is gradually added to above solution, a precipitate is formed which dissolves with further addition of the acid. What is/are present in the solid?

Give equations to explain the apperance of the precipitate and its dissolution.

114. A black coloured compound (A) on reaction with dil. H_2SO_4 gives a gas (B) which on passing in a solution of an acid (C) gives a white turbidity (D). Gas (B) when passed in acidified solution of (E) gives a precipitate (F) soluble in dilute HNO_3. After boiling this solution when excess of NH_4OH is added, a blue coloured compound (G) is formed. To this solution on addition of acetic acid and aqueous potassium ferrocyanide a chocolate coloured precipitate (H) is obtained. On addition of an aqueous solution of $BaCl_2$ to an aqueous solution of (E), a white precipitate insoluble in HNO_3 is obtained. Identify (A) to (H).

115. The gas liberated on heating a mixture of two salts with NaOH, gives a reddish brown precipitate with an alkaline solution of K_2HgI_4. The aqueous solution of mixture on treatment with $BaCl_2$ gives a white precipitate which is sparingly soluble in conc. HCl. On heating the mixture with $K_2Cr_2O_7$ and conc. H_2SO_4, red vapours (A) are produced. The aqueous solution of mixture gives a deep blue colouration (B) with potassium ferricyanide. Identify the radicals in given mixture and write the balanced equation for formation of (A) and (B).

116. (i) A powdered substance (A) on treatment with fusion mixture gives a green coloured compound (B).

(ii) The solution of (B) in boiling water on acidification with dilute H_2SO_4 gives a pink coloured compound (C).

(iii) The aqueous solution of (A) on treatment with excess of NaOH and bromine water gives a compound (D).

(iv) A solution of (D) in conc. HNO_3 on treatment with lead peroxide at boiling temperature produced a compound (E) which was of the same colour as that of (C).

(v) A solution of (A) in dilute HCl on treatment with a solution of barium chloride gave a white precipitate of compound (F) which was insoluble in conc. HNO_3 and conc. HCl.

Identify (A) to (F) and give balanced chemical equations for the reactions at steps (i) to (v).

117. When a green solid (A) is fused with sodium carbonate in open air, a coloured salt (B) is formed. A solution of pure (B) gives a coloured precipitate with solutions of lead and barium salts. On extracting (B) with water and adding sulphuric acid, a further colour change takes place and crystals of (C) are obtained on evaporation. When warmed with alcohol, an acidified solution of (C) emits a characteristic smell. On warming (C) with ammonium sulphate a colourless gas is produced and (A) is reformed. Identify (A), (B) and (C) and explain the reactions involved.

118. A white compound (A) on red heating decomposes to give a residue (B) and two gases (C) and (D). The gas (C) burns with a blue flame. The gas (D) turns lime water milky. The residue (B) on dissolving in water gives heat and solution is alkaline to litmus. The compound (A) imparts brick-red colour to flame. Identify (A) to (D).

119. Explain with proper reasoning :

(i) The aqueous solution of ferric chloride cannot be stored.

It is always acidified with hydrochloric acid.

(ii) The aqueous solution of $FeCl_3$ possesses yellow colour, but the colour becomes green on passing H_2S gas.

(iii) The aqueous solution of $K_2Cr_2O_7$ is orange. On adding an alkali, it turns yellow.

(iv) In the test of oxalate, the evolved gas burns with blue flame only initially.

(v) Why yellow ammonium sulphide is used in group (II) sulphides separation?

(vi) Why zinc sulphide is not precipitated when H_2S is passed through $ZnCl_2$ solution?

(vii) $CaSO_4$ is insoluble but is not precipitated when excess of $(NH_4)_2SO_4$ is added to $CaCl_2$.

(viii) Why ($NH_4Cl + NH_4OH$) and not [$(NH_4)_2SO_4 + NH_4OH$] is used in group (III) analysis?

(ix) Why is it necessary to add few drops of conc. HNO_3 to the filtrate of group (II) before the use of $NH_4Cl + NH_4OH$?

(x) Why NaOH cannot be used to separate $Al(OH)_3$ and $Zn(OH)_2$?

120. Which of the following precipitates is obtained on adding HCl to a solution of Sb_2S_3 in yellow ammonium sulphide?

(a) Sb_2S_3 (b) Sb_2S_5

(c) SbS_2 (d) SbS

121. Cobalt salt + KNO_2 + $CH_3COOH \rightarrow$ Yellow ppt. The yellow precipitate is

(a) Potassium cobaltonitrite

(b) Potassium cobaltinitrite

(c) Cobalt nitrite

(d) Cobalt nitrate

122. Which of the following statements is correct?

(a) Manganese salts give a violet borax bead test in reducing flame

(b) From a mixed precipitate of AgCl and AgI, ammonia solution dissolves only AgCl

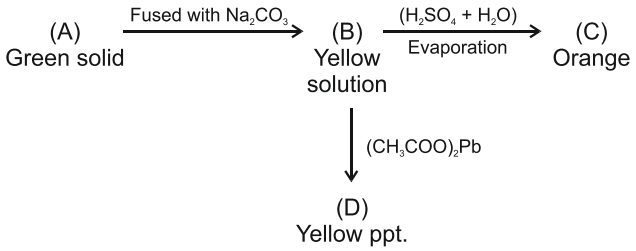
(c) Ferric ions give a deep green precipitate, on adding potassium ferrocyanide solution

(d) On boiling the solution having K^+, Ca^{2+} and HCO_3^- ion we get a precipitate of $K_2Ca(CO_3)_2$

123. $Mn^{2+} + PbO_2 \xrightarrow{\text{conc. } HNO_3}$ light purple coloured solution contains.

(a) $HMnO_4$ (b) $Pb(NO_3)_2$

(c) $Mn(NO_3)_2$ (d) H_2MnO_4

124. 

Here, A, B, C and D are respectively.

	A	B	C	D
(a)	$FeSO_4$	$FeCO_3$	$Fe(OH)_3$	$PbCO_3$
(b)	Cr_2O_3	Na_2CrO_4	$Na_2Cr_2O_7$	$PbCrO_4$
(c)	$FeCl_2$	$FeSO_4$	$PbSO_4$	$Fe(OH)_3$
(d)	$FeSO_4$	$FeCl_3$	$Fe(OH)_3$	$PbCl_2$

125. Match the observations of List-I with inferences in List-II.

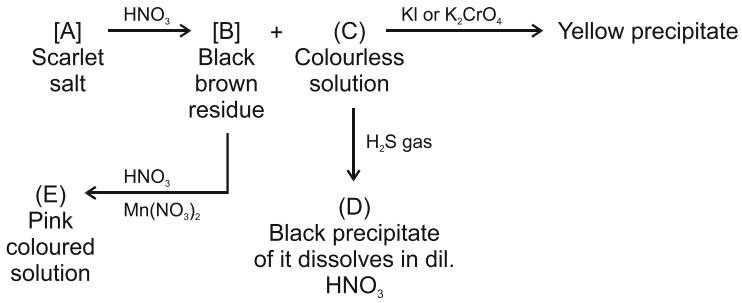
List-I	**List-II**
(a) Canary yellow precipitate with ammonium molybdate.	(p) PO_4^{3-}
(b) Ring test (Brown)	(q) As^{3+}
(c) Acid radical decomposed by dil H_2SO_4	(r) NO_3^-
(d) Acid radical decomposed by conc. H_2SO_4	(s) NO_2^-

126. Match the reagents (List-I) with their uses (List-II).

List-I (Reagents)	**List-II (Uses)**
(a) Nessler's reagent	(p) Chromyl chloride test
(b) $K_2Cr_2O_7$	(q) Prussian blue test of Fe^{3+}
(c) NaOH	(r) Chocolate brown test of Cu^{2+}
(d) $K_4[Fe(CN)_6]$	(s) NH_4^+ ion

127. A gas 'X' occupies 22.4 litre at S.T.P., its weight is 64 g. The gas turns acidified $K_2Cr_2O_7$ green and decolourise acidified $KMnO_4$ solution. White turbidity appears. When H_2S gas is passed through aqueous solution of gas. What is the gas 'X'? Explain the reactions involved.

128.



What are the compound A to E in above road map prolem? Give the reactions involved.

129. Match Column-I with Column-II.

Column-I (Reagents)	**Column-II (Uses)**
(a) $K_2Cr_2O_7$	(p) Chromyl chloride test
(b) $K_4Fe(CN)_6$	(q) Prussian blue test of Fe^{3+}
(c) NaOH	(r) Test of sulphite
(d) $Pb(CH_3COO)_2$	(s) Chocolate ppt. test of Cu^{2+}
	(t) NH_4^+ ion

❑ ❑ ❑

Book
Crunch

Mathematics

MATHEMATICS

1. Find domain (write conditions only) $f(x) = \sqrt{\ln(\sin x)} + \sin^{-1}\left(\sqrt{\ln x}\right)$

2. Define the following analytical function using basic functions:

$$f(x) = \frac{\sqrt{e^x}}{1+x^2},\ x \le 0$$

$$= \frac{\frac{1}{x} + \ln x}{e^x + \ln^2 x},\ x > 0$$

3. Simplify expression $\log_2\left(\frac{\sqrt[3]{4}\sqrt{2\sqrt[5]{16}}}{\sqrt{2}}\right) - \log_{\frac{1}{2}}\sqrt[3]{\frac{4}{\sqrt{2}}} + \log_{\frac{1}{\sqrt{3}}}(9\sqrt[3]{3})$

4. Simplify expression $\sqrt{\log_{\sqrt{2}}\sqrt{\sqrt{2}\sqrt{\sqrt{2}}} + \log_{\sqrt{\sqrt{2}}}\sqrt[4]{\sqrt{2\sqrt{2}}}}$

5. Simplify expression $\sqrt{\left(\frac{1}{\sqrt{(27)}}\right)^{2-\frac{(\log_5 13)}{2\log_5 9}}}$

6. Simplify expression $9^{2\log_3 2 + 4\log_{81} 2} \cdot \sqrt{3^{2+\frac{1}{2}\log_3 16}}$

7. Simplify expression $\left(\sqrt{2}\right)^{3\log_{\sqrt{2}} 5 - 2\log_{\sqrt{2}} 25 - \log_{\sqrt{2}} 10 + 2\log_{\sqrt{2}}\sqrt{5}}$

8. Simplify expression $(\log 2 + \log 5 + \log 300 - \log 3) \cdot 3^{\frac{1}{5\log_5 3}}$

9. Simplify expression $\frac{\log_2\sqrt[3]{\frac{2}{3}}}{\log_2^2\sqrt{7}} - \frac{2\log_{\sqrt{7}}\sqrt[3]{\frac{2}{3}}}{\log_{\sqrt{3}}\sqrt{7}} - \log_{\sqrt[6]{7}}\sqrt[3]{\frac{2}{3}} \cdot \log_{\sqrt{7}}\sqrt[3]{\frac{2}{3}}$

10. Given that $\log_6 15 = a$ and $\log_{12} 18 = b$, find $\log_{25} 24$ in terms of a and b.

11. Given $f(x) = x^2, \quad x \ge 0$
$= x, \quad x < 0$

and $g(x) = \frac{1}{x}, \quad x \geq 1$

$= 1, \quad x < 1$

Determine the following functions:

(i) $h(x) = x \times g(x)$

(ii) $\phi(x) = f(x) + g(x)$

(iii) $\psi(x) = f(x) \times g(x)$

12. Given

$f(x) = 1 + x, \quad -2 \leq x < 0$

$= 0, \quad x = 0$

$= x - 1, \quad 0 < x \leq 2,$

$\phi(x) = f(f(x))$ and $F(x) = 2\phi\left(\frac{x}{2}\right)$. Determine the function $g(x) = \phi(F(x))$ and plot its graph and write its domain and range.

13. $\lim\limits_{x\to 0} \dfrac{e^x - \frac{x^3}{6} - \frac{x^2}{2} - x - 1}{\cos x + \frac{x^2}{2} - 1}$

14. $\lim\limits_{x\to 0} \dfrac{\ln(1+x)^4 - 4x + 2x^2 - \frac{4}{3}x^3 + x^4}{6\sin x - 6x + x^3}$

15. $\lim\limits_{x\to 0} \dfrac{\sqrt[3]{1+x} - \sqrt[4]{1+2x}}{x}$

16. $\lim\limits_{x\to 2} \dfrac{\sqrt{x+7} - 3\sqrt{2x-3}}{\sqrt[3]{x+6} - 2\sqrt[3]{3x-5}}$

17. $\lim\limits_{x\to +\infty} \dfrac{\sqrt[5]{x^7+3} + \sqrt[4]{2x^3-1}}{\sqrt[6]{x^8+x^7+1} - x}$

18. $\lim\limits_{x\to\infty} \sqrt[3]{(x+1)^2} - \sqrt[3]{(x-1)^2}$

19. $\lim\limits_{x\to 0} \dfrac{\sqrt{1+x\sin x} - \sqrt{\cos 2x}}{\tan^2 \frac{x}{2}}$

20. $\lim\limits_{x\to \pm\infty} x\left(\sqrt{x^2 + \sqrt{x^4+1}} - x\sqrt{2}\right)$

21. $\lim\limits_{x\to\infty} \cos\sqrt{x+1} - \cos\sqrt{x}$

22. $\lim_{x\to 0}\left\{\ln(1+\sin^2 x)\right\}\left\{\cot\ln^2(1+x)\right\}$

23. $\lim_{x\to 0}\dfrac{\sqrt[3]{1+\tan^{-1}3x}-\sqrt[3]{1-\sin^{-1}3x}}{\sqrt{1-\sin^{-1}2x}-\sqrt{1+\tan^{-1}2x}}$

24. $\lim_{x\to\infty}\sqrt[3]{(a+x)(b+x)(c+x)}-x$

25. $\lim_{x\to 0}\dfrac{\sqrt{a^2+ax+x^2}-\sqrt{a^2+ax}}{\ln\left(\cos\dfrac{x}{a}\right)}$

26. $\lim_{x\to\infty}(x^n+a_1x^{n-1}+\ldots+a_n)^{\frac{1}{n}}-(x^n+b_1x^{n-1}+\ldots+b_n)^{\frac{1}{n}}$

27. $\lim_{n\to\infty}\dfrac{n^K\cos n!}{n+1}(0<K<1)$

28. $\lim_{x\to 0}\left[\dfrac{\cos x+\cos 2x+\ldots+\cos nx}{n}\right]^{\frac{1}{x^2}}$

29. $\lim_{x\to\infty}\sqrt{a^2x^2+ax+1}-\sqrt{a^2x^2+1}$

30. $\lim_{x\to\infty}\left(1+\dfrac{1}{x^a}\right)^x$

31. Find $\lim_{n\to\infty}n^2\sqrt{\left(1-\cos\dfrac{1}{n}\right)\sqrt{\left(1-\cos\dfrac{1}{n}\right)\sqrt{\left(1-\cos\dfrac{1}{n}\right).....\infty}}}$

32. Given $f(x)=\lim_{n\to\infty}\tan^{-1}nx$; $g(x)=\lim_{n\to\infty}\sin^{2n}x$ and $h(x)=\dfrac{1}{2}[\cos(\pi g(x))+\cos(2f(x))]$. Find the function $h(x)$.

33. Given the function

$f(x)=\dfrac{a^{\sin x}-a^{\tan x}}{\tan x-\sin x}$, $x>0$

$=\dfrac{\ln(1+x+x^2)+\ln(1-x+x^2)}{\sec x-\cos x}$, $x<0$

If $f(x)$ is continuous at $x=0$, find the value of a. Now $g(x)=\ln\left(2-\dfrac{x}{a}\right)\cdot\cot(x-a)$, $x\neq a$. If $g(x)$ is continuous at $x=a$, show that $g\left(\dfrac{1}{e}\right)=-e$.

34. Check the continuity of the function $f(x)=\lim_{n\to\infty}\dfrac{\log(2+x)-x^{2n}\sin x}{1+x^{2n}}$.

35. Show that the equation $\frac{a_1}{x-\lambda_1}+\frac{a_2}{x-\lambda_2}+\frac{a_3}{x-\lambda_3}=0$, where $a_1 > 0$, $a_2 > 0$, $a_3 > 0$ and $\lambda_1 < \lambda_2 < \lambda_3$, has two real roots lying in the intervals (λ_1, λ_2) and (λ_2, λ_3)

36. Prove that if the function $f(x)$ is continuous in the interval (a, b) and $x_1, x_2, ..., x_n$ are any values in this open interval, then we can always find a real number c in this open interval such that

$$f(c)=\frac{f(x_1)+f(x_2)+.....+f(x_n)}{n}$$

37. If $f(x)$ and $h(x)$ are continuous function and $g(x)=\lim_{m\to\infty}\frac{x^m f(x)+h(x)+1}{2x^m+3x+3}$, $x\neq 1$ and $g(1)=\lim_{x\to 1}(\ln ex)^{\frac{2}{\ln x}}$ and $g(x)$ is continuous at $x = 1$, then find the value of $2g(1) + 2f(1) - h(1)$.

38. If

$f(x) = -1 + | x - 1 |, \quad -1 \le x \le 3$

$g(x) = 2 - | x + 1 |, \quad -2 \le x \le 2$

then calculate (fog)x and (gof)x. Draw their graphs. Discuss the continuity of (fog)x at $x = 1$ and differentiability of (gof)x at $x = 1$.

39. Given

$$f(x) = b\sin^{-1}\left(\frac{x+c}{2}\right); \quad -\frac{1}{2}<x<0$$

$$= \frac{1}{2}, \quad x = 0$$

$$= \frac{e^{\frac{ax}{2}}-1}{x}, \quad 0<x<\frac{1}{2}$$

If $f(x)$ is differentiable at $x = 0$ and $|c| < \frac{1}{2}$, then find the value of a and prove that $64b^2 = (4 - c^2)$.

40. Differentiate $y=\frac{2}{3}\tan^{-1}x+\frac{1}{3}\tan^{-1}\frac{x}{1-x^2}$

41. Differentiate $y=\ln\left\{e^x\left(\frac{x-2}{x+2}\right)^{\frac{3}{4}}\right\}$

42. Differentiate $y=\sqrt{(a-x)(x-b)}-(a-b)\tan^{-1}\sqrt{\frac{a-x}{x-b}}$

43. Differentiate $y=\sqrt{x^2+1}-\ln\left(\frac{1}{x}+\sqrt{1+\frac{1}{x^2}}\right)$

44. Differentiate $y=\ln\sqrt[4]{\frac{x^2+x+1}{x^2-x+1}}+\frac{1}{2\sqrt{3}}\left(\tan^{-1}\frac{2x+1}{\sqrt{3}}+\tan^{-1}\frac{2x-1}{\sqrt{3}}\right)$

45 Find the derivative w.r.t. x of the function $(\log_{\cos x}\sin x)(\log_{\sin x}\cos x)^{-1}+\sin^{-1}\frac{2x}{1+x^2}$ at $x=\frac{\pi}{4}$,

46. Let

$f(x) = \sqrt{x}\left(1 + x\sin\frac{1}{x}\right), \quad x > 0$

$= -\sqrt{-x}\left(1 + x\sin\frac{1}{x}\right), \quad x < 0$

$= 0, \quad x = 0$

Show that $f'(x)$ exists everywhere and is finite except at $x = 0$.

47. Show that the function

$f(x) = \sin x, \quad x$ is a rational number.

$= x, \quad x$ is a irrational number

has positive derivative at $x = 0$ but $f(x)$ is not increasing at $x = 0$.

48. Show that the function

$f(x) = \frac{x}{2} + x^2 \sin\frac{1}{x}, \quad x \neq 0$

$= 0, \quad x = 0$

is continuous and differentiable in any neighbourhood of $x = 0$ and $f'(0)$ is positive but $f(x)$ is not increasing at $x = 0$.

49. Given $g(x) = f(x^2 - x - 10) + f(14 + x - x^2)$ and $f''(x) > 0$ for all real x, except at finite number of real values of x for which $f''(x) = 0$. Discuss the monotonicity of the function $g(x)$.

50. Given

$f(x) = |x - 2| + \ln(a^2 - 1), \quad x < 2$

$= 3x + 5, \quad x \geq 2$

Find the values of a for which $f(x)$ has local minima at $x = 2$.

51. The function $f(x) = x^4 - 62x^2 + ax + 9$ attains its maximum value on the interval [0, 2] at $x = 1$. Find the value of a.

52. Find the intervals of concavity of the following functions

(i) $f(x) = x^6 - 10x^4$.

(ii) $f(x) = (x + 1)^4 + e^x$

(iii) $f(x) = x + x^{\frac{5}{3}} + 1$

(iv) $f(x) = x^3, \quad x \leq 1$

$= x^2, \quad x > 1$

53. Given

$f(x) = e^x, \quad x \geq 0$

$= ax^3 + bx^2 + cx + d, \quad x < 0$

Find the constants a, b, c, d if $f''(0)$ exists and $f(x)$ has a point of inflection at $x = -1$.

54. If $f''(x)$ exists for all points in $[a, b]$ and $\frac{f(c) - f(a)}{c - a} = \frac{f(b) - f(c)}{b - c}$, where $a < c < b$, then there is a number α such that $a < \alpha < b$ and $f''(\alpha) = 0$.

55. If $f(x)$ is differentiable and $\lim_{x \to \infty} f(x)$ is finite and $\lim_{x \to \infty} f'(x)$ is finite, then show that $\lim_{x \to \infty} f'(x) = 0$.

56. Suppose that $f(x)$ and $g(x)$ are non-constant differentiable real valued functions on R. If for every $x, y \in R$, $f(x + y) = f(x)f(y) - g(x)g(y)$ and $g(x + y) = g(x)f(y) + f(x)g(y)$ and $f'(0) = 0$, then prove that maximum and minimum values of the function $f^2(x) + g^2(x)$ are same for all real x.

57. For what integral value of n, the function $f(x) = \cos nx \sin\frac{5x}{n}$ is periodic with period 3π?

58. Plot the graphs of the functions $\frac{f(x)+|f(x)|}{2}$, $\frac{f(x)-|f(x)|}{2}$ and $\frac{|f(x)|}{f(x)}$ from the graph of $f(x)$ for the follolwing functions:

(i) $f(x) = \ln x$

(ii) $f(x) = x^3$

(iii) $f(x) = \sin x$

(iv) $f(x) = \tan x$

(v) $f(x) = \sin^{-1} x$

(vi) $f(x) = \tan^{-1} x$

59. Plot the graphs of the functions

(i) $f(x) = \max\{\sin t : 0 \le t \le x\}, \quad x \ge 0$
$= \max\{\sin t : x \le t \le 0\}, \quad x < 0$

(ii) $f(x) = \min\{\cos t : 0 \le t \le x\}, \quad x \ge 0$
$= \min\{\cos t : x \le t \le 0\}, \quad x < 0$

(iii) $f(x) = \min\{\sin t : 0 \le t \le x\}, \quad x \ge 0$
$= \min\{\sin t : x \le t \le 0\}, \quad x < 0$

(iv) $f(x) = \max\{\cos t : 0 \le t \le x\}, \quad x \ge 0$
$= \max\{\cos t : x \le t \le 0\}, \quad x < 0$

(v) $f(x) = \max\{\tan t : 0 \le t \le x\}$

(vi) $f(x) = \min\{\tan t : 0 \le t \le x\}$

60. Plot graph of the function

$f(x) = (|x| - 1)^2 e^{\frac{1}{|x|-1}}, \quad x \ne \pm 1$

$= 0, \quad x = \pm 1$

Write

(i) All the points of discontinuity of $f(x)$

(ii) All the points where $f(x)$ is not differentiable

(iii) All the stationary points of $f(x)$

(iv) Intervals of monotonicity of $f(x)$

(v) All the critical points of $f(x)$

(vi) All the points of maxima of $f(x)$

(vii) All the points of minima of $f(x)$

(viii) Intervals of concavity of $f(x)$

(ix) All the points of inflection of $f(x)$

(x) Range of $f(x)$

(xi) Greatest and least value of $f(x)$

61. $\lim_{x\to 0}\dfrac{\tan\left(\left[-2\pi^2\right]x^2\right)-\left(\tan\left[-2\pi^2\right]\right)x^2}{\sin^2 x}$ (where [ ] is greatest integer function)

62. Show that $[x]+\left[x+\dfrac{1}{2}\right]=[2x]\ \forall x$ (where [ ] is greatest integer function)

63. Show that $\left[x+\dfrac{1}{2}\right]-\left[x-\dfrac{1}{2}\right]=1\ \forall x$ (where [ ] is greatest integer function)

64. If

$f(x) = 0, \quad x \in I$

$= x^2, \quad x \notin I$

Discuss the continuity and differentiability of $[f(x)]$ and $f([x])$ (where [ ] is greatest integer function).

65. For what values of a, $[x + a] - [x - a]$ = constant $\forall x$ (where [ ] is greatest integer function).

66. Find the minimum value of $\dfrac{\left(x+\dfrac{1}{x}\right)^6-\left(x^6+\dfrac{1}{x^6}\right)-2}{\left(x+\dfrac{1}{x}\right)^3+x^3+\dfrac{1}{x^3}}$ for $x > 0$.

67. If $f(x) = (ax^2 + b)^3$, then find the function g such that $f(g(x)) = g(f(x))$.

68. If $b \geq 1$ and $f(x)=\dfrac{1}{|x|}$, then show that the conditions of L. M. V. theorem are not satisfied in the interval $[-1, b]$ but the conclusion of the theorem is true iff $b>1+\sqrt{2}$.

69. Find the nature of roots of the polynomial $P(x) = 3x^4 + 12x^2 + 5x - 4$.

70. Find the nature of roots of the polynomial $P(x) = x^9 + 2x^5 + 3x^3 + x$.

71. Find the least possible number of imaginary roots of the polynomial $P(x) = x^9 - x^5 + x^4 + x^2 + 1$.

72. Simplify $(x - 1)^3 + (2x + 3)^3 = 27x^3 + 8$.

73. Simplify $2x^4 - x^3 - 9x^2 + 13x - 5 = 0$.

74. Simplify $(x^2 + x + 1)(2x^2 + 2x + 3) = 3(1 - x - x^2)$.

75. Simplify $\sqrt{x^2+x-5}+\sqrt{x^2+8x-4}=5$.

76. Simplify $\sqrt[3]{x+5}+\sqrt[3]{x+6}=\sqrt[3]{2x+11}$.

77. Simplify $\sqrt[3]{1+\sqrt{x}}+\sqrt[3]{1-\sqrt{x}}=2$.

78. Simplify $\sqrt{3}\cdot\left(3^{\frac{x}{1+\sqrt{x}}}\right)\cdot\left(\dfrac{1}{3}\right)^{\frac{2+\sqrt{x}+x}{2(1+\sqrt{x})}}=81$.

79. Simplify $6\cdot\sqrt[x]{9}-13\cdot\sqrt[x]{6}+6\cdot\sqrt[x]{4}=0$.

80. Simplify $27^x+12^x=2.8^x$.

81. Simplify $\left(\sqrt{5+2\sqrt{6}}\right)^x + \left(\sqrt{5-2\sqrt{6}}\right)^x = 10$.

82. Simplify $\log_{3x+7}(9+12x+4x^2) = 4 - \log_{2x+3}(6x^2+23x+21)$.

83. Simplify $\log_x(x^2+1) = \sqrt{\log_{\sqrt{x}}(x^2(1+x^2))+4}$.

84. Simplify $\log(6\cdot 5^x + 25\cdot 20^x) = x + \log 5$.

85. Simplify $|x-1|^{\log^2 x - \log x^2} = |x-1|^3$.

86. Simplify $4^{x^2} - 3\cdot 2^{x^2} + 1 \geq 0$.

87. Simplify $\sqrt{13^x - 5} \leq \sqrt{2(13^x+12)} - \sqrt{13^x+5}$.

88. Simplify $\sqrt{2(5^x+24)} - \sqrt{5^x-7} \geq \sqrt{5^x+7}$.

89. Simplify $(x^2-x-1)^{x^2-1} < 1$.

90. Solve the equation $(a-1)x^2 + 2(2a+1)x + (4a+3) = 0$

91. Solve the equation $\frac{x^2+1}{a^2x-2a} - \frac{1}{2-ax} = \frac{x}{a}$.

92. For $a \geq 0$, $b \geq 0$ and $x > y > 0$, prove that $(a^x + b^x)^{\frac{1}{x}} \leq (a^y + b^y)^{\frac{1}{y}}$.

93. Simplify $x^4 + y^4 = 706$, $x + y = 8$.

94. Simplify $\frac{x^2}{y} + \frac{y^2}{x} = 18$, $x + y = 12$.

95. Simplify $\left(3 - \frac{6y}{x+y}\right)^2 + \left(3 + \frac{6y}{x-y}\right)^2 = 82$, $3x + 7y = 26$.

96. Simplify $xy + 3y^2 - x + 4y - 7 = 0$, $2xy + y^2 - 2x - 2y + 1 = 0$.

97. Simplify $(3x)^{\log 3} = (4y)^{\log 4}$, $4^{\log x} = 3^{\log y}$.

98. Find the positive solutions of the system of equations $x^{x+y} = y^n$, $y^{x+y} = x^{2n}y^n$, where $n > 0$.

99. $xy = x + y$, $yz = 2(y + z)$, $zx = 3(z + x)$.

100. $(x-2)^2 + (y-3)^2 + (z-1)^2 = 24$, $xy + yz + zx = 63$, $2x + 3y + z = 30$.

101. Solve the equation $\left[\frac{3x-1}{4}\right] + \left[\frac{3x+1}{4}\right] + \left[\frac{3x-1}{2}\right] = \frac{6x+3}{5}$ ([ ] denotes greatest integer function).

102. For every positive integer n, prove that $\sqrt{4n+1} < \sqrt{n} + \sqrt{n+1} < \sqrt{4n+2}$. Hence, prove that $\left[\sqrt{n} + \sqrt{n+1}\right] = \left[\sqrt{4n+1}\right]$, where [ ] denotes greatest integer function.

103. Given $(2011a + 2012b)(2012c - 2011b) = (2011b + 2012c)(2012b - 2011a)$ and $0 < a < b < c$ and $\sqrt{c}\cdot a^x + \sqrt{b}\cdot c^x < (\sqrt{c}+\sqrt{b})\cdot b^x$, then find the interval of solution of x.

104. For each integer $n \geq 1$, define $a_n = \left[\frac{n}{[\sqrt{n}]}\right]$, where [ ] denotes the greatest integer function. Find the number of all n in the set {1, 2, 3, ..., 2010} for which $a_n > a_{n+1}$.

105. If α and β are the roots of $x^2 + px + 1 = 0$ and γ, δ the roots of $x^2 + qx + 1 = 0$, show that $q^2 - p^2 = (\alpha - \gamma)(\beta - \gamma)(\alpha + \delta)(\beta + \delta)$.

106. If the roots of the equation $ax^2 + bx + c = 0$ are of the form $\frac{k+1}{k}$ and $\frac{k+2}{k+1}$, prove that $(a + b + c)^2 = b^2 - 4ac$.

107. Find the value of m for which the equation $\frac{a}{x+a+m} + \frac{b}{x+b+m} = 1$ has roots equal in magnitude but opposite in sign.

108. Find all numbers a for which the least value of the quadratic function $4x^2 - 4ax + a^2 - 2a + 2$ in the interval [0, 2] is equal to 3.

109. For what values of a does the inequality $4^x - a \cdot 2^x - a + 3 \leq 0$ has at least one solution?

110. Sum of certain consecutive odd positive integers is $57^2 - 13^2$. Find them.

111. Along a road lie an odd number of stones placed at intervals of 10 meters. These stones have to be assembled around the middle stone. A person can carry only one stone at a time. A man carried the job with one of the end stones by carrying them in succession. In carrying all the stones he covered a distance of 3 km. Find the number of stones.

112. Balls are arranged in rows to form an equilateral triangle. The first row consists of one ball, the second row of two balls and so on. If 669 more balls are added then all the ball can be arranged in the shape of a square and each of the sides then contains 8 balls less than each side of the triangle did. Determine the initial numbers of balls.

113. Let the sequence $a_1, a_2, a_3, ..., a_n$ from an A.P. and let $a_1 = 0$, prove that

$$\frac{a_3}{a_2} + \frac{a_4}{a_3} + \frac{a_5}{a_4} + ... + \frac{a_n}{a_{n-1}} - a_2\left(\frac{1}{a_2} + \frac{1}{a_3} + ... + \frac{1}{a_{n-2}}\right) = \frac{a_{n-1}}{a_2} + \frac{a_2}{a_{n-1}}.$$

114. Sum the series $1 + \frac{\sqrt{2}-1}{2\sqrt{3}} + \frac{3-2\sqrt{2}}{12} + \frac{5\sqrt{2}-7}{24\sqrt{3}} + \frac{17-12\sqrt{2}}{144} + ...\infty$.

115. Let r be the common ratio of the G.P. $a_1, a_2, ...$, show that

$$\frac{1}{a_1^m + a_2^m} + \frac{1}{a_2^m + a_3^m} + ... + \frac{1}{a_{n-1}^m + a_n^m} = \frac{1 - r^{m(1-n)}}{a_1^m (r^m - r^{-m})}.$$

116. Find $\lim_{n\to\infty} n^{-n^2}\left((n+1)\left(n+\frac{1}{2}\right)\left(n+\frac{1}{2^2}\right)...\left(n+\frac{1}{2^{n-1}}\right)\right)^n$.

117. An A.P., a G.P. and a H.P. have a and b for their first two terms. Show that their $(n + 2)$th terms will be in G.P. if $\frac{b^{2n+2} - a^{2n+2}}{ab(b^{2n} - a^{2n})} = \frac{n+1}{n}$.

118. If a, b, c are in A.P., α, β, γ in H.P., $a\alpha$, $b\beta$, $c\gamma$ in G.P., (with common ratio not equal to 1.), then prove that $a : b : c = \frac{1}{\gamma} : \frac{1}{\beta} : \frac{1}{\alpha}$.

119. Consider n A.P.s whose first terms are 1, 2, 3, ..., n and their common differences are 1, 2, 3, ..., n respectively. If $S_{i,\,i}$ denotes sum of i terms of ith A.P., then find $S_{1,\,1} + S_{2,\,2} + S_{3,\,3} + ... + S_{n,\,n}$.

120. On the ground are placed n stones, the distance between the first and second is one yard, between the 2nd and 3rd is 3 yards, between the 3rd and 4th, 5 yards, and so on. How far will a person have to travel who shall bring them one by one to a basket placed at the first stone?

121. Sum to n terms the series and sum of infinite series $\ln\left(1-\frac{1}{2^2}\right)+\ln\left(1-\frac{1}{3^2}\right)+\ln\left(1-\frac{1}{4^2}\right)+.....$

122. Prove that $1-\frac{1}{n+1}<\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+.....+\frac{1}{n^2}<2-\frac{1}{n}$.

123. Prove that each number is the square of an odd integer in the sequence 49, 4489, 444889, ... in which every number is formed by inserting 48 in the middle of the previous number as indicated.

124. If $a > b > 0$ and $n \in N$, prove that $a^n - b^n \geq n(a-b)(ab)^{\frac{n-1}{2}}$.

125. Given $|x| < 1$, sum to infinite terms $\frac{1}{(1-x)(1-x^3)}+\frac{x^2}{(1-x^3)(1-x^5)}+\frac{x^4}{(1-x^5)(1-x^7)}+.....$.

126. Let $f_1(x)=\frac{x}{3}+10$ for all $x \in R$ and $f_n(x)=f_1(f_{n-1}(x))$ for $n \geq 2$. Then find $f_n(x)$.

127. Evalute the determinant $\begin{vmatrix} 1^2 & 2^2 & 3^2 & 4^2 \\ 2^2 & 3^2 & 4^2 & 5^2 \\ 3^2 & 4^2 & 5^2 & 6^2 \\ 4^2 & 5^2 & 6^2 & 7^2 \end{vmatrix}$

128. Prove that $\begin{vmatrix} (x-2)^2 & (x-1)^2 & x^2 \\ (x-1)^2 & x^2 & (x+1)^2 \\ x^2 & (x+1)^2 & (x+2)^2 \end{vmatrix} = -8$

129. If $2s = a + b + c$, prove that $\begin{vmatrix} a^2 & (s-a)^2 & (s-a)^2 \\ (s-b)^2 & b^2 & (s-b)^2 \\ (s-c)^2 & (s-c)^2 & c^2 \end{vmatrix} = 2s^3(s-a)(s-b)(s-c)$.

130. Show that the system of equations

$3x - y + 4z = 3$

$x + 2y - 3z = -2$

$6x + 5y + \lambda z = -3$

has at least one solution for any real number λ. Find the set of solutions if $\lambda = -5$.

131. Prove that $\frac{\cos(A+B+C)+\cos(-A+B+C)+\cos(A-B+C)+\cos(A+B-C)}{\sin(A+B+C)+\sin(-A+B+C)-\sin(A-B+C)+\sin(A+B-C)}=\cot B$

132. Prove that $\frac{\sec 8A-1}{\sec 4A-1}=\frac{\tan 8A}{\tan 2A}$

133. Prove that $\tan\alpha+\tan\beta+\tan\gamma-\tan\alpha\tan\beta\tan\gamma=\dfrac{\sin(\alpha+\beta+\gamma)}{\cos\alpha\cos\beta\cos\gamma}$

134. Prove that $\sin^3\alpha+\sin^3 2\alpha+\sin^3 3\alpha+.....$ to n terms

$=\dfrac{3}{4}\sin\dfrac{(n+1)\alpha}{2}\sin\dfrac{n\alpha}{2}\operatorname{cosec}\dfrac{\alpha}{2}-\dfrac{1}{4}\sin\dfrac{3(n+1)\alpha}{2}\sin\dfrac{3n\alpha}{2}\operatorname{cosec}\dfrac{3\alpha}{2}$.

135. If $A=\dfrac{\pi}{2^{n+1}}$, $B=\dfrac{\pi}{2^{n+2}}$, then prove that

$(\cos A+\cos B)(\cos 2A+\cos 2B)(\cos 2^2A+\cos 2^2B).....(\cos 2^nA+\cos 2^nB)=\dfrac{1}{2^{n+1}}\left(\cos\dfrac{\pi}{2^{n+2}}-\cos\dfrac{\pi}{2^{n+1}}\right)^{-1}$.

136. Prove that $\tan\dfrac{\pi}{7}\tan\dfrac{2\pi}{7}\tan\dfrac{3\pi}{7}=\sqrt{7}$

137. Prove that $\left(\tan^2\dfrac{\pi}{7}+\tan^2\dfrac{2\pi}{7}+\tan^2\dfrac{3\pi}{7}\right)\left(\cot^2\dfrac{\pi}{7}+\cot^2\dfrac{2\pi}{7}+\cot^2\dfrac{3\pi}{7}\right)=105$

138. Prove that $\sin\dfrac{2\pi}{7}+\sin\dfrac{4\pi}{7}+\sin\dfrac{8\pi}{7}=\dfrac{\sqrt{7}}{2}$.

139. If $c\cos^3\theta+3c\cos\theta\sin^2\theta=m$ and $c\sin^3\theta+3c\cos^2\theta\sin\theta=n$, prove that $(m+n)^{\frac{2}{3}}+(m-n)^{\frac{2}{3}}=2c^{\frac{2}{3}}$.

140. If $\dfrac{\tan(\theta+x)}{a}=\dfrac{\tan(\theta+y)}{b}=\dfrac{\tan(\theta+z)}{c}$ prove that $\dfrac{a+b}{a-b}\sin^2(x-y)+\dfrac{b+c}{b-c}\sin^2(y-z)+\dfrac{c+a}{c-a}\sin^2(z-x)=0$.

141. If $\dfrac{\cos x}{a}=\dfrac{\cos(x+y)}{b}=\dfrac{\cos(x+2y)}{c}=\dfrac{\cos(x+3y)}{d}$, prove that $b(b+d)=c(c+a)$.

142. If $a\cot^2\alpha+b\cot^2\beta=1$, $a\cos^2\alpha+b\cos^2\beta=1$ and $a\sin\alpha=b\sin\beta$, then prove that $(a^2-b^2)^2+ab=0$.

143. Let $\cos\alpha=\cos\beta\cos\phi=\cos\gamma\cos\theta$ and $\sin\alpha=2\sin\dfrac{\phi}{2}\sin\dfrac{\theta}{2}$, prove that $\tan^2\dfrac{\alpha}{2}=\tan^2\dfrac{\beta}{2}\tan^2\dfrac{\gamma}{2}$.

144. If $\cos\theta=\cos\alpha\cos\beta$ and $\cos\phi=\cos\gamma\cos\beta$ where $\cos\beta\neq 0$ and $\tan\dfrac{\beta}{2}=\tan\dfrac{\theta}{2}\tan\dfrac{\phi}{2}$, prove that $\sin^2\beta=(\sec\alpha-1)(\sec\gamma-1)$.

145. If cos x = tan y, cos y = tan z, cos z = tan x, prove that sin x = sin y = sin z = 2sin 18º.

146. If $\sqrt{2}\cos A=\cos B+\cos^3 B$ and $\sqrt{2}\sin A=\sin B-\sin^3 B$, show that $\sin(A-B)=\pm\dfrac{1}{3}$.

147. If an angle θ be divided into two parts such that the tangent of one part is m times the tangent of the other, then prove that their difference ϕ is obtained by the equation $\sin\phi=\dfrac{m-1}{m+1}\sin\theta$.

148. Given that the angles α, β, γ are connected by the relation

$2\tan^2\alpha\tan^2\beta\tan^2\gamma+\tan^2\alpha\tan^2\beta+\tan^2\beta\tan^2\gamma+\tan^2\gamma\tan^2\alpha=1$, find the value of $\sin^2\alpha+\sin^2\beta+\sin^2\gamma$.

149. Prove that $x^2 - x\cos(A+B)+1$ is a factor of

$2x^4 + 4x^3 \sin A \sin B - x^2(\cos 2A + \cos 2B) + 4x\cos A\cos B - 2$. Also find the other factor.

150. Show that for all values of θ, the expression $a\sin^2\theta + b\sin\theta\cos\theta + c\cos^2\theta$ lies between

$\frac{a+c}{2} - \frac{\sqrt{b^2+(a-c)^2}}{2}$ and $\frac{a+c}{2} + \frac{\sqrt{b^2+(a-c)^2}}{2}$.

151. Prove that $\frac{\sin\alpha}{\sin\beta} + \frac{\sin\beta}{\sin\alpha} = \frac{(\sin\alpha - \sin\beta)^2}{\sin\alpha\sin\beta} + 2$. Hence deduce that if $0 < \alpha, \beta < \pi$, $\frac{\sin\alpha}{\sin\beta} + \frac{\sin\beta}{\sin\alpha} \geq 2$.

152. Prove that the expression $\cos\theta(\sin\theta + \sqrt{\sin^2\alpha + \sin^2\theta})$ lies between $-\sqrt{1+\sin^2\alpha}$ and $\sqrt{1+\sin^2\alpha}$, $0 < \theta < \frac{\pi}{2}$.

153. If $0 < \theta < \pi$, prove that $\cot\frac{\theta}{4} - \cot\theta > 2$ and $\cot\frac{\theta}{2} - \cot\theta \geq 1$.

154. Solve $8\sin^6 x + 3\cos 2x + 2\cos 4x + 1 = 0$

155. Solve $\sin^2 x\cos^2 x - 10\sin x\cos^3 x + 21\cos^4 x = 0$

156. Solve $3\tan\frac{x}{2} + \cot x = \frac{5}{\sin x}$

157. Solve $4\sin^4 3x - 3\cos x + 5 = 0$

158. Solve $32\cos^6 x - \cos 6x = 1$

159. Solve $\tan x + \cot x - \cos 4x = 3$

160. Solve $\sin^8 2x + \cos^8 2x = \frac{41}{128}$

161. Solve $\sin^{10} x + \cos^{10} x = \frac{29}{64}$

162. Solve $\sin^{10} x + \cos^{10} x = \frac{29}{16}\cos^4 2x$

163. Solve $\sqrt{-3\sin 5x - \cos^2 x - 3} + \sin x = 1$

164. Solve $\tan x + \frac{1}{9}\cot x = \sqrt{\frac{1}{\cos^2 x} - 1} - 1$

165. Solve $(1+\cos x)\sqrt{\tan\frac{x}{2}} - 2 + \sin x = 2\cos x$

166. Solve $6\tan x + 5\cot 3x = \tan 2x$

167. Solve $\tan\left(x - \frac{\pi}{4}\right)\tan x\tan\left(x + \frac{\pi}{4}\right) = \frac{4\cos^2 x}{\tan\frac{x}{2} - \cot\frac{x}{2}}$

168. Solve $\sin^2 5x\left(\sin 7x\cos x-\sin\frac{x}{2}\cos\frac{3x}{2}\right)=\frac{\sin\frac{3x}{2}\cos\frac{x}{2}+\sin x\cos 7x}{1+\cot^2 5x}$

169. Solve $\sin^6 x+\sin^4 x+\cos^6 x+\cos^4 x+\sin\frac{x}{2}=3$

170. Solve $\sin\frac{x}{4}+2\cos\frac{x-2\pi}{3}=3$

171. Solve $\cot\left(\frac{\pi}{3}\cos 2\pi x\right)=\sqrt{3}$

172. Solve $2\sin^2\left(\frac{\pi}{2}\cos^2 x\right)=1-\cos(\pi\sin 2x)$

173. If α and β are distinct roots of the equation $a\cos\theta+b\sin\theta=c$, between 0 and 2π, and if $\alpha+\beta$ also satisfies the equation, show that $a = c$.

174. If θ_1 and θ_2 are two distinct values of θ, $0\le\theta_1,\ \theta_2\le 2\pi$, satisfying the equation $\sin(\theta+\alpha)=\frac{1}{2}\sin 2\alpha$ prove that $\frac{\sin\theta_1+\sin\theta_2}{\cos\theta_1+\cos\theta_2}=\cot\alpha$.

175. Solve $\sin^{-1}x+\sin^{-1}(1-x)=\cos^{-1}x$.

176. Solve $\sin^{-1}\frac{x}{2}+\cos^{-1}\left(x+\frac{\sqrt{3}}{2}\right)=\frac{\pi}{6}$

177. Solve $\cot x+\frac{\sin x}{\cos x-2}\ge 0$

178. Solve $8\sin^2\frac{x}{2}+3\sin x-4>0$

179. Solve $\tan x+\tan 2x+\tan 3x>0$

180. Solve $\sin 2x\sin 3x-\cos 2x\cos 3x>\sin 10x$

181. Solve $\cot x+\cot\left(x+\frac{\pi}{2}\right)+2\cot\left(x+\frac{\pi}{3}\right)>0$

182. Solve $\frac{\cos^2 2x}{\cos^2 x}\ge 3\tan x$

183. Solve $\tan^2(\sin^{-1}x)>1$

184. Prove that $\sin\theta\sec 3\theta=\frac{1}{2}(\tan 3\theta-\tan\theta)$ and hence find the sum to *n* terms of the series

$\sin\theta\sec 3\theta+\sin 3\theta\sec 3^2\theta+\sin 3^2\theta\sec 3^3\theta+.....$

185. If $U_n = \sin n\theta \sec^n \theta$, $V_n = \cos n\theta \sec^n \theta$, n = 0, 1, 2, ..., prove that $V_n - V_{n-1} = -U_{n-1}\tan\theta$. Hence deduce that $U_1 + U_2 + ... + U_n = \cot\theta \sec^{n+1}\theta(\cos^{n+1}\theta - \cos(n+1)\theta)$.

186. Find the intervals of monotonicity of the function $f(x) = 3\cos^4 x + 10\cos^3 x + 6\cos^2 x - 3$, $0 \le x \le \pi$.

187. Prove that $\dfrac{(a+b+c)^2}{a^2+b^2+c^2} = \dfrac{\cot\frac{A}{2} + \cot\frac{B}{2} + \cot\frac{C}{2}}{\cot A + \cot B + \cot C}$

188. Prove that $a^3\cos(B-C) + b^3\cos(C-A) + c^3\cos(A-B) = 3abc$

189. Prove that $a^2\cos 2B + b^2\cos 2A + 2ab\cos(A-B) = c^2$

190. Prove that $\dfrac{1+\cos(A-B)\cos C}{1+\cos(A-C)\cos B} = \dfrac{a^2+b^2}{a^2+c^2}$

191. Prove that $\dfrac{(a+b+c)(b+c-a)(c+a-b)(a+b-c)}{4b^2c^2} = \sin^2 A$.

192. Prove that $2abc\cos\dfrac{A}{2}\cos\dfrac{B}{2}\cos\dfrac{C}{2} = 2s\Delta$

193. In any triangle, if $\tan\dfrac{A}{2} = \dfrac{5}{6}$ and $\tan\dfrac{B}{2} = \dfrac{20}{37}$, find $\tan\dfrac{C}{2}$ and prove that in this triangle $a + c = 2b$.

194. The sides of a triangle are in A.P. and the greatest and least angles are θ and ϕ; prove that

$4(1-\cos\theta)(1-\cos\phi) = \cos\theta + \cos\phi$

195. If the angles of a triangle are in the ratio 1 : 2 : 4, then prove that $a^2b^2c^2 = (b^2-a^2)(c^2-b^2)(c^2-a^2)$.

196. If $B = 3C$, prove that $\cos C = \sqrt{\dfrac{b+c}{4c}}$, $\sin C = \sqrt{\dfrac{3c-b}{4c}}$ and $\sin\dfrac{A}{2} = \dfrac{b-c}{2c}$.

197. *ABCD* is a trapezium such that *AB* is parallel to *CD* and *CB* is perpendicular to them. If $\angle ADB = \theta$, $BC = p$ and $CD = q$, show that $AB = \dfrac{(p^2+q^2)\sin\theta}{p\cos\theta + q\sin\theta}$.

198. In any triangle *ABC* if *D* be any point of the base *BC*, such that *BD* : *DC* = *m* : *n*, and if $\angle BAD = \alpha$, $\angle DAC = \beta$, and $\angle CDA = \theta$ and *AD* = *x*, prove that $(m+n)\cot\theta = m\cot\alpha - n\cot\beta = n\cot B - m\cot C$ and $(m+n)^2 . x^2 = (m+n)(mb^2 + nc^2) - mna^2$.

199. A ring, 10 cm in diameter, is suspended from a point 12 cm above its center by 6 equal strings attached to its circumference at equal intervals. Find the cosine of the angle between consecutive strings.

200. The side of a base of a square pyramid is *a* metre and it's vertex is at a height of *h* metre above the center of the base. If θ and ϕ be respectively the inclinations of any face to the base and of any two faces to one another, prove that $\tan\theta = \dfrac{2h}{a}$ and $\cot\dfrac{\phi}{2} = \sqrt{1+\dfrac{a^2}{2h^2}}$.

201. If p_1, p_2, p_3 are altitudes of a triangle *ABC* from the vertices *A*, *B*, *C* and Δ the area of a triangle, then prove that $p_1^{-2} + p_2^{-2} + p_3^{-2} = \dfrac{a^2+b^2+c^2}{4\Delta^2}$.

202. If $\frac{a}{1+m^2n^2} = \frac{b}{m^2+n^2} = \frac{c}{(1-m^2)(1+n^2)}$, then prove that $\tan\frac{A}{2} = \frac{m}{n}$, $\tan\frac{B}{2} = mn$ and $\Delta = \frac{mnbc}{m^2+n^2}$.

203. The sides a, b, c of a triangle are the roots of the equation $x^3 - px^2 + qx - r = 0$. Prove that

$\Delta = \frac{1}{4}\sqrt{p(4pq - p^3 - 8r)}$.

204. If D is the mid-point of BC and AD is perpendicular to AC, then prove that $\cos A\cos C = \frac{2(c^2 - a^2)}{3ac}$.

205. The base angles of a triangle are $22\frac{1}{2}^\circ$ and $112\frac{1}{2}^\circ$. Show that the base is equal to twice the height.

206. Prove that the perpendicular from A divides BC into portions which are proportional to the cotangents of the adjacent angles and that it divides the angle A into portions whose cosines are inversely proportional to the adjacent sides.

207. If p_1, p_2, p_3 are altitudes of a triangle ABC from the vertices A, B, C and Δ the area of the triangle, then prove that $p_1^{-1} + p_2^{-1} - p_3^{-1} = \frac{s-c}{\Delta}$.

208. If p, q, r are the lengths of internal bisectors of the angles A, B, C respectively, then prove that

$\frac{1}{p}\cos\frac{A}{2} + \frac{1}{q}\cos\frac{B}{2} + \frac{1}{r}\cos\frac{C}{2} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$.

209. In a right angled triangle ABC, the bisector of the right angle C divides AB into segments p and q and if $\tan\frac{A-B}{2} = t$, then show that $p : q = (1 - t) : (1 + t)$.

210. Prove that

(i) $OP^2 = 9R^2 - (a^2 + b^2 + c^2)$

(ii) $OG^2 = R^2 - \frac{1}{9}(a^2 + b^2 + c^2)$

211. Prove that $a\cot A + b\cot B + c\cot C = 2(R + r)$

212. If $C = 90^\circ$, prove that $R + r = \frac{1}{2}(a + b)$ and $\frac{c}{r} = \frac{c+a}{b} + \frac{c+b}{a}$.

213. If x, y, z are respectively the distance of the vertices from orthocentre, prove that $x + y + z = 2(R + r)$.

214. If x, y, z are the distances of the vertices of a triangle from the corresponding points of contact with the incircle, prove that $\frac{xyz}{(x+y+z)} = r^2$.

215. If R_1, R_2 and R_3 are respectively the radii of the circumcircles of the triangle IBC, ICA and IAB, prove that $R_1R_2R_3 = 2R^2r$.

216. If x, y, z be the lengths of the tangents drawn to the incircle parallel to the sides BC, AC, AB respectively and intercepted between the other two sides, then prove that $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

217. Prove that $r_2r_3 + r_3r_1 + r_1r_2 = s^2$.

218. Prove that $\frac{1}{r_1}+\frac{1}{r_2}+\frac{1}{r_3}=\frac{1}{r}$.

219. Prove that $r_1 + r_2 + r_3 - r = 4R$.

220. Prove that $\left(\frac{1}{r}-\frac{1}{r_1}\right)\left(\frac{1}{r}-\frac{1}{r_2}\right)\left(\frac{1}{r}-\frac{1}{r_3}\right)=\frac{16R}{r^2(a+b+c)^2}$

221. Prove that $R=\frac{(r_1+r_2)(r_2+r_3)(r_3+r_1)}{4(r_1r_2+r_2r_3+r_3r_1)}$

222. Two circles, of radii *a* and *b*, cut each other at an angle θ. Prove that the length of the common chord is $\frac{2ab\sin\theta}{\sqrt{a^2+b^2+2ab\cos\theta}}$

223. Three equal circles touch one another. Find the radius of the circle which touches all three.

224. Three circles, whose radii are *a*, *b* and *c* touch one another externally and the tangents at their points of contact meet in a point. Prove that the distance of this point from either of their points of contact is $\left(\frac{abc}{a+b+c}\right)^{\frac{1}{2}}$.

225. Of two regular polygons of *n* sides, one circumscribed and the other is inscribed in a given circle, prove that the perimeters of the circumscribing polygon, the circle and the inscribed polygon are in the ratio $\sec\frac{\pi}{n}:\frac{\pi}{n}\text{cosec}\frac{\pi}{n}:1$ and that the areas of the polygons are in the ratio $\cos^2\frac{\pi}{n}:1$.

226. A pyramid stands on a regular hexagon as base. The perpendicular from the vertex of the pyramid on the base passes through the center of the hexagon and its length is equal to that of a side of the base. Find the tangent of the angle between the base and any face of the pyramid and also of half the angle between any two side faces.

227. A regular pyramid has for its base a polygon of *n* sides and each slant face consists of an isosceles triangle of vertical angle 2α. If the slant faces are each inclined at an angle β to the base and at an angle 2γ to one another, show that $\cos\beta=\tan\alpha\cot\frac{\pi}{n}$ and $\cos\gamma=\sec\alpha\cos\frac{\pi}{n}$.

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